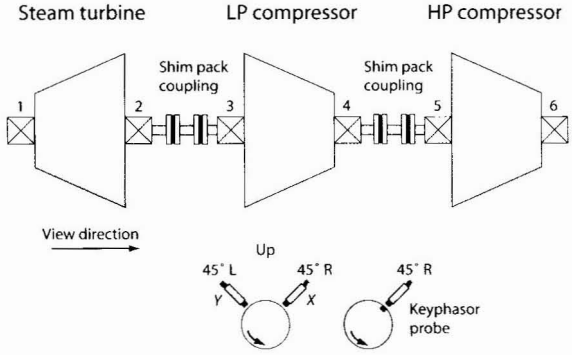



**Figure 25-1.** The synthesis gas (syngas) compressor train. An HP/LP steam turbine drives LP and HP barrel compressors in tandem.

**Table 25-1.** Operating parameters for the syngas compressor train.

| | |
|---|---|
| **Steam Turbine** | |
| Power | 22 MW (30 000 hp) |
| Inlet press. | 52 bar (750 psia) |
| Inlet temp. | 400 °C (750 °F) |
| Exhaust press. | 1 bar (15 psia) |
| Exhaust temp. | 130 °C (270 °F) |
| Trip speed | 11 200 rpm |
| First critical | 3650 rpm |
| Max cont. speed | 10 800 rpm |
| Bearing type | 5 pad tilting pad |
| Diametral clearance | 300 µm (12 mil) |
| **LP Compressor** | |
| Gas | Reformed natural gas |
| Capacity | 300 $m^3$/min (10 500 $ft^3$/min) |
| Speed | 10 800 rpm |
| Inlet conditions | 1.7 bar @ 38 °C (25 psia @ 100 °F) |
| Discharge conditions | 17 bar @ 107 °C (250 psia @ 225 °F) |
| First resonance | 5200 rpm |
| Bearing type | 5 pad tilting pad |
| Diametral clearance | 225 µm (9 mil) |
| **HP Compressor** | |
| Gas | Reformed natural gas |
| Capacity | 130 $m^3$/min (4500 $ft^3$/min) |
| Speed | 10 800 rpm |
| Inlet conditions | 12 bar @ 120 °C (175 psia @ 250 °F) |
| Discharge conditions | 31 bar @ 220 °C (450 psia @ 425 °F) |
| First resonance | 4300 rpm |
| Bearing type | 5 pad tilting pad |
| Diametral clearance | 225 µm (9 mil) |

reaching 50 μm pp (2.0 mil pp) at 10,200 rpm. At that point, operators slowed the machine train down to avoid damage. At 9500 rpm, vibration was acceptable, but plant output was limited to 85% of capacity.

The problem had to be resolved quickly. The plant could not operate at full capacity until the compressor was repaired. Any shutdown for repairs would be expensive; each day the machine train was shut down, they would have to vent product to a flare stack and stop downstream production. Deadlines were approaching for both the owner's acceptance of the plant and for delivery contracts. The machine had to be repaired quickly, in a single shutdown. The contractor asked Bently Nevada machinery specialists to document the machine behavior and help determine the cause of the compressor vibration.

### Steady State Analysis

Bently Nevada specialists set up their equipment while the machine was running. There were only eight channels available in the data acquisition equipment, enough for only two machines. Because coupling unbalance was suspected, they decided to concentrate on the steam turbine and the LP compressor. When they were ready, the compressor was brought back up to 10,385 rpm and they collected steady state machinery data and began to analyze the problem.

Orbit plots of data from the inboard turbine bearing (bearing 2) and the inboard LP compressor bearing (bearing 3) showed similar behavior (Figure 25-2). At both locations, the shaft moved in large, nearly circular orbits, with a predominant, 1X frequency component. However, the 1X vibration amplitude at bearing 3 was more than twice the vibration amplitude at bearing 2.

The compressor manufacturer (OEM) and the plant construction engineers suspected that the replacement coupling was unbalanced because

1. Compressor vibration amplitudes had been significantly lower in factory tests than on-site. The turbine and LP and HP compressors were shop tested prior to shipment and ran very well; vibration amplitudes were on the order of 4 - 6 μm pp (about 0.2 mil pp).

2. The vibration measured on-site had properties that were typical of an unbalanced shaft: a circular orbit, predominantly 1X vibration, and vibration amplitude that appeared to increase in proportion to the square of rotor speed.

The OEM and construction engineers attributed the difference in vibration amplitudes to the great difference in the masses of the turbine and compressor

rotors. The mass of the turbine rotor was approximately eight times greater than that of the compressor rotor.

Based on this analysis, a decision was made to balance the coupling, and they developed a plan to minimize downtime. Before shutdown, they prepared the tooling and calculated the balance weights they would need. At the same time, the Bently Nevada specialists prepared to capture shutdown (transient) data.

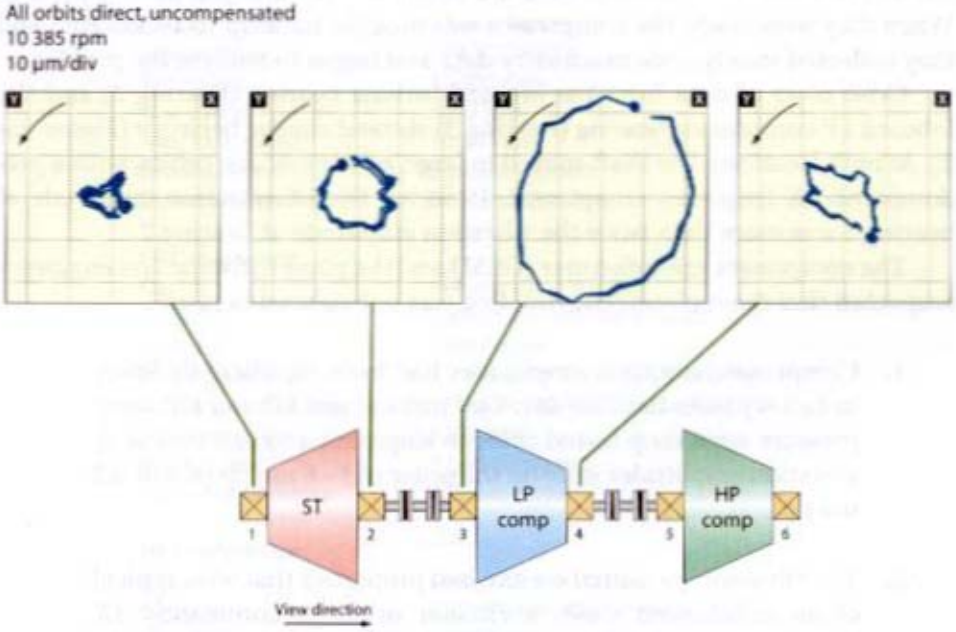



**Figure 25-2.** Steady state orbits at full speed before shutdown. The uncompensated, direct data shows that the orbits at bearings 2 and 3 are predominantly 1X and circular. The bearing 3 orbit is around 50 µm pp (2.0 mil pp), suggesting an unbalance problem.

## Transient Data Analysis

Transient data was collected during the machine shutdown. While maintenance crews prepared to balance the coupling, the Bently Nevada specialists and engineers reviewed the new data.

With slow roll data available, the high-speed orbits could be waveform compensated. The orbits then revealed a different behavior (Figure 25-3). The *compensated* vibration from the inboard turbine bearing (bearing 2) was actually very small. Most of the uncompensated vibration at bearing 2 was actually due to runout. The compensated orbits showed that most of the vibration was centered at the LP compressor bearing 3. Also, the bearing 4 compensated orbit showed a flattening in the lower left quadrant that suggested a high radial load.

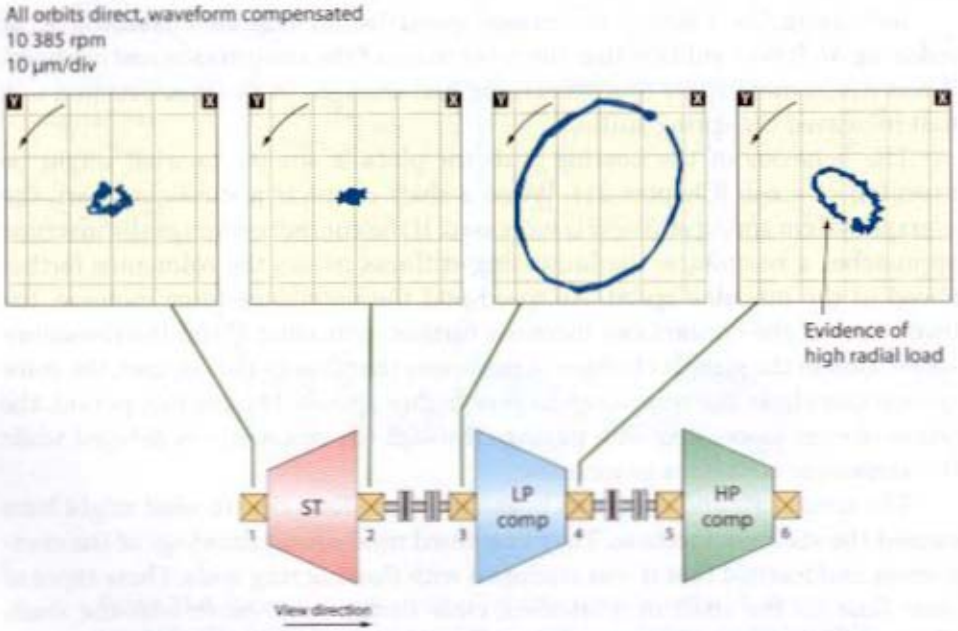



**Figure 25-3.** Waveform compensated orbits. Transient data allowed capture of slow roll waveforms, which were used to compensate the full speed orbits. A very different dynamic picture emerged. The compensated orbit at bearing 2 is quite small, which does not suggest an unbalance problem. The orbit at bearing 3 is little changed, while the bearing 4 orbit is partially flattened, suggesting the presence of a high radial load.

A more important piece of information was discovered. According to factory test data, the LP rotor first balance resonance should have been between 4900 and 5400 rpm. Polar and Bode plots of the shutdown data (Figure 25-4) showed some evidence of a resonance in this speed range. However, above the resonance region, the amplitude increased approximately as the square of rotor speed, while the phase lag continued to change at a constant rate. The behavior of the vibration amplitude implied that the resonance had shifted to somewhere near or above the 10,480 rpm operating speed.

A machine's balance resonance speed is a function of the rotor mass and spring stiffness:

$$\Omega_{res} = \sqrt{\frac{K}{M}} \tag{25-1}$$

where $K$ is the rotor system spring stiffness, and $M$ is the rotor mass (see Chapter 11).

Increasing the balance resonance speed would require increasing $K$ or reducing $M$. It was unlikely that the rotor mass of the compressor had changed; it was much more likely that something had changed in the machine in a way that increased the spring stiffness.

The behavior of the bearing 3 vector plots is similar to what might be expected of a rub (Chapter 21). When a shaft contacts a stationary part, the average system spring stiffness is increased. If the contact occurs as the machine approaches a resonance, the increasing stiffness moves the resonance farther ahead of the machine speed. As speed and the unbalance force increase, the dwell time of the contact can increase, further increasing $K$ and the resonance speed. Unless the system changes in some way that breaks the contact, the rotor system can chase the resonance to ever higher speeds. During this period, the phase change associated with passage through the resonance is delayed while the amplitude continues to increase.

The specialists and engineers began to search for a clue to what might have caused the stiffness increase. They examined mechanical drawings of the compressor and learned that it was equipped with floating ring seals. These types of seals float on the shaft in lubricated, close contact and move with the shaft. These seals have been known to lock up, preventing them from floating properly. The locked seal then acts much like a very close clearance bearing, constraining shaft motion and increasing the rotor system spring stiffness.

The specialists theorized that the floating seal near bearing 3 had locked up, resulting in behavior like a rub. The increased spring stiffness could have shift-

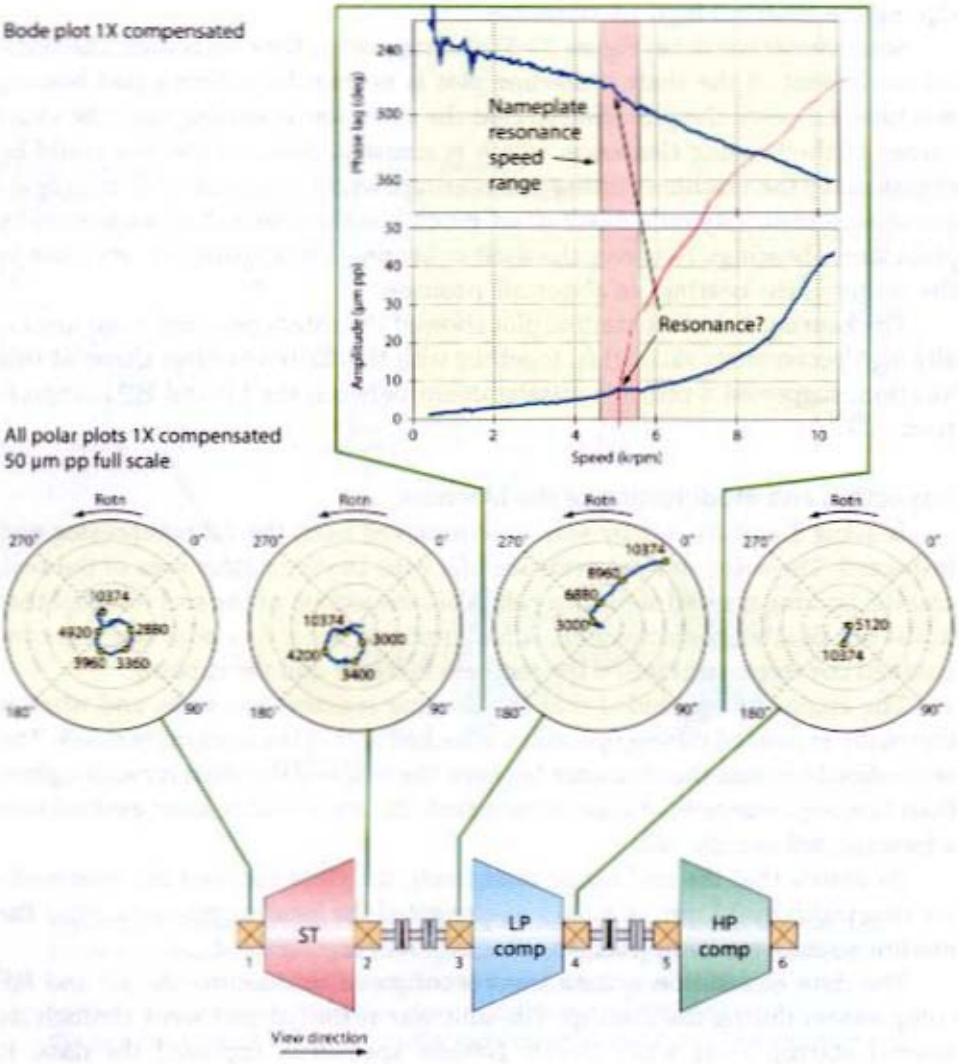



**Figure 25-4.** Transient shutdown data. The polar plots show the 1X, compensated, shutdown data from the *X* probes at bearings 1 through 4, and the Bode plot shows bearing 3 data. The data shows some evidence of a resonance in the nameplate speed range (nominally 5200 rpm). Above the resonance region, though, the amplitude increases dramatically, and the phase lag continues to change at a constant rate. This behavior implies that the resonance has shifted to somewhere near or above the 10 480 rpm operating speed.

ed the compressor's balance resonance to near its operating speed. The nearby resonance would amplify the normal unbalance response of the machine, producing the observed high 1X vibration.

Shaft centerline data (Figure 25-5) also supported the conclusion. The vertical movement of the shaft centerline plot is normal for a tilting pad bearing machine; however, the plot showed that the rotor was operating near the exact center of the bearing clearance, which is unusual. Some of the rise could be explained by the machine's tilting pad bearings, which can cause a shaft to operate at an eccentricity ratio of 0.2 or so, much less than for a shaft supported by plain sleeve bearings. However, the shaft in bearing 3 was operating very close to the center of the bearing, an abnormal position.

The bearing 4 shaft centerline plot showed the rotor operating at an unusually *high* eccentricity ratio; this, together with the flattened orbit shape at this location, suggested a possible misalignment between the LP and HP compressors.

### Inspection and Modification of the Machine

Bearing 3 and its nearby seal were removed from the LP compressor and inspected. The rotor showed evidence of a wipe in the contact area of the seal, and the bearing showed signs of a rub. Also, inspection of the seal revealed that it was not floating correctly; thus, it had probably acted as a poor bearing, constrained the rotor, and caused the stiffness increase and the rubbing.

The engineers concluded that the floating seal was too wide, and when it thermally expanded during operation, it locked within the stationary sleeve. The seal rubbed because the clearance between the seal and the shaft is much tighter than bearing clearances. As speed increased, the rotor/seal contact evolved into a forward, full annular rub.

To ensure that the seal would float freely, the OEM reduced the intermediate ring width by 0.1 mm (4 mils) and elongated the locating pin slots. After the modifications were completed, the machine was reassembled.

The data acquisition system was reconfigured to examine the LP and HP compressors during the startup. The unit was restarted and went through its normal startup cycle while Bently Nevada specialists captured the data. It showed that the LP compressor balance resonance was now between 4900 and 5400 cpm (Figure 25-6), consistent with factory test data. The phase change through resonance was now normal at the inboard LP compressor bearing. At full speed, vibration at bearing 3 had decreased, from 44 μm pp (1.7 mil pp) to 3 μm pp (0.1 mil pp). Note that, above the resonance, the Bode plot shows a small jump in both amplitude and phase near 5700 rpm that suggests the presence of slight rub in the machine, perhaps at another location.

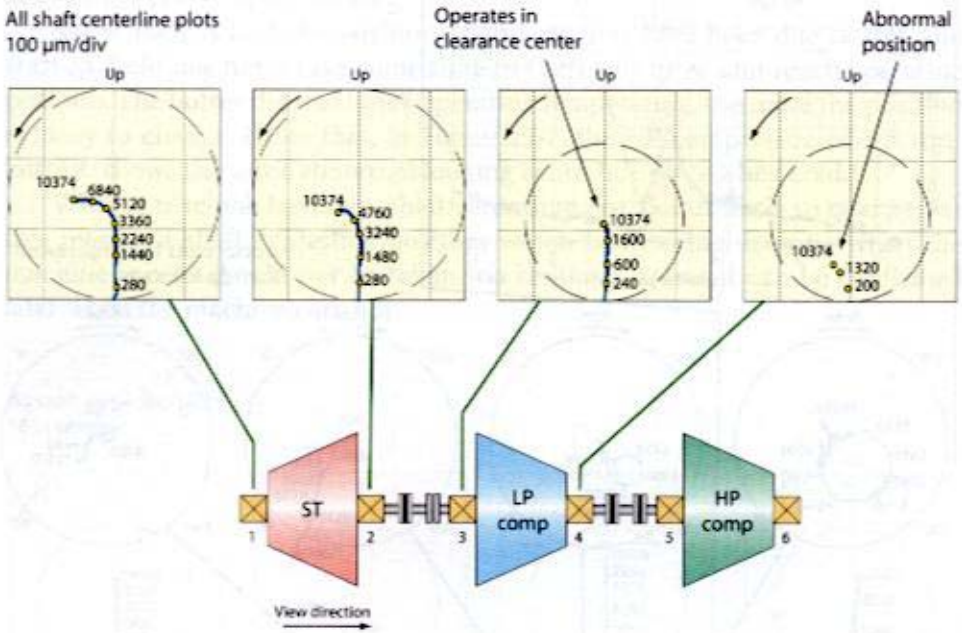



**Figure 25-5**. Shutdown shaft centerline plots. The vertical movement of the shaft centerline is normal for a tilting pad bearing machine. At bearing 3, the rotor operates near the exact center of the bearing clearance, which is unusual. At bearing 4, the rotor operates at an unusually *high* eccentricity ratio; this and the flattened orbit shape at this location suggests a possible misalignment between the LP and HP compressors.

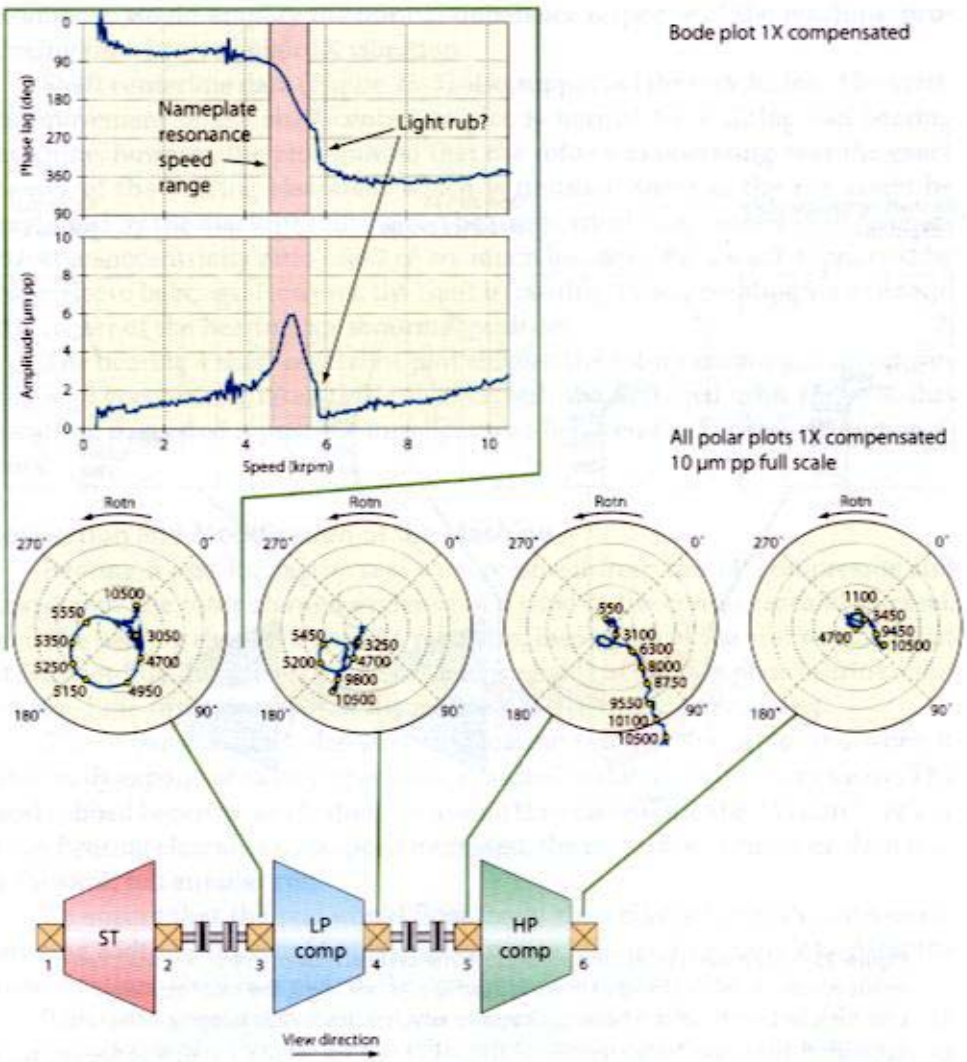



**Figure 25-6**. Startup data after repair of the seal near bearing 3. The data is for bearings 3 through 6, and the scale is now 10 µm pp full scale. The LP rotor exhibits a normal looking, in-phase resonance, and high-speed vibration levels are normal. Immediately above the resonance, the Bode plot shows a jump in both amplitude and phase that suggests that a slight rub still exists in the machine.

Although vibration amplitudes at bearing 3 were low at operating speed, the shaft centerline plots (Figure 25-7) still indicated abnormal shaft positions at running speed. The HP compressor rotor appeared to be operating too high in the bearings, and LP bearing 3 also showed a high rotor position. It is possible that the actual bearing clearance was greater than reported by the OEM, in which case the data plots would not accurately represent the rotor position relative to the center of the bearing.

More likely is that the position anomalies may have been due to the cold startup. Cold machines take some time to thermally grow and reach operating position. The hotter the machine's operating temperature, the more the position is likely to change. (Note that, in Figure 25-7, the HP compressor, which runs hottest, shows the most abnormal looking centerline plots when cold.)

While a machine heats up, shaft centerline position is likely to change; for this reason, a shaft centerline position which is abnormal looking when the machine is cold should not be taken too seriously unless it can be confirmed later when the machines are hot.

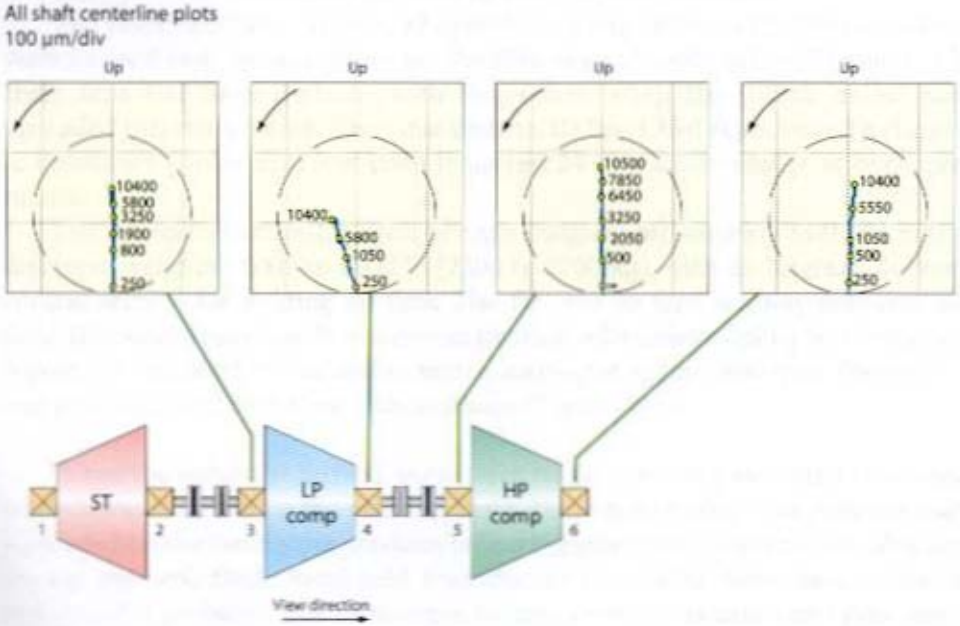



**Figure 25-7**. Cold startup shaft centerline plots after the seal repair. The rotor positions in bearings 3 and 4 have changed and are higher (compare to Figure 24-5). The rotor also appears to be too high in bearings 5 and 6. All of this data is somewhat suspect because the machines have not had time to reach operating temperature.

## Summary

Transient data is almost always needed for machinery diagnostics, although it is usually more difficult to obtain than steady state data. However, steady state data contains significantly less information, and it can be misleading because of the relatively small amount of information available. You should always be careful when attempting to diagnose a machine problem using only steady state data.

In this case, the steady state data and some historical information seemed to indicate that the problem was a coupling unbalance. Because slow roll data was not available, the uncompensated orbit from the inboard turbine bearing was interpreted as being due to unbalance. The transient data revealed much more information: the compensated orbits showed different behavior, and the Bode and polar plot data pointed toward a rub of some sort. Rub is always a secondary malfunction that is caused by some other root cause; in this case, the underlying problem was the locked floating seal.

Chapter 26

# Chronic High Vibration in a Draft Fan

A BENTLY NEVADA MACHINERY SPECIALIST WAS ASKED to balance an induced draft (ID) fan at a grain distillery. The fan was located on the sixth floor of a sixty-year-old power house, a very flexible support structure.

The plant had three ID fans, all operating at the same speed. The other two were located next to each other on the fifth floor directly below ID Fan 1. All three fans had been replaced with larger fans when the plant's boiler was upgraded two years earlier. Since installation, ID Fan 1 had experienced a chronic unbalance problem; it had been balanced 24 times, an average of once per month.

The fan was a horizontal, squirrel cage design approximately 15 ft (4.6 m) in diameter, weighing 5000 to 6000 lb (2300 to 2700 kg), with an internal, hollow, conical section for guiding air flow. The fan was located midway between its fluid-film sleeve bearings. It was driven through a flexible coupling by a 3-phase, 8-pole, 400 hp (300 kW) induction motor, turning at around 890 rpm. The motor was also equipped with fluid-film bearings (Figure 26-1).

When the specialist arrived, there were no permanently mounted vibration transducers installed on the machine train. Hand-held velocity transducers had been used on the fan during previous balancing attempts. This is undesirable for several reasons. First, hand-held transducers can suffer from measurement repeatability problems due to changes in measurement location and poor contact between the transducer and machine casing. Second, fluid-film bearings generally do not transmit shaft relative vibration to the casing very well. For this kind of machine, casing vibration amplitudes are typically less than one third of shaft relative vibration amplitudes, and the fluid-film bearing can produce a

phase delay in the casing vibration; thus, casing mounted velocity probes usually provide a poor picture of what is happening with the shaft and make accurate balancing more difficult. Third, because of the structural flexibility of the building, the two fans on the floor below (which operated at the same speed) could transmit vibration through the building structure to the casing of the fan being balanced, where it could be detected by casing-mounted velocity probes.

Before balancing, the fan was sandblasted clean to remove any accumulated fouling. *XY* shaft relative proximity transducers were installed on the fan at bearings 3 and 4 at 15° L and 75° R, and a Keyphasor transducer was installed to provide a phase reference.

Moving-coil, *XY* velocity transducers (Seismoprobes) were also installed, using magnetic bases, on the motor at 0° and 90° R near bearings 1 and 2, and on the fan at 15° L and 75° R near bearings 3 and 4. These transducers were installed because of the possibility of significant casing motion, which might affect shaft relative measurements. However, the frequency of the fan, about 890 cpm, was close to the minimum rated frequency response (600 cpm) of the velocity transducers. At 600 cpm, the amplitude would be down by 30%, and the phase would lead vibration by 90°. At running speed, a significant, phase-lead error of more than 50° could be expected. Thus, vibration data from these transducers would have to be used cautiously.

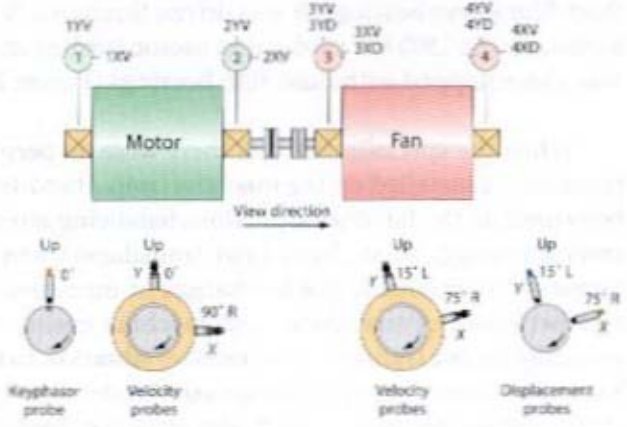



**Figure 26-1.** Machine train diagram. Only velocity transducers were installed on the motor at 0° and 90° R. Both shaft relative and velocity transducers were installed on the fan at 15° L and 75° R.

After the transducers were installed, the transducer suite was connected to a Bently Nevada, computer-based, data acquisition system. The velocity transducers on the fan measured nearly 5 integrated mil pp (130 μm pp), mostly vertical vibration, *with the fan stopped!* This was vibration being transmitted from the other two fans through the building structure. The shaft relative transducers showed no vibration.

The specialist prepared to get baseline data after the fan was brought up to full speed. With an induction motor, the startup is usually so rapid that you can't get good data (a tape recorder can help here); during coastdown, the speed usually changes more slowly, which results in better data. For that reason, power to the motor was cut, and data was taken during the coastdown.

1X, compensated polar plots of the shutdown are shown in Figure 26-2. The bottom row shows shaft relative vibration for all four fan transducers. The top row shows casing vibration, integrated to displacement, for the associated velocity transducers. The approximately 180° phase change in the velocity data is due to the *transducer* resonance at its rated frequency and does not accurately reflect casing vibration. Given the probable transducer phase error at running speed, better estimates of amplitude and phase values are indicated by the blue dots.

The two sets of polar plots show that, for each transducer location, the shaft relative and casing vibration are about the same amplitude; the (adjusted) relative phase shows that they are within 90° of each other. Thus, the two vibrations will add in a way that produces a very high shaft absolute vibration. The high level of 1X vibration suggests a large amount of unbalance.

### The First Balancing Attempt

Ideally, given the large amount of casing vibration present, it might have been best to use shaft absolute vibration for balancing. However, there was significant phase error associated with the velocity data; also, the casing vibration was approximately in phase with the shaft relative vibration. For these reasons, the specialist decided to use the shaft relative data for balancing purposes.

Since the shaft relative vibration amplitudes from the $X$ transducers were higher than from the $Y$ transducers, the specialist decided to use the 1X, compensated $X$ data for balancing the fan. The original vibration vectors, $\boldsymbol{O}$, measured at running speed, were:

$$\boldsymbol{O}_{3X} = 13.0 \text{ mil pp } \angle 259° \text{ (330 μm pp } \angle 259°)$$
$$\boldsymbol{O}_{4X} = 14.8 \text{ mil pp } \angle 282° \text{ (376 μm pp } \angle 282°)$$

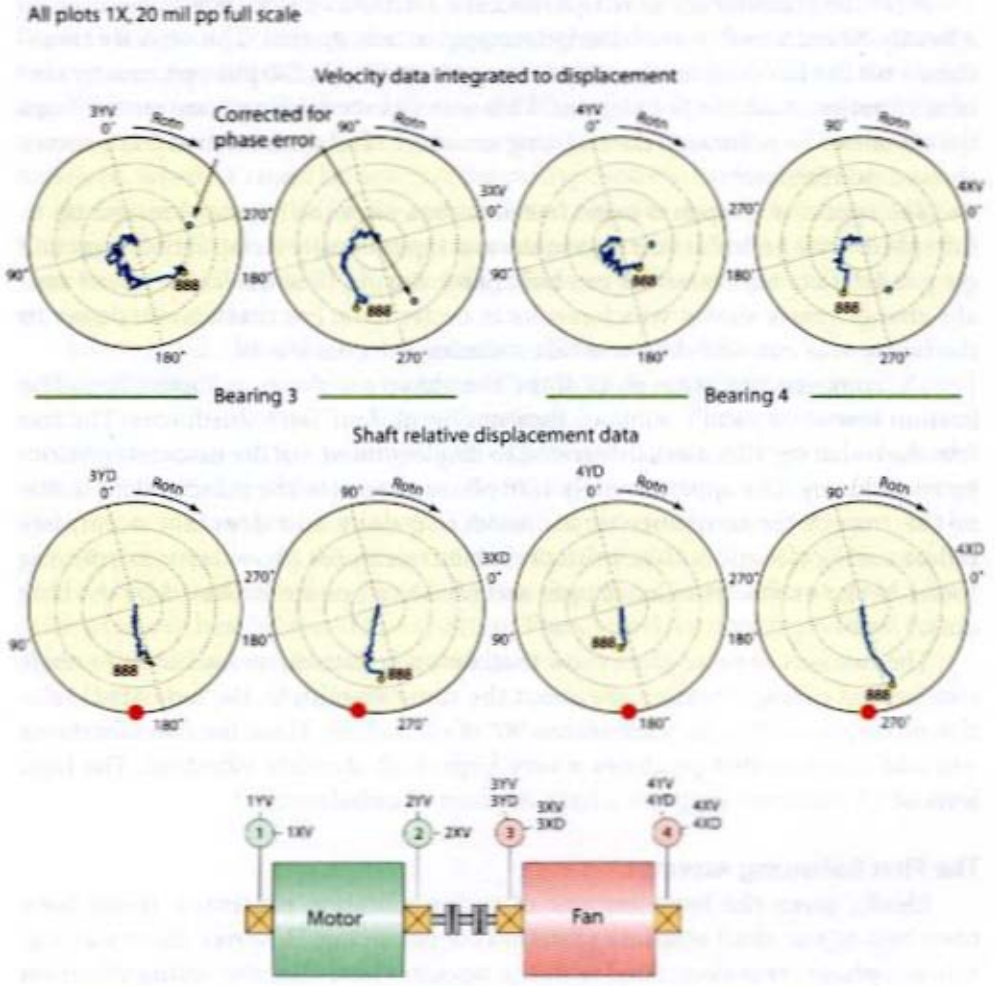



**Figure 26-2.** Original shutdown data for the fan. The bottom row shows shaft relative vibration for all four fan transducers. The top row shows velocity data, integrated to displacement, from the associated velocity transducers. The blue dots show better estimates for the running speed vibration vectors, compensating for the probable velocity transducer phase error. The two sets of polar plots show that, for each transducer location, the shaft relative and casing vibration at running speed are approximately in phase and about the same amplitude. The heavy spots indicated by the displacement data are shown as red dots.

The behavior of the shaft relative data suggests that the fan was operating below the first balance resonance. This made location of the heavy spot a relatively simple task, and the estimated heavy spot locations are shown as red dots on the shaft relative polar plots. The indicated heavy spot direction suggested adding weight approximately opposite the vibration readings, at 60° to 90° relative to the $X$ transducers. Based on previous experience with this kind of machine and the large amount of unbalance present, the specialist decided to add a calibration weight, $\boldsymbol{m}_{cal} = 12$ oz $\angle 75°$ (340 g $\angle 75°$). At this location, the weight would be opposite the indicated heavy spot and would likely reduce the unbalance. At the same time, it would serve as a calibration weight for calculating influence vectors. The weight was installed at a location close to the midspan and near the outer perimeter of the fan. Thus, the radius of attachment was nearly the fan radius, about 7.5 ft (2.3 m). A weight at this radius would produce a force of 1500 lb (6.7 kN) at running speed and affect both measurement planes nearly equally. The fan was restarted, and a new set of coastdown data was obtained.

The results were very surprising (Figure 26-3, top row). The vibration should have decreased; instead, the vibration actually *increased*, and the indicated heavy spot location moved only about 40° in the direction of rotation. The situation is summarized in the middle row of Figure 26-3. The new vibration vectors, $\boldsymbol{T}$, represent the total response of the fan to the original unbalance and the calibration weight:

$$\boldsymbol{T}_{3X} = 27.0 \text{ mil pp } \angle 213° \text{ (686 } \mu\text{m pp } \angle 213°)$$
$$\boldsymbol{T}_{4X} = 27.8 \text{ mil pp } \angle 229° \text{ (706 } \mu\text{m pp } \angle 229°)$$

The calibration weight produced a change in the vibration, $\boldsymbol{C} = \boldsymbol{T} - \boldsymbol{O}$, of

$$\boldsymbol{C}_{3X} = 20.3 \text{ mil pp } \angle 186° \text{ (516 } \mu\text{m pp } \angle 186°)$$
$$\boldsymbol{C}_{4X} = 22.3 \text{ mil pp } \angle 197° \text{ (566 } \mu\text{m pp } \angle 197°)$$

Since the fan was operating below the first balance resonance, the change in vibration should have been toward $\boldsymbol{m}_{cal}$, not away from it.

The influence vector, $\boldsymbol{H}$, can be calculated using this equation (from Chapter 16):

$$\boldsymbol{H} = \frac{\boldsymbol{C}}{\boldsymbol{m}_{cal}} \tag{26-1}$$

This yields

$$\boldsymbol{H}_{3X} = 1.69 \text{ mil pp/oz } \angle 111° \text{ (1.52 μm pp/g } \angle 111°)$$
$$\boldsymbol{H}_{4X} = 1.86 \text{ mil pp/oz } \angle 122° \text{ (1.66 μm pp/g } \angle 122°)$$

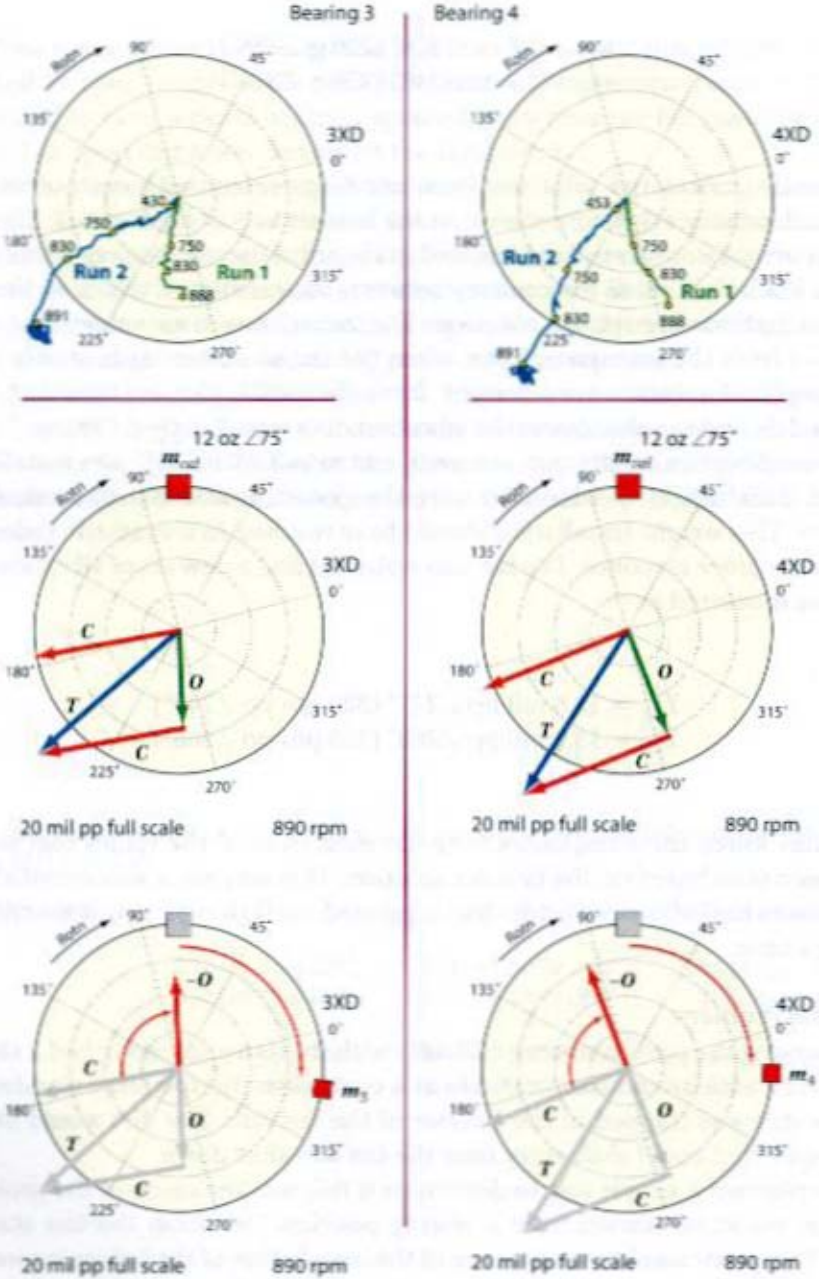
The phase of an influence vector is directly related to the heavy spot/high spot relationship in the rotor system. If a machine was operating well below the resonance, as the shutdown data had been suggesting, then the influence vector phase should have been closer to 0°. The influence vectors for both measurement planes are very similar, *but the phase angle suggests operation slightly above the resonance.* This was the first sign that something was not happening as expected. The specialist rechecked his calculations and obtained the same result. He decided to continue and calculate a balance solution for the next run:

$$\boldsymbol{m} = \frac{-\boldsymbol{O}}{\boldsymbol{H}} \tag{26-2}$$

**Figure 26-3.** Results of the first calibration run. At top are the polar plots from the fan's two *X* shaft relative probes. The original run is shown in green. A 12 oz calibration weight was added to the fan at 75° relative to the *X* probes. The next shutdown is shown in blue. The middle pair of plots show the graphical construction for 890 rpm, using the vector values from the polar plots above. The $\boldsymbol{C}$ vector shows the change in vibration from the first run to the second. The bottom row shows the graphical balance solution. The calibration weight is modified and rotated to produce $-\boldsymbol{O}$.

1X compensated, 20 mil pp full scale

Figure 26-3



When he did this for each measurement plane, he obtained

$$\boldsymbol{m}_3 = 7.7 \text{ oz } \angle 328° \text{ (220 g } \angle 328°)$$
$$\boldsymbol{m}_4 = 8.0 \text{ oz } \angle 340° \text{ (230 g } \angle 340°)$$

The two calculated solutions show good agreement with each other. The graphical balance solution is shown in the bottom row of Figure 26-3. The calibration weight was rotated and reduced in size until the new response was equal to $-\boldsymbol{O}$. There is a small discrepancy between the calculated values in the text and the indicated graphical solution. The calculation uses values that were obtained from the instrumentation when the fan was running at steady state. The graphical solution was derived from the polar plot vectors that were obtained during the shutdown; the vibration data was slightly different.

The calibration weight was removed, and $\boldsymbol{m} = 8.4$ oz $\angle 332°$ was installed in the fan. This weight and location were the closest available to the calculated solution. This weight installation should have resulted in a dramatic reduction of the unbalance response. The fan was restarted and a new set of vibration vectors was measured as

$$\boldsymbol{T}_{3X} = 12.8 \text{ mil pp } \angle 247° \text{ (325 } \mu\text{m pp } \angle 247°)$$
$$\boldsymbol{T}_{4X} = 13.9 \text{ mil pp } \angle 268° \text{ (353 } \mu\text{m pp } \angle 268°)$$

While lower, these responses were not even close to the values that should have been seen based on the balance solution. This was not a simple unbalance problem; as the influence vectors had suggested, something else was wrong with this machine.

### The Real Problem

Knowing the past balancing difficulty with this fan and having had a similar experience with an induced draft fan at a coal mine, the specialist wondered if dust or dirt was trapped in the interior of the fan hub. The dirt would form a heavy spot that could shift every time the fan was shut down.

He planned a simple test to determine if this was the cause of the problem. The fan would be started from a resting position 180° from the last starting point. This point was known because of the installation of the balancing weight. If the vibration was the same after starting from a different position, then dirt

was probably not the problem. If, however, there was a significant change, this would indicate that dirt might be shifting and affecting the balance state of the fan.

Power was cut to the fan motor, and the fan coasted down. When the fan reached 10 rpm, maintenance personnel removed the access cover. As the fan continued to slow, a cloud of dust appeared every time the fan passed a certain point. The specialist knew he was on the right track.

Further investigation revealed that an open area surrounding the anchor bolts let dust into the hub of the fan. Every time the fan was shut down, the dirt would shift to a different position in the hub and change the unbalance. To solve the problem, two holes were cut at the outer diameter of the hub, and the contaminating dirt was blown out with compressed air. After cleaning, all of the holes were sealed, and the balancing procedure was repeated.

After two balancing runs, the 1X vibration was reduced to less than 1.5 mil pp (38 μm pp). Table 26-1 shows the measurements, weights, and influence vectors for the fan. Note that the phases of the influence vectors were now consistent with what would be expected for a machine operating below the resonance.

**Table 26-1**. ID Fan 1 balancing data after cleaning and sealing. The final balancing weight was 38 oz ∠6° (1.1 kg ∠6°).

| Probe location | *O* | *H* | Final vibration |
|---|---|---|---|
| 3XD | 22.0 mil pp ∠229°<br>559 μm pp ∠229° | 0.56 mil pp/oz ∠46°<br>0.50 μm pp/g ∠46° | 1.4 mil pp ∠169°<br>36 μm pp ∠169° |
| 4XD | 22.4 mil pp ∠248°<br>569 μm pp ∠248° | 0.60 mil pp/oz ∠63°<br>0.54 μm pp/g ∠63° | 0.8 mil pp ∠114°<br>20 μm pp ∠114° |

The process was successfully repeated with the other two fans, which suffered the same problem. All of the fans were running more than two years later with no unbalance problem or interruption in service.

The root cause of the problem was dirt that was trapped in the hub of the fan. When the fan stopped rotating, the dirt would settle to the bottom of the approximately 5 ft (1.5 m) radius hub. When the fan was started, the dirt would be held in position by centrifugal force. This caused the fan to have an ever changing source of unbalance. Repeated balancing of this machine had not and would never have solved the problem. Whenever a machine fails to respond as expected to repeated balancing attempts, there are two possibilities: the method being used is incorrect, or there is another problem in the machine that must be discovered.

Chapter 27

# A Generator Vibration Puzzle

Bently Nevada machinery specialists traveled to a power plant to document the startup of a steam turbine generator set following a rotor overhaul. They were asked to document the vibration response characteristics during startup, load tests, and shutdown. They were also asked to review the dynamic data and evaluate the current mechanical condition of the unit based on the vibration measurements.

The machinery specialists arrived at the site and set up their computer-based data acquisition equipment. The machine was a 150 MW, 3600 rpm, steam turbine generator in a tandem-compound, double flow configuration (Figure 27-1). The opposed flow HP/IP turbine was connected to a single, dual flow LP unit and a generator. The generator was connected to an 897 rpm exciter through a flexible coupling and gearbox. The turbine and generator bearings (1 through 6) were instrumented with *XY* proximity probes. The gearbox and exciter bearings were not instrumented. Table 27-1 shows the machine train parameters.

The existing Keyphasor probe observed a split hub on the machine. This arrangement made it difficult to obtain a reliable, once-per-turn signal because the transducer picked up the split line on both sides of the hub as well as the Keyphasor slot. This arrangement caused problems with the Keyphasor signal, which resulted in a partial loss of startup data.

During the startup, the unit was brought up to a speed of 2300 rpm and maintained near that speed for a five hour heat soak. After the heat soak, the unit was brought up to full speed and run at no load for the next twelve hours. During the startup, machine behavior was normal, with direct vibration reaching a maximum of around 3.6 mil pp (90 μm pp) at bearing 4.

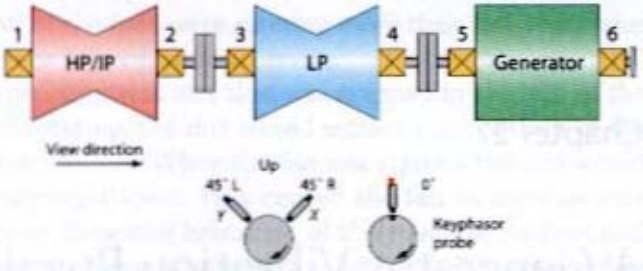



**Figure 27-1.** Machine train diagram. The rigidly coupled HP/IP and dual flow LP steam turbines drive a 150 MW, 60 Hz generator. Rotation is *Y* to *X* (CW).

**Table 27-1.** Machine train operating parameters.

| | |
|---|---|
| **Steam Turbines** | |
| Power | 150 MW |
| Speed | 3600 rpm |
| Inlet press. | 100 bar (1450 psig) |
| Inlet temp. | 540 °C (1000 °F) |
| Reheat temp. | 540 °C (1000 °F) |
| Exhaust press. | 64 mm Hg abs (2.5 in Hg abs) |
| **Generator** | |
| Apparent power | 183 824 kVA |
| Power | 156 250 kW |
| Voltage | 18 000 V |
| Current | 5896 A |
| Power factor | 0.85 |
| Frequency | 60 Hz |
| Hydrogen press. | 3 bar (45 psig) |
| **Exciter** | |
| Power | 980 kW |
| Voltage | 375 V |
| Current | 2615 A |
| Speed | 897 rpm |

## Unusual Vibration Behavior

The next morning, the machinery specialists prepared to take load test data. After starting data acquisition, they noticed that the vibration in bearing 5, the generator inboard bearing, was acting strangely. Immediately after startup, the 1X vibration measured by the *X* transducer at bearing 5 had been 1.4 mil pp ∠238° (36 µm pp ∠238°). Now the vibration amplitude was slowly changing, alternating between 3.5 and 0.8 mil pp (90 and 20 µm pp). The vibration amplitude changed from high to low and back to high, completing a cycle in about 2 hours (Figure 27-2). Direct orbits from bearing 5 showed that, when the amplitude was highest, the vibration was predominantly 1X.

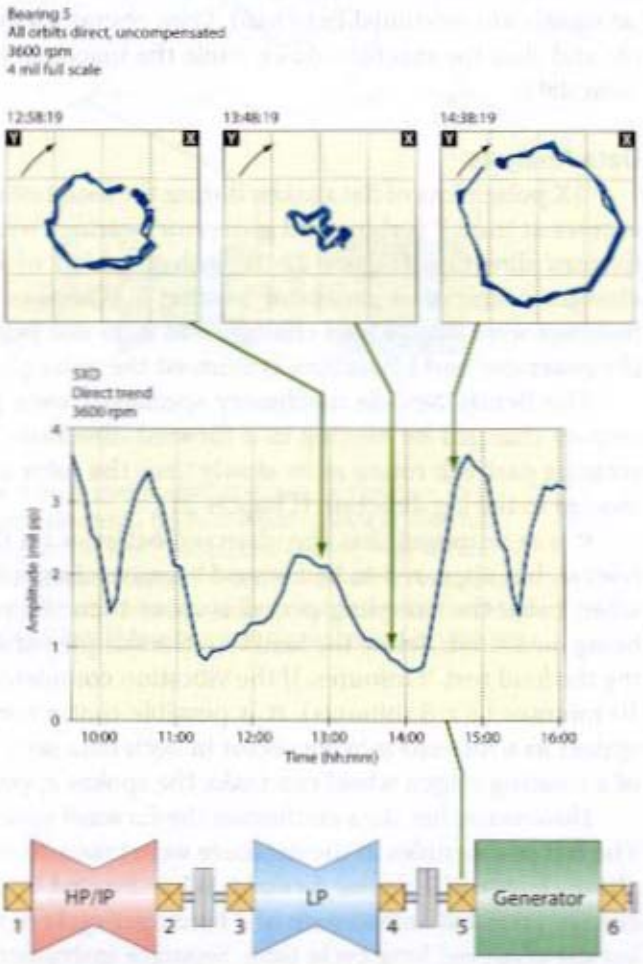



**Figure 27-2.** Modulation of vibration amplitude. The vibration at bearing 5 slowly changes in amplitude, alternating between 3.5 and 0.8 mil pp (90 and 20 µm pp), completing a cycle in about 2 hours. Direct, uncompensated orbits from bearing 5 show that, when the amplitude is highest, the vibration is predominantly 1X.

This vibration was studied for the next 6 hours by plant operators, OEM representatives, and the Bently Nevada specialists. During this period, the OEM representatives decided to change the oil temperature and pressure in the generator hydrogen seal to see if that would eliminate or reduce the vibration. Nothing worked, and the vibration continued its slow cycling.

The vibration behavior was puzzling, but, since the amplitudes were not high enough to constitute an immediate problem, plant operators decided to proceed with the load tests. The machinery specialists prepared to take more data during the load test and shutdown. Perhaps the changing load would shed some light on the problem.

Over a 4 hour period, operators increased the load to 151 MW. During this time, the periodically changing vibration continued in much the same way, with no significant relationship to load. Then operators shed the load on the generator and shut the machine down while the machinery specialists captured transient data.

### Data Analysis

1X polar plots of data taken during the load tests showed that the vibration vectors at the LP turbine and generator bearings were very slowly rotating in a *forward* direction (Figure 27-3), with a period of over an hour. The largest changes occurred at generator bearing 5. (Changes in vibration in the HP/IP bearings were due to load changes and were not periodic.) The 1X orbits from the generator and LP turbine confirmed the polar plot data.

The Bently Nevada machinery specialists were puzzled. The 1X vibration vectors changed by moving in a forward direction. This was unusual; a loose rotating part will rotate more slowly than the rotor and produce a 1X vibration change in the lag direction (Chapter 21).

It was proposed that the observed behavior on the data plots was actually *reverse*, but appeared to be forward because of sample aliasing. Aliasing occurs when twice the sampling period is more than the period of the phenomenon being measured. Before the load test, the sample period was 5 minutes and, during the load test, 9 minutes. If the vibration completed a full change in less than 10 minutes (2 x 5 minutes), it is possible that a reverse moving vector would appear as a forward moving vector in both data sets, much the same as a movie of a rotating wagon wheel can make the spokes appear to move backwards.

However, other data confirmed the forward nature of the vibration change. The last two samples in the database were taken 30 seconds apart and showed a phase change of 1 degree *forward*. While limited by poor resolution, this phase change provided an estimate of 3 hours per cycle of vibration change, confirming the observed long cycle time. Separate instrumentation also confirmed the

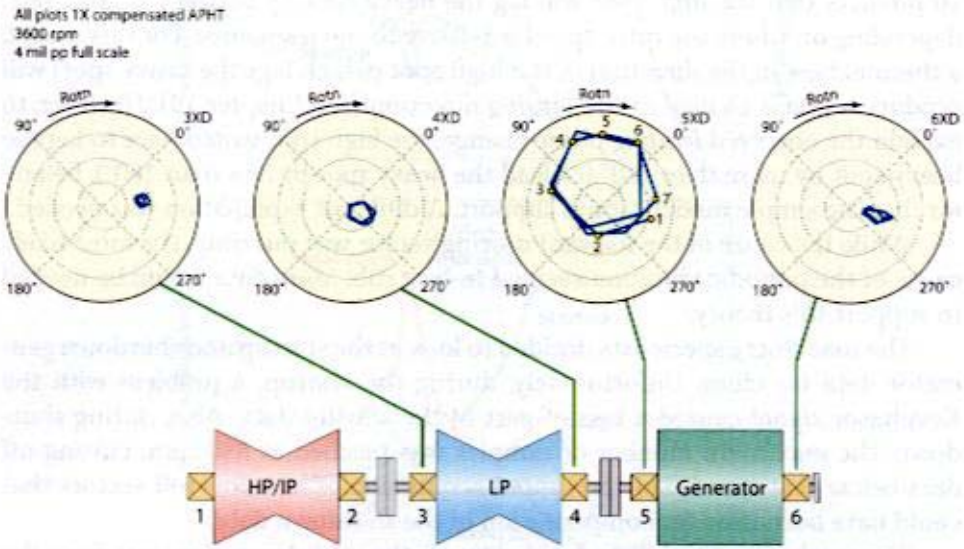



**Figure 27-3**. Steady state operation with changing 1X vibration. Polar APHT plots show approximately 2 hours of vibration data at 3600 rpm, sampled 9 minutes apart. The first 7 samples for bearing 5 are numbered and show that the 1X vibration vector rotates in a forward direction, completing 1 revolution in about 1 hour. The largest change in vibration amplitude occurs at bearing 5, but all four bearings show changes in vibration.

observation of forward movement. An oscilloscope and a tape recorder used during the data acquisition process provided continuous data that supported the periodically sampled data.

A 1X rub at the location of the rotor high spot can produce a phase change. The rub causes local heating and a thermal bow, which produces a new, effective heavy spot, shifting the high spot response. The simple rotor model of Chapter 10 predicts that the high spot will lag the heavy spot by between 0° and 180°, depending on where the rotor speed is relative to the resonance. For this model, a thermal bow in the direction of the high spot (which lags the heavy spot) will produce a phase change in the *lagging* direction (see Chapter 19). However, to explain the observed *leading* phase change, the high spot would have to lag the heavy spot by more than 180° (or lead the heavy spot by less than 180°), behavior that the simple model cannot support. A different explanation was needed.

While the cause of the forward moving vector was puzzling, the most likely cause of the periodic vibration seemed to be a rub. More data would be needed to support this theory.

The machinery specialists decided to look at the startup and shutdown generator data for clues. Unfortunately, during the startup, a problem with the Keyphasor signal caused a loss of part of the startup data. Also, during shutdown, the maximum number of samples was reached at 880 rpm, cutting off data below that speed. This prevented acquisition of hot slow roll vectors that could have been used for compensation of the shutdown data.

The machinery specialists decided to use the cold slow roll vectors from the startup data to compensate the hot shutdown data. This was done cautiously, because they did not know if thermal growth had caused the slow roll vectors to change. The two sets of compensated data are plotted with the same amplitude and speed scales in Figure 27-4. The results were surprising. The startup data (blue) was very different from the shutdown data (red). In the figure, the data from the *X* probe at bearing 5 is shown, but data from the other generator probes showed similar differences in behavior.

The shutdown data showed a clear resonance near 1400 rpm. The startup showed a very different picture; there was no clearly defined resonance below 2400 rpm, the speed where the data was cut off, and the amplitude and phase lag increased steadily. The data suggested that a balance resonance existed somewhere near 2400 rpm.

The startup and shutdown phase changes have a similar overall pattern, but the pattern during startup appears to be shifted to a higher frequency. Also, the shutdown phase lag is shifted by about 120° (compare left and right scales), indicating that a significant change in the location of the heavy spot has taken place.

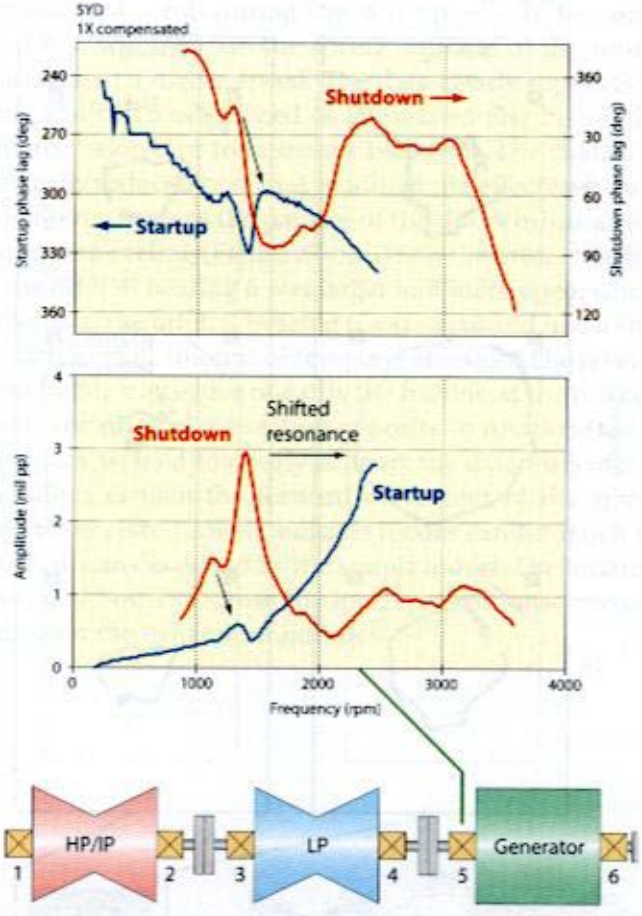



**Figure 27-4.** Startup and shutdown data for bearing 5. The shutdown data (red) shows a clear resonance at the inboard end of the generator at 1400 rpm. The startup shows a very different picture, with no clearly defined resonance below 2400 rpm. The startup (blue) data suggests that a balance resonance exists somewhere near 2400 rpm, indicating the presence of a rub during the startup. The phase data shows very different behavior, with the phase shifted nearly 120° between startup (left scale) and shutdown (right scale).

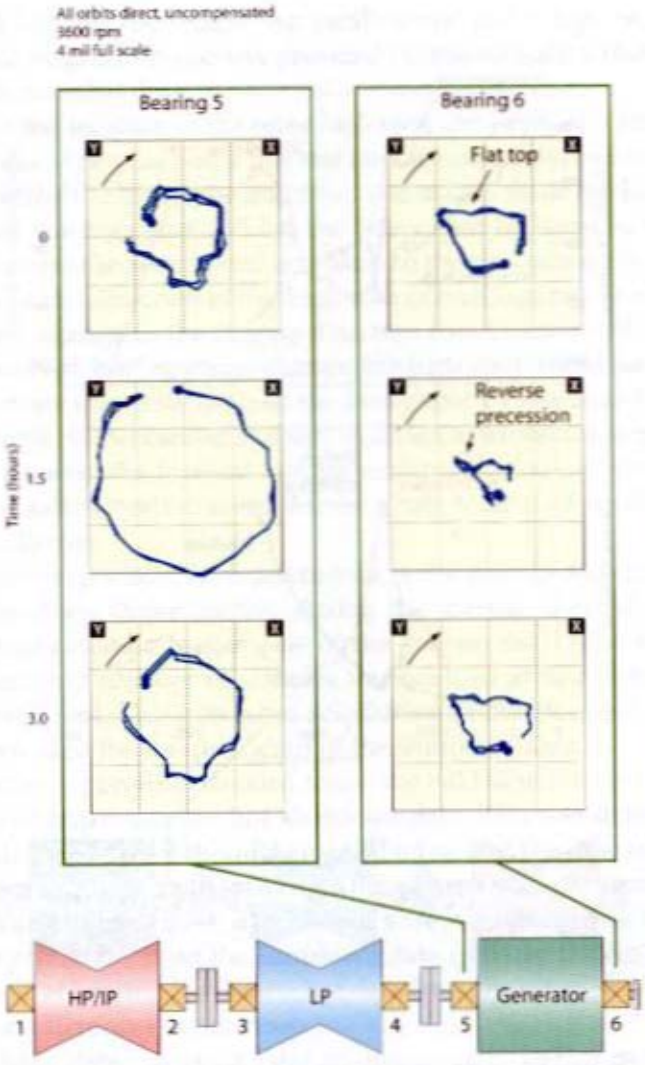



**Figure 27-5.** Orbit changes during steady state operation. Direct, uncompensated orbits for bearings 5 and 6 show modulation of vibration amplitude and phase. The bearing 6 orbit (top right) has a flat top that suggests a constraint. After a period of 1 1/2 hours (middle row), the bearing 5 orbit is large and circular, while the bearing 6 orbit shows evidence of partial reverse precession. After 3 hours have passed, the behavior is similar to that at the beginning.

The data from both generator bearings was very similar and strongly suggested the presence of a rub during the startup. A rub, by constraining the motion of the rotor, will increase the spring stiffness of the rotor system and move the resonance to a higher speed. The data clearly supports the rub diagnosis. Evidently, the rub was reduced or eliminated just before the final shutdown, allowing the resonance to return to 1400 rpm. The change in phase also suggests a thermally induced bow that modified the effective heavy spot.

Even more interesting was the pattern of the direct orbits at bearings 5 and 6 during the vibration cycling (Figure 27-5). When the orbit at bearing 5 was relatively small, the orbit at bearing 6 was larger and more open; when the orbit at bearing 5 was largest, the orbit at bearing 6 was small and had a sharp corner at the upper left with a small amount of reverse precession. The reverse precession at bearing 6 was highly suggestive of a rub; the friction at the rub contact acts to briefly accelerate the rotor in a direction opposite to rotation (see Chapter 21).

Overall, the data seemed to clearly support the diagnosis of a rub. But how could the specialists explain the forward movement of the vibration vector? They knew that rotor systems with multiple modes exhibit much more complicated behavior than can described by the simple model. The mixture of multiple mode responses and bow can cause the high spot response vector to have any phase, depending on the dynamic conditions.

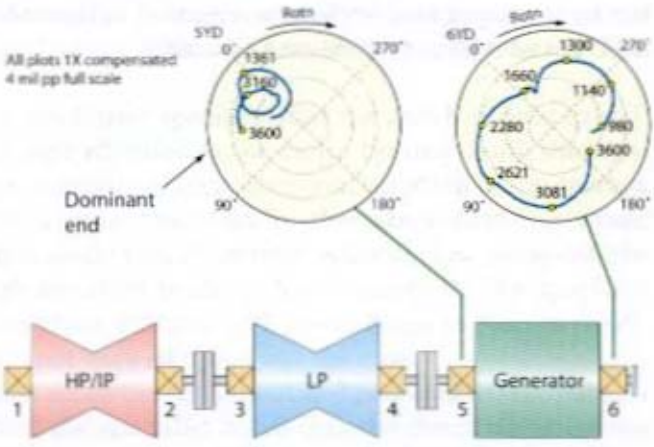



**Figure 27-6.** Shutdown data for both generator bearings. The generator operates above the second mode, and bearing 5 is the dominant end of the machine, as shown by the nested polar loops. At running speed, the high spot location leads the indicated first mode heavy spot. See also the bearing 5 shutdown data in Figure 27-4.

They proposed this explanation: most large generators operate above the second mode. A generator with a *dominant end*, where the first and second mode heavy spots are in the same quadrant, could produce a high spot response that lags the indicated heavy spots by more than 180°. If this were to happen, a rub-induced thermal bow could produce a new heavy spot that would shift *both* modal heavy spots in the forward direction. The magnitude of the new vibration vector would depend on the relative magnitudes of the original heavy spots and the bow. In this scenario, a forward rotating response vector would be possible.

Polar plots of the shutdown (Figure 27-6) revealed that the generator was operating above the second mode, that bearing 5 was the dominant end of the machine, and that, at running speed, the high spot location was leading the indicated first mode heavy spot. This information supported the theory.

## Conclusions

All the data clearly suggested to the machinery specialists that a rub occurred in the generator, near bearing 5. The most likely location for such a rub would be the carbon ring of the hydrogen seal, which could have been installed incorrectly, resulting in an internal misalignment between the ring and the shaft. Even though the seal was lubricated, direct contact between the shaft and the carbon ring could produce high friction in the area of the rub. The shutdown data suggested that the rub had opened up the clearance, which eliminated the rub, sometime before the shutdown.

The forward-moving, 1X vibration vector was most likely caused by rub-induced local heating of the surface of the generator rotor, which was operating above the second resonance.

After the machine was restarted, the unusual vibration behavior reappeared, but later disappeared without any further action. Most likely, the rub worked itself out as it increased the seal clearance.

Chapter 28

# High Vibration in an Electric Motor

Operators of a synchronous-motor-driven, syngas compressor had been experiencing episodes of high vibration on the motor. Occasionally, the vibration would suddenly become high enough that the unit would trip. The lost production, and concerns about the health of the machine train, prompted the plant operators to ask Bently Nevada to obtain vibration data and help diagnose the problem with the machine.

When the Bently Nevada machinery specialist arrived at the plant, the machine was operating. The machine train (Figure 28-1) consisted of a 10 000 hp (7450 kW), 1800 rpm, synchronous motor driving a syngas barrel compressor at 4459 rpm through a speed-increasing gearbox, using flexible couplings. Fortunately, the machine train was well instrumented: *XY* proximity probes were installed at all eight fluid-film bearings, and Keyphasor probes observed both the low- and high-speed rotors. Table 28-1 lists the nameplate data for this machine train.

The operators told the machinery specialist that the unit had been experiencing sporadic periods of high vibration in the motor bearings, with the highest on bearing 2. During the episodes of high vibration, vibration in the gearbox and compressor remained normal. Occasionally, the motor vibration would become so high that the unit would trip. After trips, while at slow roll speed, the monitors would show high vibration readings from bearing 2. The specialist asked whether the vibration was correlated with any changes in process conditions, but the operators had not noticed if this was the case.

The machinery specialist set up his equipment and prepared to take data. At the start of data acquisition, the specialist noted that the direct vibration was about 2 mil pp (50 μm pp) on bearing 1 and a little over 3 mil pp (75 μm pp) on

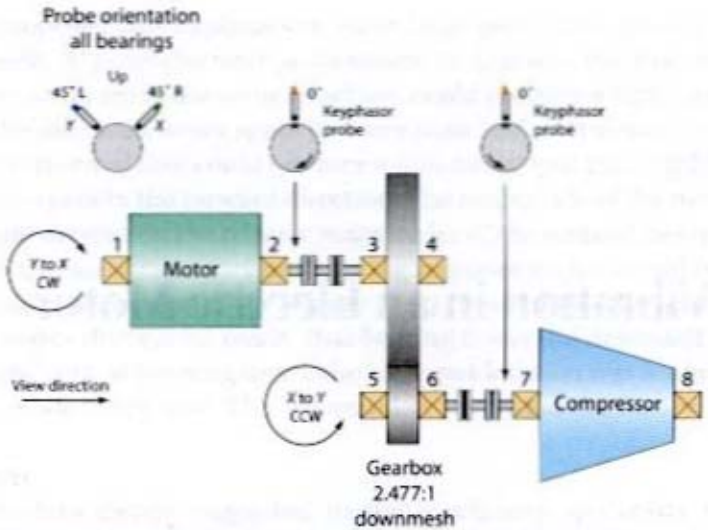



**Figure 28-1.** Machine train diagram. A 10 000 hp (7450 kW), 1800 rpm, synchronous motor drives a 4459 rpm syngas barrel compressor through a speed-increasing gearbox and flexible couplings. Shaft relative proximity probes are installed at all eight fluid-film bearings, and Keyphasor probes observe both the low- and high-speed rotors.

**Table 28-1.** Machinery nameplate data.

| | |
|---|---|
| **Synchronous motor** | |
| Poles | 4 |
| Speed | 1800 rpm |
| Voltage | 13 200 V |
| Current | 370 A |
| Apparent power | 8460 kVA |
| Power factor | 0.90 |
| Power | 10 200 hp (7614 kW) |
| Service factor | 1.0 |
| **Gearbox** | |
| Input speed | 1800 rpm |
| Output speed | 4459 rpm |
| Ratio | 2.4771 |
| Service power | 8500 hp |
| Service factor | 1.4 |
| **Compressor** | |
| Gas | Synthetic gas |
| Mole weight | 15.94 |
| **Rated conditions** | |
| Inlet pressure | 21.3 bar (309 psi) |
| Inlet temperature | 536 °F (280 °C) |
| Volume | 23 900 acfm (678 $m^3$/min) |
| Discharge pressure | 367 psi (25.3 bar) |
| Max. cont. speed | 4680 rpm |
| 1st critical | 2400 rpm |

bearing 2. After a period of time, the vibration in both motor bearings started to rise; at bearing 2 it suddenly increased to 9 mil pp (230 μm pp), and the machine tripped. The specialist continued to capture data during the coastdown.

### Data Analysis

The trend plot of direct vibration at the motor bearings, starting about 10 minutes before trip, is shown in Figure 28-2, along with selected orbits. The motor vibration was steady for most of the data collection period, but started increasing dramatically about 4 minutes before the trip. The gearbox and compressor data did not show any increase in vibration.

Both motor bearings appeared to be heavily loaded; the first set of direct orbits were flattened on top, showing clear evidence of constrained rotor motion. Both orbits showed reverse precession.

Bearing 1 orbits (blue) maintained approximately the same shape from the start of the data until 01:12:45, just before the trip, at which time the orbit became significantly distorted, approaching a figure eight shape. After the trip, the orbit became banana shaped and switched to forward precession below 1400 rpm.

Orbits from bearing 2 also showed evidence of a large radial load. A small external loop developed by 1:09:17. By 1:12:45, just seconds before the trip, an external loop had developed in the bottom of the orbit, which had significantly higher amplitude. Precession switched to forward at this time. Direct vibration at bearing 2 approached 9 mil pp (230 μm pp) just before the trip, probably close to the available bearing clearance.

The direct orbits at the moment of trip are shown for the entire machine train in Figure 28-3. It was clear that the vibration problem was in the motor; the gearbox and compressor bearings showed very little vibration.

The 1X APHT plots (Figure 28-4) showed significant changes in 1X vectors at the motor bearings before the trip; the 1X vector movement was much larger on bearing 2. APHT plots for other bearings in the machine train showed no change in the 1X vectors during this period.

The shaft centerline plots for the motor also showed evidence of a rub. As vibration amplitude increased, the shaft centerlines at both bearings shifted up and to the right by nearly a mil (25 μm), with more vertical change occurring in bearing 2.

The shaft centerline position in the motor showed more unusual behavior during the shutdown (Figure 28-5). The centerline positions in both bearings initially moved down to the left; this is not the behavior one would expect from fluid-film bearings with $Y$ to $X$ shaft rotation. Below about 700 rpm, the shaft centerline behavior appeared normal until the speed reached about 70 rpm, the

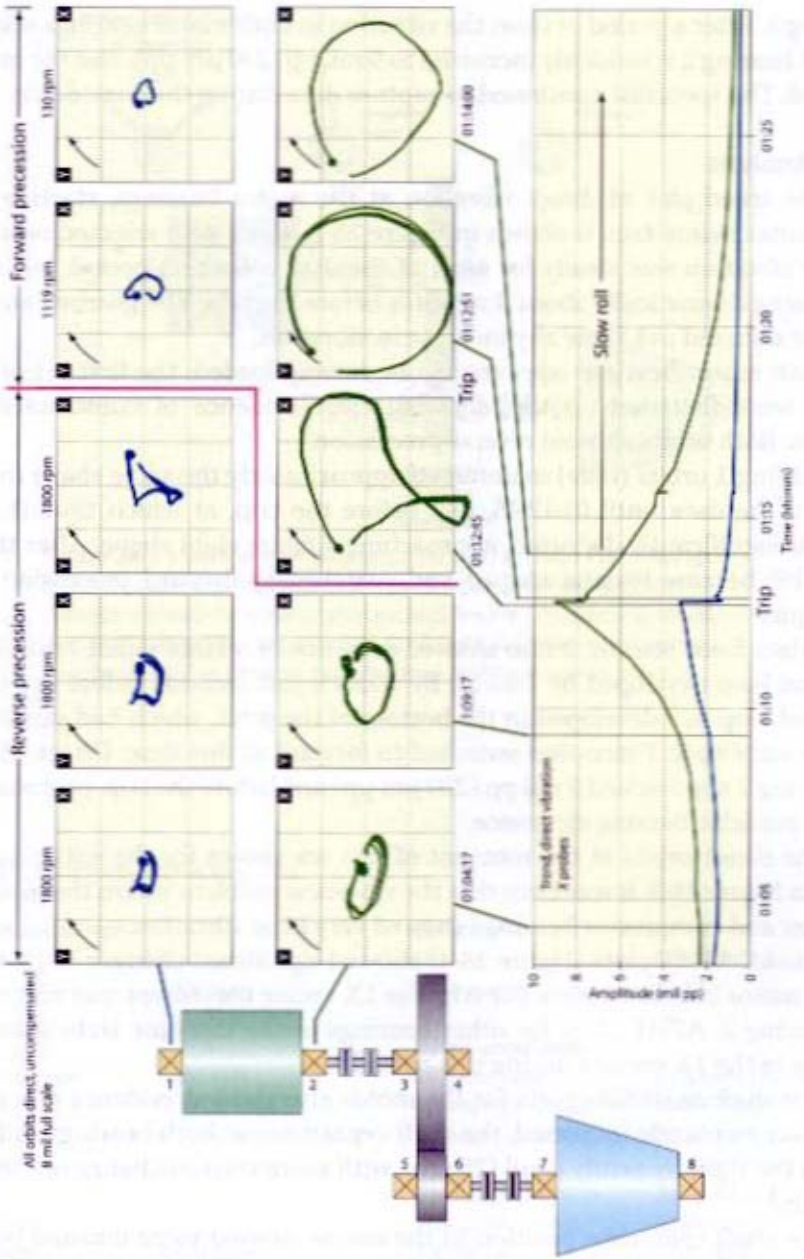



**Figure 28-2**. Direct vibration trend plots and orbits for the motor. Bearing 1 data is shown in blue; bearing 2 in green. See the text for a full description.

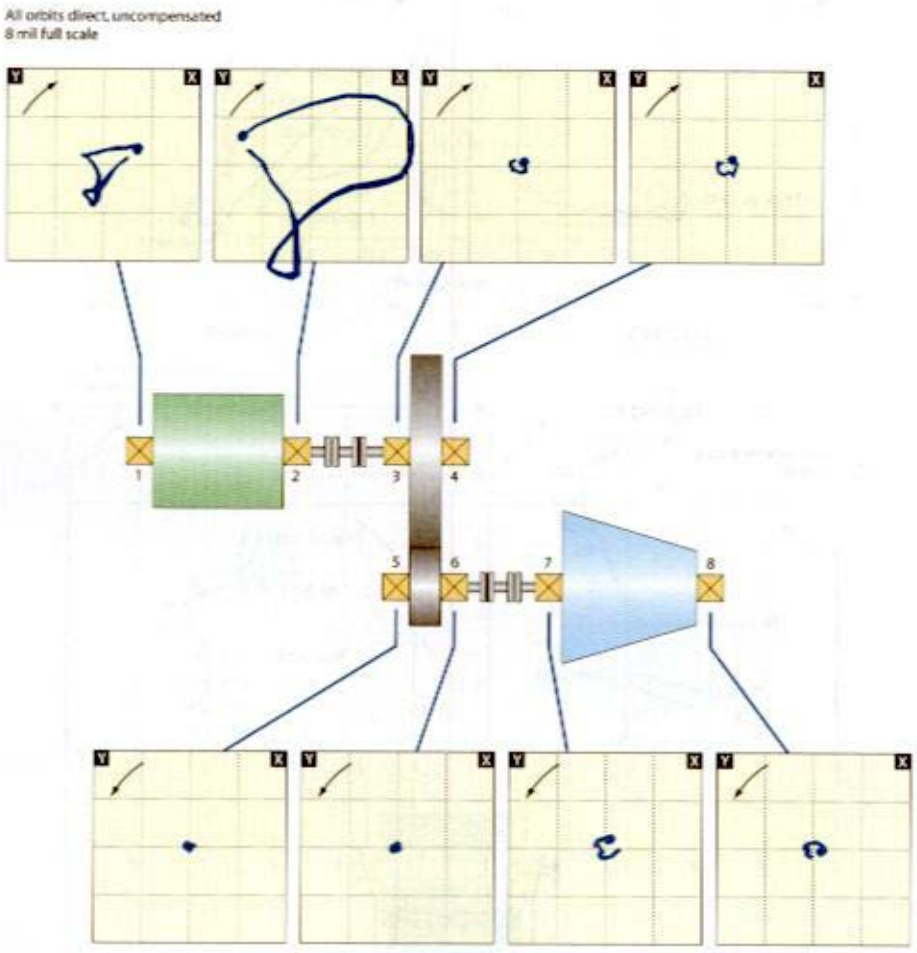



**Figure 28-3**. Direct, uncompensated orbits along the machine train just before the trip. It is clear that the vibration problem is in the motor; the gearbox and compressor bearings show very little vibration.

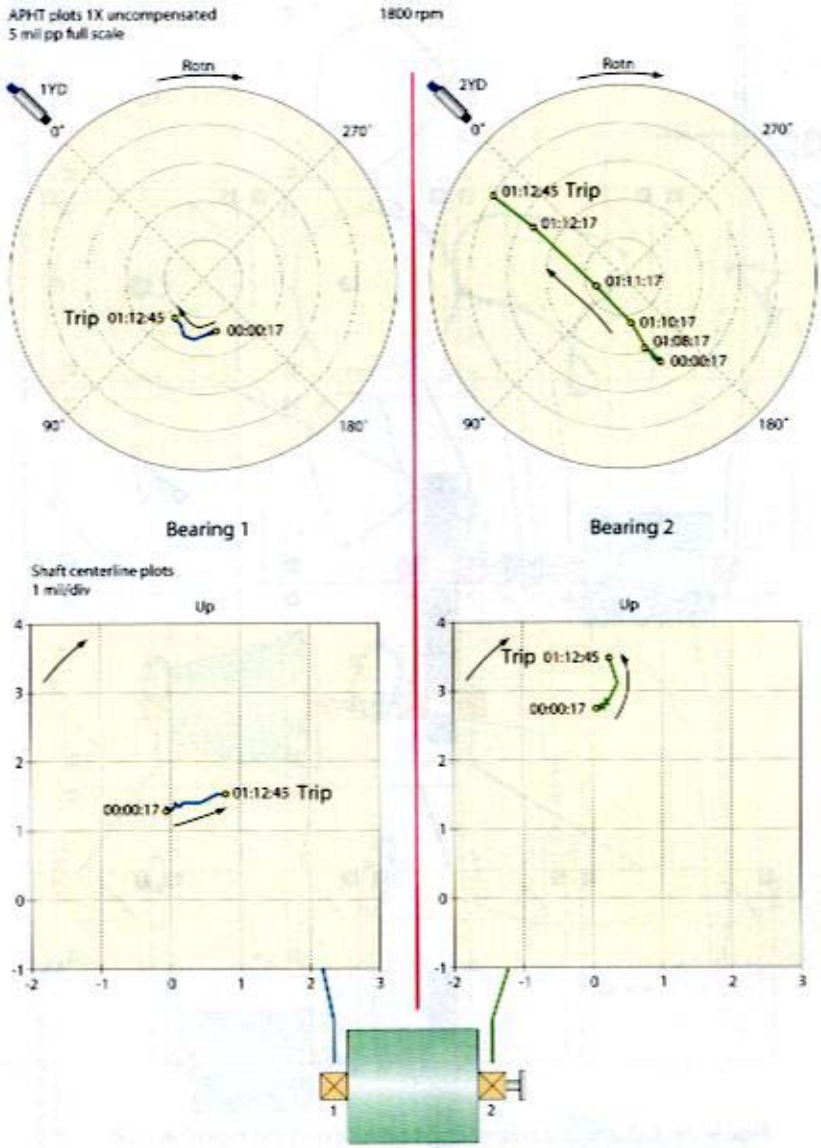



**Figure 28-4.** Motor bearing APHT and shaft centerline plots at 1800 rpm. The arrows indicate the direction of change during this period. There are significant changes in 1X vectors, especially at bearing 2. As vibration amplitude increases, the shaft centerline at bearing 2 shifts up and to the right by nearly a mil (25 µm) and slightly more at bearing 1.

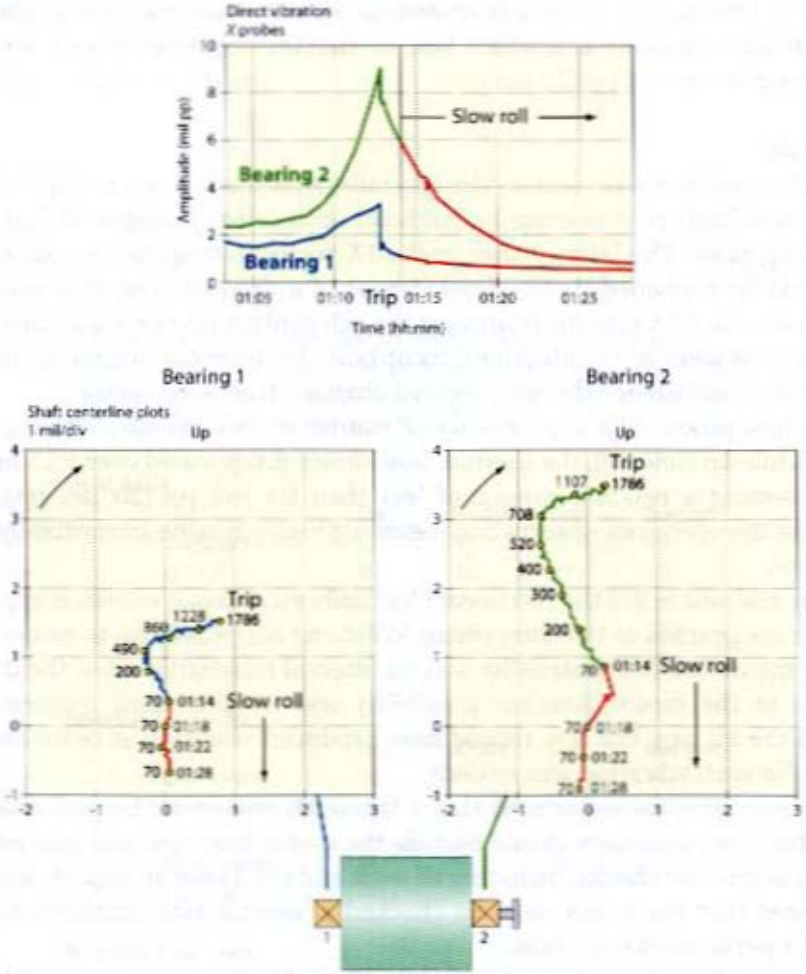



**Figure 28-5.** Motor bearing shaft centerline position during shutdown. The centerline positions both initially move down to the left. Below around 700 rpm, shaft centerline behavior appears normal until about 70 rpm, the slow roll speed. Once on slow roll, the shaft continues to move down about one mil (25 µm) in bearing 1 and almost two mils (50 µm) in bearing 2 over the next 15 minutes. During this period, the direct vibration trend plot (top) shows that the shaft slow roll vibration at bearing 2, which is mostly 1X, decreases from 5.6 mil pp (140 µm pp) to 0.8 mil pp (20 µm pp).

slow roll speed. Once on slow roll, over the next 15 minutes *the shaft continued to move down* another mil (25 μm) in bearing 1 and almost two mils (50 μm) in bearing 2. During this period, as shown in the accompanying trend plot, the shaft vibration in bearing 2, which was mostly 1X, decreased from 5.6 mil pp (140 μm pp) to 0.8 mil pp (20 μm pp).

### Diagnosis

At this point, it was clear to the specialist that there was a problem in the motor, most likely near bearing 2. The direct orbits clearly suggested that a rub was taking place. The large change in the 1X vector during steady state operation could be explained by the development of a thermal bow. This would be consistent with a 1X rub; the heating at the rub contact point would cause local thermal expansion of the shaft and rotor bow. As the rotor heated during the rub, the large unbalance that was created changed the 1X vibration.

The bow produced a large, mostly 1X runout at slow roll speed. As the rotor cooled while on slow roll, the thermal bow slowly disappeared over a 15 minute period, leaving a residual runout of less than 0.8 mil pp (20 μm pp). This explained the operators' observation regarding high vibration immediately after shutdown.

Why and where did the rub occur? Virtually no unusual vibration appeared in either the gearbox or the compressor. Whatever happened had to be associated with the motor. One possibility was an internal misalignment of the dust or oil seals in the motor. Another possibility was an alignment problem that reduced the air gap, but this should have produced vibration at twice line frequency. No such vibration was evident.

The specialist recommended that a thorough inspection be performed on the motor. The inspection should include the motor bearings, and internal and external alignment checks, including all seals and the motor air gap. He also recommended that the motor shaft be checked at several axial locations for any signs of a permanent rotor bow.

The seal clearances were measured and found to be very uneven, with zero clearance on one side of the motor dust seal at bearing 1 (Table 28-2). Measurements indicated that this seal was positioned too low and too far to the right, resulting in a large clearance at the bottom and zero clearance at left. When technicians removed the seals, they found evidence of rub at both ends of the machine.

Based on this data, the machinery specialist developed a theory as to what happened. The zero clearance at the bearing 1 dust seal pointed toward that seal as the primary source of the problem. A heavy rub on the dust seal thermally bowed the rotor, causing the vibration at bearing 2 to increase, which eventual-

ly caused a rub on the bearing 2 oil seal. When this second rub occurred, vibration increased rapidly and tripped the machine. The thermal bow caused the high vibration at slow roll until the rotor cooled down.

After the seals were reassembled and clearances adjusted, the unit ran smoothly with no problems.

**Table 28-2.** Measured dust and oil seal clearances on the motor. Dimensions are in mils, and directions are as viewed from the motor looking toward the gearbox. Evidence of rub was found where noted. The clearances were set to the indicated values.

| | | | | |
|---|---|---|---|---|
| **Dust seals** | | | | |
| | **Top** | **Right** | **Left** | **Bottom** |
| **Outboard** | 12 | 35 | 0 (rub) | 96 |
| Set to | 20 | 20 | 20 | 20 |
| **Inboard** | 18 | 18 | 16 | 250 |
| Set to | 25 | 25 | 25 | 70 |
| **Bearing 1 oil seals** | | | | |
| | **Top** | **Right** | **Left** | **Bottom** |
| **Outboard** | 8 | 11 | 8 | 14 |
| No changes | - | - | - | - |
| **Motor side** | 15 | 13 | 15 | 13 |
| No changes | - | - | - | - |
| **Bearing 2 oil seals** | | | | |
| | **Top** | **Right** | **Left** | **Bottom** |
| **Motor side** | 5 (rub) | 11 | 20 | 20 |
| Set to | 12 | 11 | 11 | 14 |
| **Coupling side** | 16 | 9 | 11 | 12 |
| Set to (min) | 11 | 11 | 11 | 11 |

Chapter 29

# Problems with a Pipeline Compressor

A GAS PIPELINE OPERATOR IN SOUTH AMERICA had purchased a number of compressor packages for installation during expansion of the pipeline. Each package consisted of a gas turbine-driven compressor.

At several of the compressor stations, high vibration trips and/or high vibration alarms occurred during the startup test runs. Initial data acquired by field personnel at one station indicated the trips were due to the onset of high amplitude, subsynchronous vibration at both compressor bearings. If the machines did not trip, then the subsynchronous vibration would subside as the discharge pressure increased.

The pipeline operator asked Bently Nevada to acquire additional startup, shutdown, and steady state vibration data under various process conditions and to assist field personnel with identification of the root cause of the problem. A Bently Nevada machinery specialist traveled to the site to examine the compressor package.

## The Machine Train

The machine train consisted of a 3200 kW (4300 hp), industrial gas turbine that was used as a gas generator. An overhung, aerodynamically coupled power turbine, rotating at around 15 000 rpm, drove the compressor through a flexible coupling (Figure 29-1).

The compressor was a modification to an older, 7-stage, single flow design; for the pipeline application, a compressor with comparable flow and half the head was needed. The modification replaced the center stage with a labyrinth seal and the final three wheels with the same wheels used in the first three stages. This produced two, 3-stage compressors on one shaft, supported by two

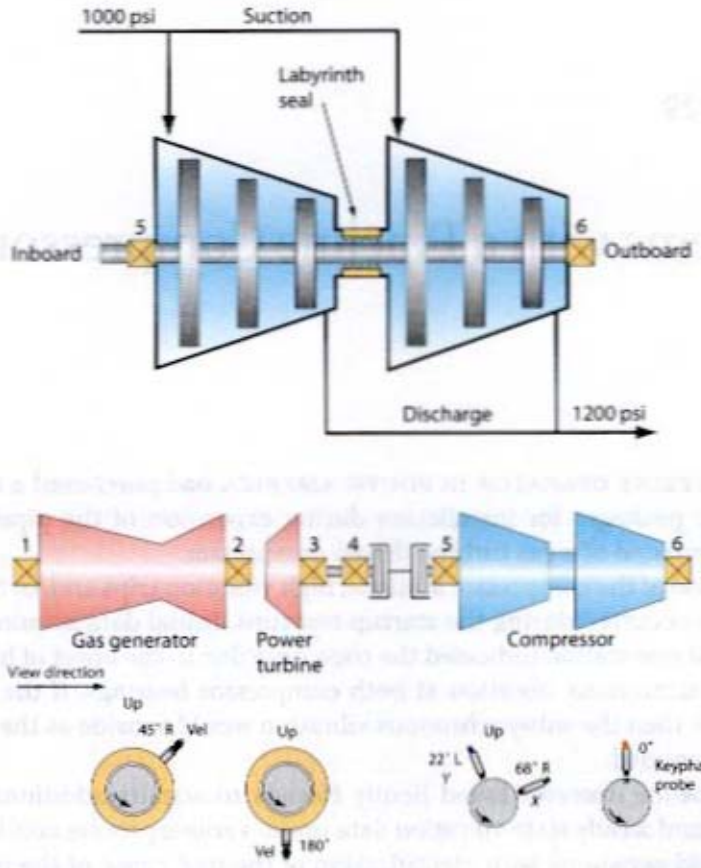



**Figure 29-1**. Gas turbine compressor train diagram. A 3200 kW (4300 hp), industrial gas turbine is used as a gas generator. An overhung, aerodynamically coupled power turbine drives the compressor at around 15 000 rpm through a flexible coupling. A seven-stage compressor was modified to produce two, three-stage compressors on one shaft, supported by two tilting pad bearings. The two compressors are operated in parallel with identical suction and discharge pressures. A midspan labyrinth seal separates the suction and discharge sides.

tilting pad bearings. The two compressors were operated in parallel, with identical suction and discharge pressures. The midspan labyrinth seal separated the discharge end of the first compressor from the suction end of the second compressor.

With this modification, the discharge pressure at rated load was 83 bar (1,200 psi) when the suction pressure was 70 bar (1,000 psi), for a pressure ratio of 1.2 and a differential pressure of 14 bar (200 psi).

The compressor was equipped with *XY* proximity probes on bearings 5 and 6 and with a Keyphasor probe, while the power turbine and gas generator each had a single velocity transducer mounted on the outer casing. The machinery specialist connected his data acquisition equipment to the buffered monitor outputs and prepared to take data.

The specialist was told that the compressor had floating ring seals at each end. The seals were designed for axially balanced thrust loading of the seal faces. The balanced design was intended to maintain a free floating seal so that it would not lock up and constrain the rotor. These seals also had an antirotation pin to prevent spinning. The pipeline operator suspected that the seals might be the source of the problem; perhaps the antirotation pins were causing the seals to lock up, or the seals were triggering a fluid-induced instability that was the source of the subsynchronous vibration.

The operators also suspected that the machine had insufficient damping, which was contributing to the instability problem. This last theory seemed improbable to the Bently Nevada specialist; fluid-induced instability problems usually have low sensitivity to damping and are primarily caused by circulating fluid trapped between the rotor and stator somewhere in the machine (see Chapter 22).

To discover the root cause of the vibration, the operators scheduled a series of tests based on different mechanical configurations of the compressor. The first series would be with the machine in its current configuration: pinned, balanced floating seals and the standard tilting pad bearings. Then, if the vibration problem still occurred, a second series of tests would use seals with a design that excluded the antirotation pin. The final series would use fluid-film bearings surrounded by a squeeze film damper to increase the damping in the machine. Because the vibration had been observed to disappear when the discharge pressure built up, each series of tests would include changing the suction or discharge pressures.

### Tests With Pinned Seals

When the cold machine with pinned seals was started, intermittent low level subsynchronous vibration appeared in both compressor bearings starting at 10,300 rpm. As the machine reached 14,300 rpm, it reappeared and grew to high amplitude as the rotor speed increased to 14,700 rpm. The machine was shut down, and the data was analyzed.

Figure 29-2 shows a cascade plot for bearing 5 (bearing 6 data was similar) and unfiltered (direct) shaft orbits at 14,700 rpm for both bearings. The multiple Keyphasor dots indicate that the vibration was at a subsynchronous frequency well below running speed. Also, the orbit shapes were fairly circular.

The cascade plot shows several important features. First, relatively high amplitude, 1X forward vibration occurred between approximately 4,000 and 7,000 rpm. This suggested a balance resonance, which could be confirmed with Bode or polar plots. Second, the subsynchronous vibration began to appear at a little over 10,000 rpm and eventually grew in amplitude with increasing rotor speed. The frequency of this vibration was about 5,500 cpm, or 3,7% of the 14,700 rpm running speed, at approximately the same frequency as the earlier high 1X vibration. This strongly suggested that the subsynchronous vibration was associated with the compressor natural frequency. The subsynchronous vibration was almost purely forward; the reverse component at that frequency was very small.

The cascade plot also shows a low level, approximately ½X vibration that tracked running speed during the startup. While barely visible during the 1X resonance, it reappeared suddenly at just over 10,000 rpm, where it coincided with the rotor natural frequency and was amplified. At that speed, the machine acceleration suddenly increased when the gas generator fired up (the very rapid acceleration smeared and laterally displaced the 1X spectral data at this time). The frequency of this low level component could not be determined exactly due to the limited resolution of the spectrum, but it was clearly forward and tracking, suggesting a fluid-induced instability.

The 1X Bode plot (Figure 29-3) confirmed the resonance and revealed some additional, important information. The amplitude behavior was abnormal during the balance resonance between 5,000 and 6,600 rpm. The growth in amplitude was truncated and was followed by a sharp drop at 6,600 rpm. The phase change through the resonance was also abnormal: the phase slope became more gradual at 5,000 rpm and showed a sharp drop at 6,600 rpm. Finally, the phase plots from each bearing were almost identical, indicating that this resonance was an in-phase mode, most likely the first bending mode of the compressor rotor.

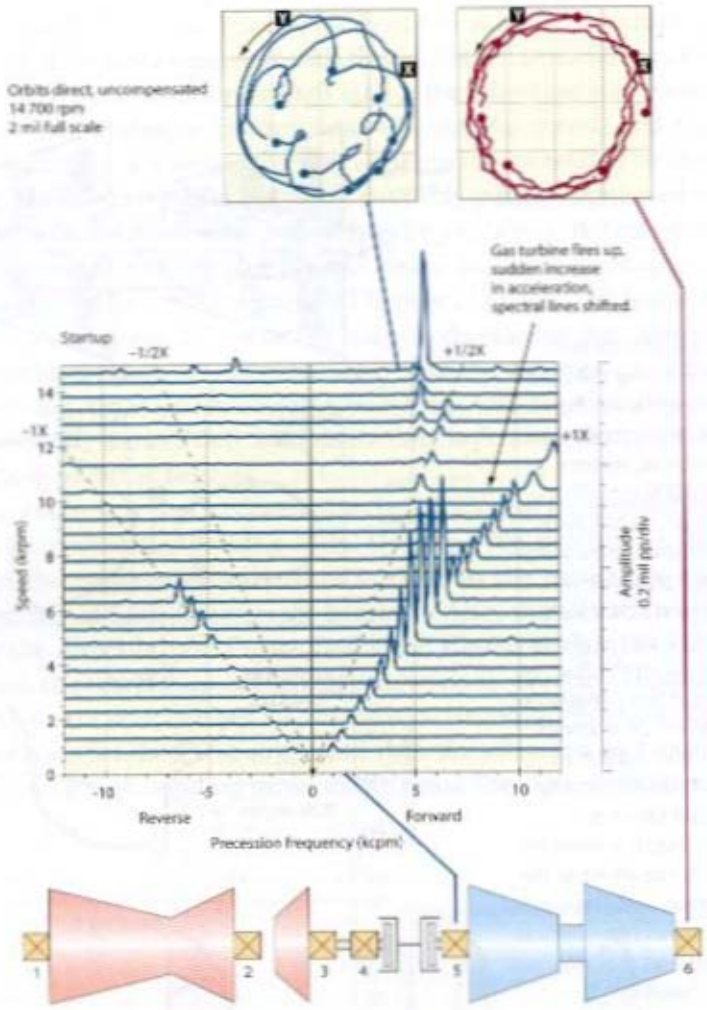



**Figure 29-2** Cold startup data from bearing 5. The multiple Keyphasor dots on the 14,700 rpm orbits from bearings 5 and 6 indicate that the vibration is at a subsynchronous frequency well below running speed. The orbit shapes are fairly circular. On the full spectrum cascade plot, relatively high amplitude, 1X forward vibration occurs between approximately 4,000 and 7,000 rpm. A low level, approximately 1/2X, forward vibration tracks running speed during the startup. While barely visible during the 1X resonance, it reappears at just over 10,000 rpm when the gas turbine fires up, where it coincides with the rotor natural frequency and is amplified. Subsynchronous, forward vibration, locked to this frequency, continues intermittently until 14,300 rpm, then suddenly grows to high amplitude at 14,700 rpm.

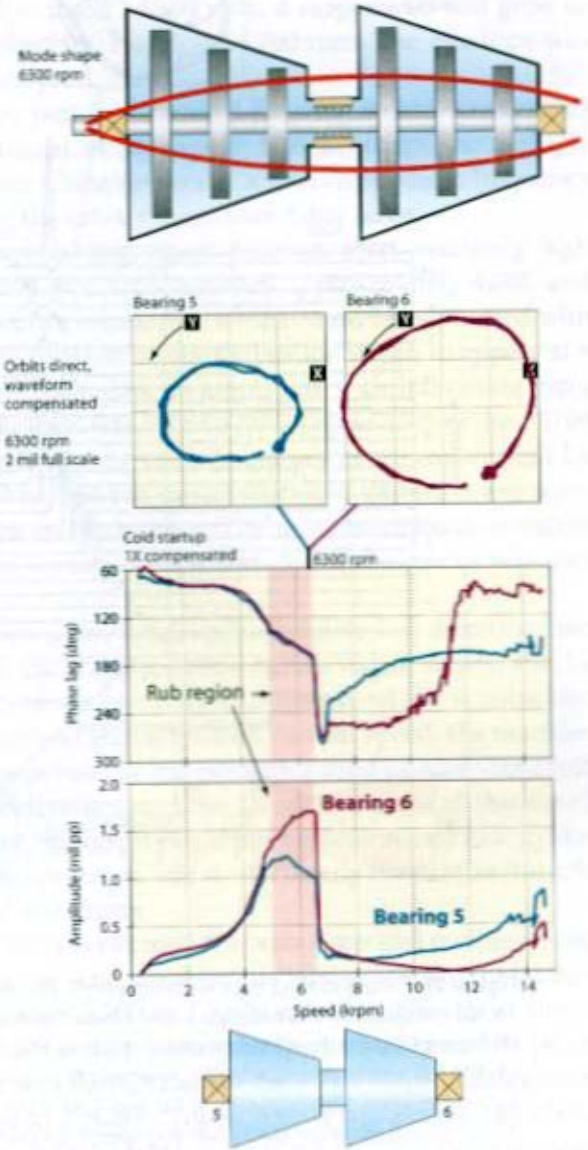



**Figure 29-3.** Compressor resonance behavior during the cold start with the original configuration. The 1X Bode plot shows an extended balance resonance region around 6000 rpm. The amplitude and phase behavior through the resonance are abnormal, indicating a probable rub. The 6300 rpm direct orbits are predominantly 1X and in phase, indicating that this is the first bending mode of the compressor rotor. Given the mode shape of the resonance, the highest 1X vibration amplitude during the resonance would occur at midspan (top); thus, it is most likely that the rub is occurring at the midspan labyrinth seal. The orbits show only mild distortion, indicating that the rub was probably not close to either bearing.

At this point, there was enough evidence to support a preliminary diagnosis. All of the data was consistent with fluid-induced instability as the source of the subsynchronous vibration at high speed: the vibration was forward, the frequency was not a simple integer ratio of running speed, and the vibration appeared suddenly at a frequency that appeared to be related to the resonance frequency. The instability did not track running speed; it appeared to lock onto the natural frequency of the rotor first bending mode, a characteristic of whip.

Also, the Bode plot amplitude and phase behavior during the resonance were classic symptoms of a 1X rub (see Chapter 21). Given the mode shape of the resonance, the highest 1X vibration amplitude during the resonance would occur at midspan; thus, it was most likely that the rub was occurring at either the midspan labyrinth seal or a nearby interstage seal. Direct orbits at both bearings (Figure 29-3) showed only mild distortion, indicating that the rub was probably not close to either bearing.

The problem now was to determine the location of the instability. The operators still suspected the floating seals. However, the machinery specialist had noted that the subsynchronous vibration due to the instability had appeared simultaneously at both compressor bearings. Also, the vibration amplitude, as shown by the direct orbits (Figure 29-2), had similar magnitudes at both bearings. Finally, the waveforms of the subsynchronous vibration (Figure 29-4) indicated an in-phase relationship for the subsynchronous vibration. This supported the cascade plot data and suggested that the vibration was associated with excitation of the first bending mode of the rotor. The data strongly suggested an

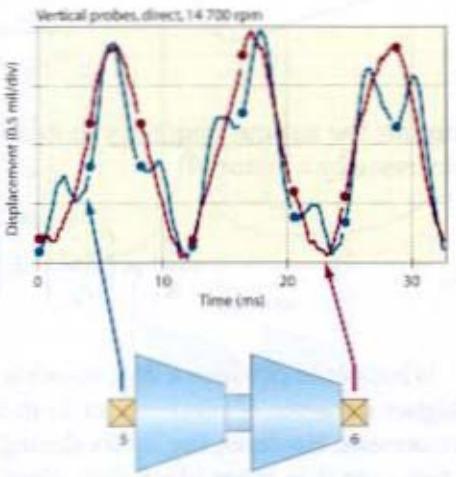



**Figure 29-4.** Waveforms of the subsynchronous vibration. Subsynchronous vibration, which is essentially in phase, dominates the direct (unfiltered) waveforms, supporting the idea that it is associated with the first bending mode of the rotor.

instability source near the rotor midspan, and the midspan labyrinth seal appeared to be the most likely location.

The operators continued with the series of tests. After restarting the machine, they changed the temperature of oil supplied to the radial bearings and the oil seals; this produced no improvement in stability.

During a hot coastdown following one of these test runs, it became evident that the compressor behavior had changed (Figure 29-5). The data on the Bode plot now appeared normal; there was no evidence of rub in either the amplitude, which now peaked at 4900 rpm, or the phase data. The vibration increase above 12 000 rpm suggested the presence of a second balance resonance somewhere above operating speed. The specialist calculated the Synchronous Amplification Factor (SAF) using the Half-power Bandwidth method as 5.2.

The frequency of the resonance peak at 4900 rpm was below the instability frequency of 5500 cpm, yet the specialist knew that fluid-induced instability is always associated with a natural frequency of the rotor system. The natural frequency, $\omega_n$, is approximately

$$\omega_n = \sqrt{\frac{K}{M}} \tag{29-1}$$

where $K$ is the rotor system spring stiffness and $M$ is the rotor mass. It is unlikely that the rotor mass changed; the observed shift in natural frequency implied a change in $K$, where

$$\frac{\omega_{n2}}{\omega_{n1}} = \sqrt{\frac{K_2}{K_1}} \tag{29-2}$$

Thus, for the natural frequency to increase from 4900 cpm to 5500 cpm, $K$ had to increase by a factor of

$$\frac{K_2}{K_1} = \left(\frac{\omega_{n2}}{\omega_{n1}}\right)^2 = \left(\frac{5500}{4900}\right)^2 = 1.26$$

What could produce a 26% increase in spring stiffness? One possibility was a higher dynamic eccentricity ratio in the bearings when the whip instability was present. However, the orbits during the instability were less than 2 mil pp. In this case it is more likely that, given the mode shape of the instability, the

rotor was using most of the available clearance at midspan, perhaps even directly contacting the midspan labyrinth seal; an increase in stiffness at midspan would have the greatest potential to modify the natural frequency of the rotor first mode. Thus, the significant change in spring stiffness supported the possibility that the source of the instability was at the midspan seal.

The operators isolated the entire station from the pipeline and used the station surge control bypass piping to run the compressor in a closed loop. This provided the opportunity to control suction and discharge pressures independently by pinching the flow at various points in the loop. During these test runs, the unit tripped on high vibration at significantly different rotor speeds. The compressor appeared to have an instability threshold that varied with differential pressure and speed.

During one closed loop test with around 14 bar (200 psi) suction and 15 bar (220 psi) discharge pressures, the machine was able to reach 100% speed without experiencing any subsynchronous vibration. The machine seemed to run

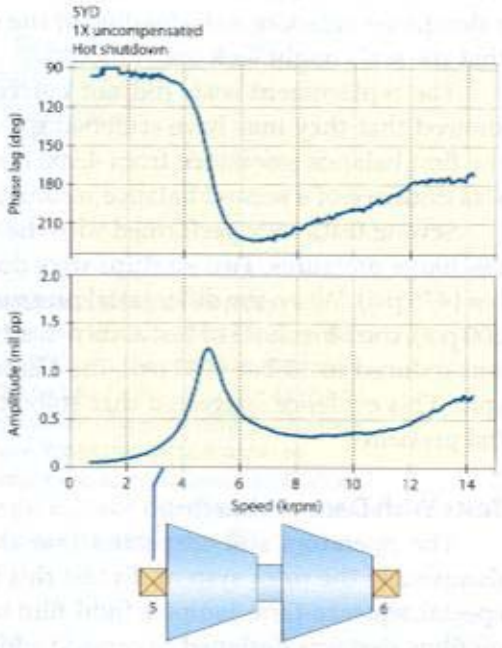



**Figure 29-5.** Bode plot of hot shutdown data from the bearing 5 vertical probe, original configuration. The 1X behavior now appears normal; the amplitude and phase show no evidence of a rub, and the resonance peak is at 4900 rpm. The vibration increase above 12 000 rpm suggests the presence of a second balance resonance somewhere above operating speed. The SAF, using the Half-power Bandwidth method, is 5.2.

fine at this low suction pressure. What would happen if suction pressure was increased while the machine kept running at full speed?

The operators increased suction pressure from 14 bar (200 psi) while the compressor ran at over 15,000 rpm in a closed loop. As suction pressure increased, discharge pressure approximately increased by the pressure ratio of 1.2, and differential pressure increased slowly. When the suction pressure reached 28 bar (410 psi), high subsynchronous vibration suddenly appeared, and the machine tripped (Figure 29-6).

During this test, the low level, ½X, forward vibration component again appeared. This component still showed evidence of tracking running speed as the machine tripped and began coasting down. A check of other transducer data showed that this component did not appear on any bearing 6 data, but it *did* appear in the power turbine casing vibration data. This suggested that the origin of this low level vibration was in the power turbine, not the compressor.

### Tests With Unpinned Seals

The operators suspected that the floating ring seals were the source of the instability, so they replaced the seals with a different design. To eliminate the possibility that the antirotation pins were preventing free seal motion, the pins were not included in the new design. The replacement seals did not have as good a design for balanced axial loading of the seal faces, so there was still some risk that the seals might lock up.

The replacement seals did not correct the problem, but a hot coastdown showed that they may have stiffened the rotor system, shifting the location of the first balance resonance from 4,900 rpm to 5,000 rpm. Also, there no longer was evidence of a second balance resonance near running speed.

Several tests were performed with the unpinned seals at various suction and discharge pressures. Two startups were done with suction pressures of about 33 bar (475 psi). When the differential pressure was between 13 and 14 bar (190 and 200 psi), the Threshold of Instability was 14,800 rpm. When differential pressure was reduced to 13 bar (180 psi), the Threshold of Instability dropped to 14,100 rpm. This evidence suggested that stability was enhanced with higher differential pressure.

### Tests With Damper Bearing

The operators still suspected that the instability was due to insufficient damping in the rotor system. To test this theory, they replaced bearing 5 with a special, squeeze-film-damped, fluid-film bearing, a bearing with two concentric oil films that was designed to provide additional damping.

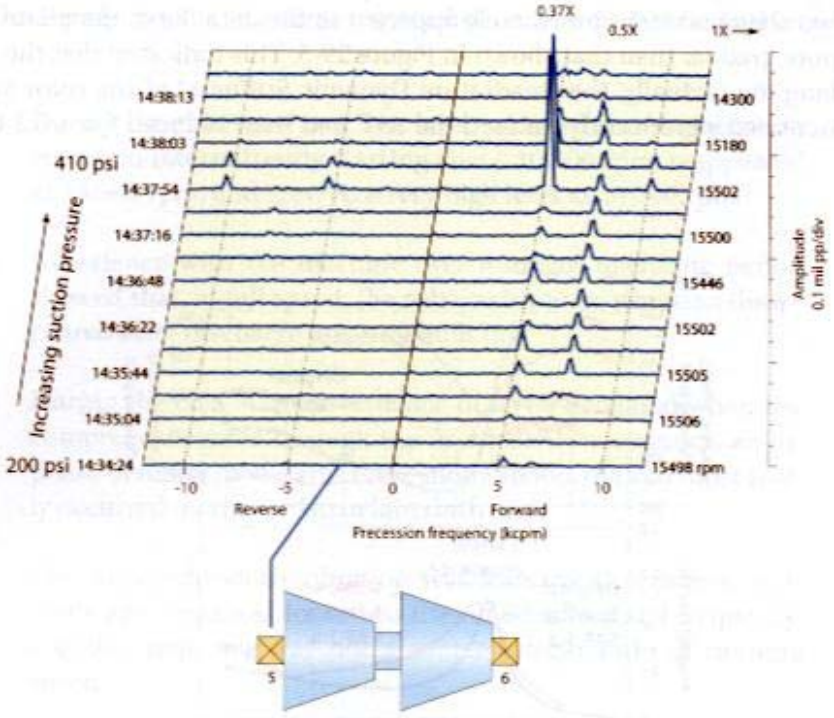



**Figure 29-6**. Full spectrum waterfall plot of closed loop tests with changing suction pressure, original configuration. While the compressor ran in a closed loop at over 15 000 rpm, the operators increased suction pressure from 14 bar (200 psi). As suction pressure increased, discharge pressure increased by a factor of approximately 1.2, slowly increasing the differential pressure. When the suction pressure reached 28 bar (410 psi), high amplitude, subsynchronous vibration suddenly appeared and tripped the machine. During this test, the low level, 1/2X forward vibration component again appeared.

During subsequent tests, the machine reached 98% of full speed at normal operating pressures without evidence of instability. Several runs were made, but the instability did not reappear. Hot shutdown data showed that the damper bearing had significantly altered the rotor system characteristics (Figure 29-7).

Two changes were immediately apparent in the data. First, the phase slope was more gradual than that shown in Figure 29-5. This indicated that the effective damping (actually, the Quadrature Dynamic Stiffness) of the rotor system had increased significantly; in fact, the SAF had been reduced from 5.2 to 2.5.

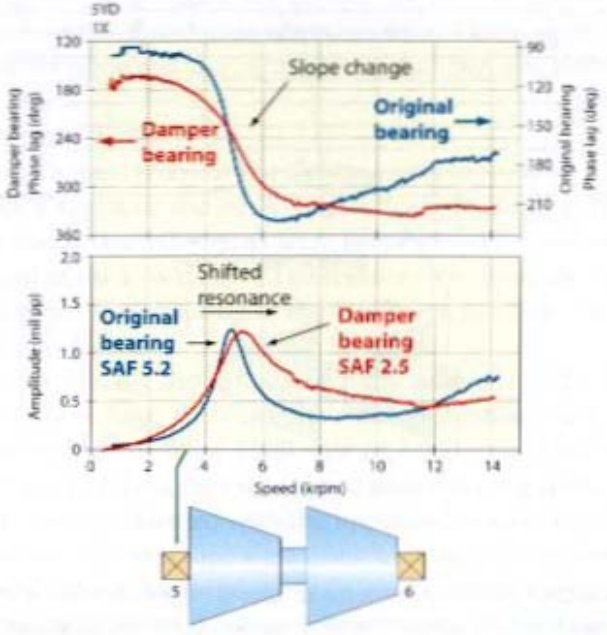



**Figure 29-7.** Comparison of hot shutdown data from original and damper bearing. Data from the original bearing (blue) is the same as in Figure 29-5. The damper bearing (red) has altered the rotor system characteristics: the phase slope is more gradual and the first balance resonance has moved from 4900 to 5300 rpm. These changes indicate that the Quadrature Dynamic Stiffness of the rotor system has increased significantly and that the damper bearing is stiffer than the original bearing.

Second, the first balance resonance had moved from 4900 to 5300 rpm. This indicated that damper bearing was stiffer than the original bearing.

### Additional Analysis

The Bently Nevada machinery specialist reviewed what he had learned so far:

1. During the cold start test, low level, subsynchronous vibration appeared intermittently starting about 10 300 rpm, reappeared at 14 300 rpm, and grew to a very high level at 14 700 rpm.

2. Experience with the machine over a longer operating period showed that, at full speed, the subsynchronous vibration disappeared after discharge pressure built up.

3. During the cold startup, evidence of a rub appeared when the compressor passed through the first balance resonance, an in-phase bending mode. Given the mode shape, the rub most likely occurred on the midspan labyrinth seal.

4. The subsynchronous vibration was forward, it appeared suddenly and remained locked to the modified natural frequency, and the frequency was not a simple integer ratio of running speed.

5. The subsynchronous vibration was approximately the same amplitude at both bearings, and it was in phase, confirming excitation of the first bending mode.

6. Changes in bearing and seal oil temperature had little effect on the subsynchronous vibration at operating speed.

7. The instability frequency was higher than the hot shutdown resonance frequency, indicating a 26% stiffening of the rotor system when the instability was active.

8. During closed loop testing at low suction and discharge pressures, the machine was able to reach 100% speed without experiencing any subsynchronous vibration.

9. During the same test, at a constant speed over 15 000 rpm, the vibration appeared when the suction pressure was increased to 28 bar (410 psi).

10. New floating ring seals without antirotation pins had no significant effect on the instability.

11. When the suction pressure was increased to a moderate value of about 33 bar (475 psi), the higher differential pressure increased the Threshold of Instability (increased stability).

12. The damper bearing installed at bearing 5 increased system damping, reducing the SAF to 2.5. It also increased rotor system spring stiffness, moving the first balance resonance from 4900 to 5300 rpm. With this bearing, the compressor was stable at 98% of full speed and load.

What was the pattern in all this information? The damper bearing seemed to have solved the problem, at least temporarily, but the Bently Nevada specialist still suspected that the midspan seal was the real root cause of the problem. He realized that two things had been changed when the damper bearing was installed: damping and stiffness.

The specialist considered the equation for the Threshold of Instability, $\Omega_{th}$, that was derived from the simple rotor model in Chapter 14:

$$\Omega_{th} = \frac{1}{\lambda}\sqrt{\frac{K}{M}} \tag{29-3}$$

where $\lambda$ is the Fluid Circumferential Average Velocity Ratio and $K$ and $M$ are the rotor system spring stiffness and mass. This expression defines the speed at which the rotor system will begin the subsynchronous vibration associated with fluid-induced instability. At speeds below this value, the machine will be stable.

To stabilize the rotor system, $\Omega_{th}$ must be increased to some value above the maximum running speed of the machine. This could be done by increasing the rotor system natural frequency, usually by increasing $K$, or by reducing the fluid circulation in the midspan seal, $\lambda$, which he suspected was responsible for the instability.

There was a way to check if the increase in $K$ due to the damper bearing was large enough to stabilize the machine. The specialist could estimate $\lambda$ from the

hot shutdown resonance speed of 4900 rpm and the observed cold startup Threshold of Instability of 14 300 rpm. Solving Equation 29-3 for $\lambda$,

$$\lambda = \frac{1}{\Omega_{th}}\sqrt{\frac{K}{M}} = \frac{1}{14\,300 \text{ rpm}} 4900 \text{ rpm} = 0.34$$

Now the specialist assumed that, after installation of the damper bearing, $\lambda$ in the seal remained the same. This was reasonable; no changes were made to the seal during the damper bearing installation. Using Equation 29-3, the system should now have a Threshold of Instability of

$$\Omega_{th} = \frac{1}{\lambda}\sqrt{\frac{K}{M}} = \frac{1}{0.34}(5300 \text{ rpm}) = 15\,600 \text{ rpm}$$

This speed was above the speeds that were reached during testing. It was quite possible that the beneficial effect of the damper bearing was due to the higher spring stiffness, $K$, of the bearing.

Could increasing the bearing damping have also improved stability? There is no damping term in Equation 29-3. This is a result of the rotor model that was used to derive the expression. However, Muszynska [1] originally developed a slightly more complex model than the one that was presented in Chapter 10. This expanded model includes the effects of damping, $D$, and a new damping term, the *system damping*, $D_S$. In the full model, the primary damping term, $D$, is associated with $\lambda$ and $\Omega$ in the region where the destabilizing tangential force ($jD\lambda\Omega\boldsymbol{r}$ in the model) originates. The system damping term, $D_S$, accounts for all other sources of damping forces in the system. (In this model, the *total* damping force becomes $-(D_S + D)\dot{\boldsymbol{r}}$.)

In Muszynska's model, the Threshold of Instability is

$$\Omega_{th} = \left(1 + \frac{D_S}{D}\right)\frac{1}{\lambda}\sqrt{\frac{K}{M}} \tag{29-4}$$

In our rotor model of Chapters 10 and 14, we assumed that $D_S = 0$; applying that here, Equation 29-4 reduces to Equation 29-3.

To see the possible effect of the damper bearing, the specialist assumed that the source of the instability was in the midspan seal, where the parameters $D$ and $\lambda$ were presumed to exist. The damping in the rest of the compressor,

including the bearings, was lumped into $D_S$. (The bearings were tilting pad designs, with inherently low $\lambda$ by design. Although the oil control rings and the floating seal rings *were* potentially capable of producing a significant destabilizing tangential force, they were ignored for this qualitative analysis.) When the damper bearing was installed, $D_S$ increased (he knew this from the change in SAF), and, according to Equation 28-4, $\Omega_{th}$ would also increase.

The question was how much? The sensitivity of $\Omega_{th}$ to changes in $D_S$ depends on the relative sizes of the two damping terms. Without knowing the values of $D$ and $D_S$, it was difficult to say how sensitive $\Omega_{th}$ was in this situation. The large change in SAF with the damper bearing argued for a large enough ratio of $D_S/D$ to produce good sensitivity. It seemed safe to conclude that, because of the system damping increase, $\Omega_{th}$ was likely to be somewhere above 15 600 rpm. Thus, it is likely that the increased stiffness and damping of the new bearing both acted to stabilize the machine.

The damper bearing testing pointed toward a possible solution to the stability problems in these compressors, but the fix was a rather complex and expensive retrofit. The specialist still believed that the primary source of the instability was the seal, and the best solution should address the root cause of the problem.

There were still unsolved puzzles about the behavior of this machine. Why didn't the instability appear during the 14 bar (200 psi) closed loop tests? Why did it appear at moderate suction pressures, but improve with increasing differential pressure?

The answer to the first question might have to do with the density of the gas in the seal. At low pressures, the density would be relatively low, perhaps too low to develop destabilizing tangential forces. That would explain the stability of the machine at low pressures.

When suction pressure was increased, discharge pressure also increased, approximately by the pressure ratio of 1.2. Thus, the average pressure and density in the seal also increased. The higher density would increase the strength of the tangential force in the seal (most likely by increasing $D$) until it destabilized the machine. This would explain why the compressor was stable until it reached moderate suction pressure.

Another factor was the fluid circulation term, $\lambda$. The value of $\lambda$ would depend partially on the internal geometry of the seal and partially on the axial flow rate of gas through the seal. As gas traversed the seal from the high-pressure side to the low-pressure side, the rotor would spin it up to some average circumferential velocity, $\lambda\Omega$. The value of $\lambda$ would partially depend on the amount of time the gas remained in the seal.

Increasing differential pressure would reduce that time. With low differential pressure, the time would be relatively long, and $\lambda$ would be relatively high. With high differential pressure, the gas would move through relatively fast, and $\lambda$ would be lower. Lower $\lambda$ would increase the Threshold of Instability, stabilizing the machine. This could be the reason why the compressors would stabilize when discharge pressure built up to maximum. The axial flow in the seal would reduce $\lambda$ and increase the Threshold of Instability to above operating speed.

In addition to these pressure effects, the rub had probably opened up the seal clearance, increasing the axial flow. It was quite possible that, as the seal clearance opened up because of rubs, axial flow velocity through the seal region would increase, $\lambda$ would drop, and compressor stability would be enhanced. There was no historical data showing whether the Threshold of Instability had increased over time, but it was quite possible that the numerous cold startups during the testing may have caused repeated seal rubs that altered the machine behavior during the course of the testing. The observed increase in the Threshold of Instability associated with the damper bearing may have been partially due to increased seal clearances.

### Conclusions and Recommendations

The Bently Nevada machinery specialist concluded that the most likely source of the instability was the midspan labyrinth seal of the compressor. This conclusion was supported by the fact that the whip vibration was similar in amplitude and in phase at both ends. If the whip were occurring at either the bearing or floating seal, it would produce larger whip amplitudes at the nearest bearing.

This conclusion was also supported by the effect that changes in average pressure and pressure differential had on stability. Gas present in the seal was capable of affecting the rotor system spring stiffness, $K$, and seal damping, $D$. At low average pressures, the density of the gas in the seal area was too low to produce instability. Equation 29-4 shows that, if $D$ were very small, then $\Omega_{th}$ would be high, resulting in a stable machine. Under this circumstance, $\Omega_{th}$ would be very sensitive to changes in $D$. As average pressure and gas density increased, the increase in $D$ would drop $\Omega_{th}$ to below operating speed, resulting in the appearance of the instability. This decrease in stability was partially offset by the effect differential pressure had on $\lambda$; as differential pressure increased, axial flow increased, reducing $\lambda$ and increasing $\Omega_{th}$. Thus, the appearance of the instability depended on two, independent factors with opposite effects, both of which were likely associated with the midspan seal.

The specialist also concluded that, during the cold startup, the compressor entered a rub condition at 5000 rpm and left the rub condition at about 6600

rpm. The Bode plot of the startup had shown classic resonance broadening due to a radial rub. In addition, the direct orbits from both compressor bearings during the rub showed only minimal distortion. For these reasons, the rub was not suspected to be near either bearing, supporting a location at the midspan seal.

The specialist recommended modification of the midspan seal rather than use of the damper bearing. Evidence strongly pointed to the seal as the source of the instability; any design fix should concentrate on the root cause of the problem. The seal could be shortened to improve axial flow, possibly sacrificing some efficiency, or a new design could be used that incorporated antiswirl injection. This would involve injecting gas at discharge pressure into the seal with a tangential velocity component *against* the direction of rotation, which would reduce $\lambda$ and stabilize the machine.

The operators conducted additional field tests and found that the damper bearings provided acceptable stability. They decided to install the damper bearings on some of the compressor packages for more testing. On some of these machines, the new bearings caused the second balance resonance to move close to running speed, causing a high sensitivity to unbalance. Thus, it became evident to the operators that the damper bearings were not a good, long term solution.

Following the specialist's original recommendation, the operators modified the midspan labyrinth seal of the compressor. They shortened the seal length and applied the antiswirl concept by injecting gas at discharge pressure into the seal. The modified seals provided a good, long-term solution to eliminating the stability problem, and the fix was incorporated in all of the compressor packages.

## References

1. Muszynska, A, "One Lateral Mode Isotropic Rotor Response to Nonsynchronous Excitation," *Proceedings of the Course on Rotor Dynamics and Vibration in Turbomachinery*, von Karman Institute for Fluid Dynamics, Belgium (September 1992): pp. 21-25.

# Appendix

Appendix 1

# Phase Measurement Conventions

THE ROTOR MODEL PRESENTED IN CHAPTER 10 uses phase and amplitude measurement conventions that are convenient for the mathematical description of the rotor system. They are different from those commonly used by machinery instrumentation systems.

When you compare the measured behavior of a machine to the prediction of a mathematical model, you must convert between the two measurement systems. This chapter will explain the differences between the systems, starting with the instrumentation convention, and show how to convert between them.

## The Instrumentation Convention

In Chapter 2 we defined phase as the time delay between two events. When event "B" occurs some time after event "A," we say that event "B" *lags* event "A." If we start a timer when event "A" occurs, the time increases until event "B" occurs. The time delay, or *lag*, between the two events is expressed in degrees of a vibration cycle, a positive number called *positive phase lag*, or typically, *phase lag*.

Most machine-oriented instrumentation systems use a positive phase lag convention. In this *instrumentation convention*, increasing time delay produces a larger positive number. When the time delay represents *absolute* phase lag, the "A" event is the Keyphasor event and the "B" event is the next positive peak of the vibration waveform; the difference is expressed as an angle. A phase lag between 180° and 360° is sometimes called a *phase lead*, and could be expressed as a negative lag number between −180° and 0°. When some machines pass through resonances, phase changes greater than 360° can be measured. However, angles in the instrumentation convention are typically converted to a

positive number between 0° and 360° for display on instrumentation or data plots.

This is equivalent to reducing the phase angle *modulo 360* (abbreviated *mod 360*): for phase numbers greater than 360°, repeatedly subtract 360° until the result falls between 0° and 360°; for negative numbers, repeatedly add 360°. For example, using this convention, a calculated phase angle of −60° would be displayed as 300°, and a phase angle of 740° would be displayed as 20°. Note that *phase should not be reduced mod 360 before performing difference calculations.*

In the instrumentation convention, the amplitude of shaft relative displacement vibration is normally displayed in peak-to-peak units (for example, 20 μm pp) and velocity and acceleration are displayed in zero-to-peak (peak, or pk) units.

As we showed in Chapter 3, the amplitude and absolute phase lag of the filtered signal describe a vibration vector that rotates at the filter frequency, in the direction of rotation. The transducer sensitive axis defines the zero degree reference position, and the Keyphasor event behaves like a strobe that illuminates the position of the rotating vector.

In the instrumentation convention, the phase lag represents the time delay between the Keyphasor event and when the vibration vector arrives at the transducer position. This phase lag angle defines the angular position of the vibration vector when the Keyphasor event occurs. If, after the Keyphasor event occurs, the vector rotates 220° in the direction of rotation to point at the transducer, then you must rotate it 220° *against* rotation, from the transducer, to see where it pointed when the Keyphasor event occurred.

Figure A1-1 (top) shows two examples of vector plotting using the instrumentation convention. The same vector (150 μm pp ∠220°) is plotted in the transducer response plane for different rotation directions. At left, the machine rotation direction is *X* to *Y* (counterclockwise); at right, it is *Y* to *X* (clockwise).

### The Mathematical Convention

In the mathematical convention, the phase is the angle of the vibration vector when the Keyphasor event occurs, measured from the *X* axis, in the positive (counterclockwise) direction. The transducer sensitive axis defines the *X* axis of a complex plane.

While it is possible to define different mathematical coordinate systems that are dependent on the direction of shaft rotation, it is simplest to use a coordinate system that is independent of this variable. In this system, the vibration vector is always assumed to rotate in the positive mathematical sense (counterclockwise), regardless of the direction of shaft rotation.

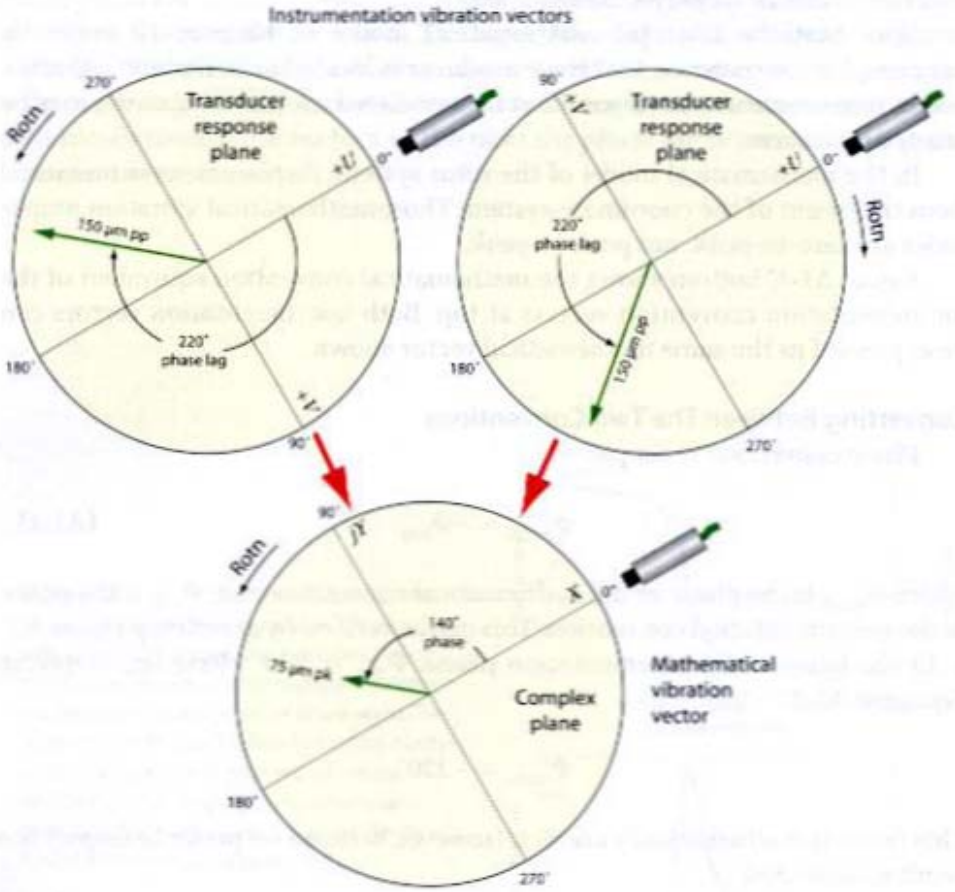



**Figure A1-1.** Phase measurement conventions. At top are two examples of vector plotting using the instrumentation convention. The same vector (150 µm pp ∠220°) is plotted in the transducer response plane for different rotation directions, *X* to *Y* (counterclockwise) at left, *Y* to *X* (clockwise) at right. In the mathematical convention (bottom), the phase is the angle of the vibration vector when the Keyphasor event occurs, measured from the *X* axis, in the positive (counterclockwise) direction. The transducer sensitive axis defines the *X* axis of a complex plane.

Mathematical phase can be measured as a positive or negative number and can exceed 360°. When subtracting phase numbers, the form that retains negative values is best. After calculations have been completed, it is recommended that the results be displayed modulo 360.

Note that the isotropic mathematical model of Chapter 10 implicitly assumes, for convenience, that the transducer is located at horizontal right; the actual measurement transducer (and its associated coordinate system) may be at any orientation.

In the mathematical model of the rotor system, displacement is measured from the origin of the coordinate system. Thus, mathematical vibration amplitudes are zero-to-peak, not peak-to-peak.

Figure A1-1, bottom shows the mathematical convention equivalent of the instrumentation convention vectors at top. Both instrumentation vectors can be expressed as the same mathematical vector shown.

### Converting Between The Two Conventions

Phase conversion is simple:

$$\Phi_{math} = -\Phi_{inst} \tag{A1-1}$$

where $\Phi_{math}$ is the phase in the mathematical convention and $\Phi_{inst}$ is the phase in the instrumentation convention. This can be verified by examining Figure A1-1. In the figure, the instrumentation phase, $\Phi_{inst} = 220°$ phase lag. Applying Equation A1-1,

$$\Phi_{math} = -220^\circ$$

This result is mathematically correct. However, because we prefer to display the result modulo 360,

$$\Phi_{math} = -220^\circ + 360^\circ = 140^\circ$$

The result agrees with that shown in Figure A1-1.

Displacement amplitude conversion is also simple. The mathematical convention amplitude (pk) is one half of the instrumentation amplitude (pp). If you discover a discrepancy of a factor of two in the displacement amplitude, it is likely this little detail was overlooked. Remember that velocity and acceleration amplitudes are already typically displayed as peak values in instrumentation.

## Phase On Bode and APHT Plots

The difference in phase measurement conventions is most obvious on a Bode (Figure A1-2) or APHT plot. The left side of the phase plot is labeled as *phase lag* in the Bently Nevada convention, where the phase lag becomes *more positive* in a downward direction. The right side of the plot is labeled as *phase* using the mathematical convention, where the phase becomes *more negative* in a downward direction. As speed increases through the resonance, the phase data moves toward the bottom of the plot; the *phase lag* is increasing, but the *phase* is decreasing.

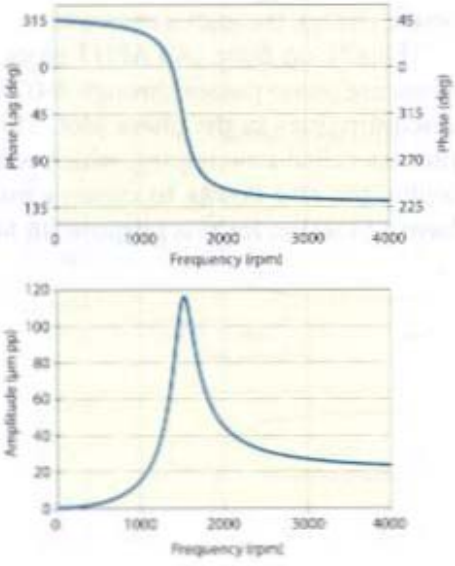



**Figure A1-2.** Bode plot showing instrumentation and mathematical phase labeling. The left side of the phase plot is labeled using the Bently Nevada convention, where phase lag becomes more positive downward. The right side of the plot is labeled using mathematical phase (mod 360), where phase becomes more negative in a downward direction. Both are correct, but the Bently Nevada convention is preferred.

Plots in technical literature sometimes show phase lag increasing upward; often this is a result of plotting phase lag data using the autoscaling features of the plotting software. This can lead to confusion and misinterpretation of the data.

We know that, normally, phase lag increases as machine speed increases through a resonance, but there are several possible ways to present this data. Figure A1-3 shows different ways of plotting the same phase data. At top, the Bently Nevada instrumentation convention is used, and this is the recommended method for general machinery work. Next are two ways of labeling the same data using the mathematical convention; one uses the continuum of real negative numbers, the other reduces the phase modulo 360. Note that the data plots the same way, with phase data moving lower with increasing speed. The bottom plot inverts instrumentation phase, so that phase lag increases in the upward direction. This is confusing because it inverts the critical visual characteristic of leading and lagging phase and is not recommended.

When examining unfamiliar data, be sure to check which phase plotting convention is being used. The easiest way to verify the data is to look for the phase change through a resonance.

Finally, on Bode and APHT plots, the phase is displayed modulo 360. If the changing phase passes through 360°, simple plotting routines can produce jump discontinuities in the phase plot. These discontinuities can be eliminated by a process called *unwrapping*, which removes the phase jump discontinuities and unites the end points to create a continuous curve. Some software packages have a function for this purpose (in MATLAB, the *unwrap* function).

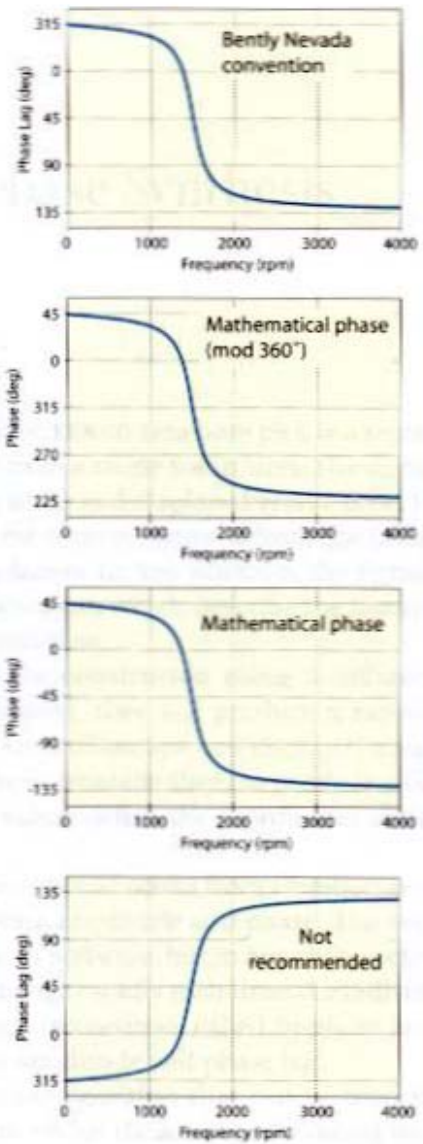



**Figure A1-3**. Different phase plotting methods. The top plot uses the Bently Nevada convention, where phase lag increases downward. The next two plots, where the phase decreases downward, use the mathematical convention. Two methods are shown: phase modulo 360° and continuous phase. All three plots present the phase data the same way; the Bently Nevada convention is preferred. In the bottom plot, where phase lag increases upward, the curve is inverted, producing a phase plot that can cause confusion. This method is not recommended.

Appendix 2

# Filtered Orbit and Timebase Synthesis

In chapter 4, we showed how the unfiltered timebase plot is a representation of the unfiltered, ac signal voltage from a single transducer. The signal is converted to the appropriate engineering units and displayed versus time. In Chapter 5, we showed how an unfiltered orbit is reconstructed from the unfiltered waveforms from two, orthogonal transducers. In that situation, the signals are converted to engineering units of displacement, which describe the instantaneous position coordinates of the shaft centerline.

A *filtered* timebase and *filtered* orbit are constructed using a different process. When electronic vector filters are used, they will produce a rapidly changing (dynamic), analog output signal. An oscilloscope can display the signals as separate filtered timebase waveforms or combine them to produce a filtered orbit, where the instantaneous signal values define the coordinates of the shaft centerline on the filtered orbit.

Computers synthesize the filtered waveforms and orbits from vibration vectors, which are complex numbers representing amplitude and phase. The vectors can be generated in hardware, firmware, or software, but in any case, vector data is essentially *static* in that it does not change rapidly with time. An individual vector can be stored in rectangular form (sometimes called *in-phase* and *quadrature* components) or in polar form as amplitude and phase lag.

This chapter will present a computational algorithm that can be used to create filtered timebase and orbit plots from vector data. In the following discussion, we will assume the vector data has been converted to engineering units and stored in polar form, using positive phase lag (the instrumentation convention, see Appendix 1).

## Timebase Synthesis

The amplitude and phase of the nX displacement vibration vector is $\boldsymbol{x}_{inst}$,

$$\boldsymbol{x}_{inst} = A_{inst} \angle\alpha_{inst} \tag{A2-1}$$

where $A_{inst}$ is the filtered vibration amplitude provided by the instrumentation, and $\alpha_{inst}$ is the phase lag in the instrumentation convention.

In order to synthesize the timebase waveform, the vector must be converted from the instrumentation convention to the mathematical convention,

$$\boldsymbol{x} = A \angle\alpha \tag{A2-2}$$

In the instrumentation convention, the amplitudes of displacement vibration vectors are in peak-to-peak (pp) units, while velocity and acceleration vector amplitudes are in zero-to-peak (pk) units. Thus, for displacement vectors with peak-to-peak amplitude,

$$A = \frac{A_{inst}}{2} \tag{A2-3}$$

and for velocity and acceleration vectors,

$$A = A_{inst} \tag{A2-4}$$

Phase in the mathematical convention is the negative of the instrumentation phase (Appendix 1):

$$\alpha = -\alpha_{inst} \tag{A2-5}$$

At a rotor speed, $\Omega$, in rpm, the nX vector frequency, $\omega$, in rad/s, is

$$\omega(\text{rad/s}) = \text{n}\frac{\pi}{30}\Omega(\text{rpm}) \tag{A2-6}$$

Using the compact and convenient exponential notation, the *rotating* vibration vector can be described as

$$\boldsymbol{x} = Ae^{j(\omega t+\alpha)} \tag{A2-7}$$

To synthesize the filtered waveform, the nX filtered vibration vector is rotated in the complex plane at the filter frequency (Figure A2-1). In the figure, the same instrumentation convention vector is plotted two ways, corresponding to different directions of shaft rotation. Regardless of the direction of shaft rotation, *the mathematical convention* nX *vector is always assumed to rotate in a positive mathematical direction* ($X$ to $Y$, or counterclockwise). This is shown in the middle of the figure.

At time $t = 0$ (the Keyphasor event), the orientation of the vector corresponds to the phase, $\alpha$, of the filtered vibration. The instantaneous projection of the rotating vector on the transducer sensitive axis corresponds to the filtered position of the shaft as seen by the transducer. This projection is the real part of Equation A2-7,

$$x = A\cos(\omega t + \alpha) \tag{A2-8}$$

When $x$ is plotted against time on the horizontal axis, this is a timebase plot. The period of one cycle of vibration is given by

$$T = \frac{2\pi}{\omega} \tag{A2-9}$$

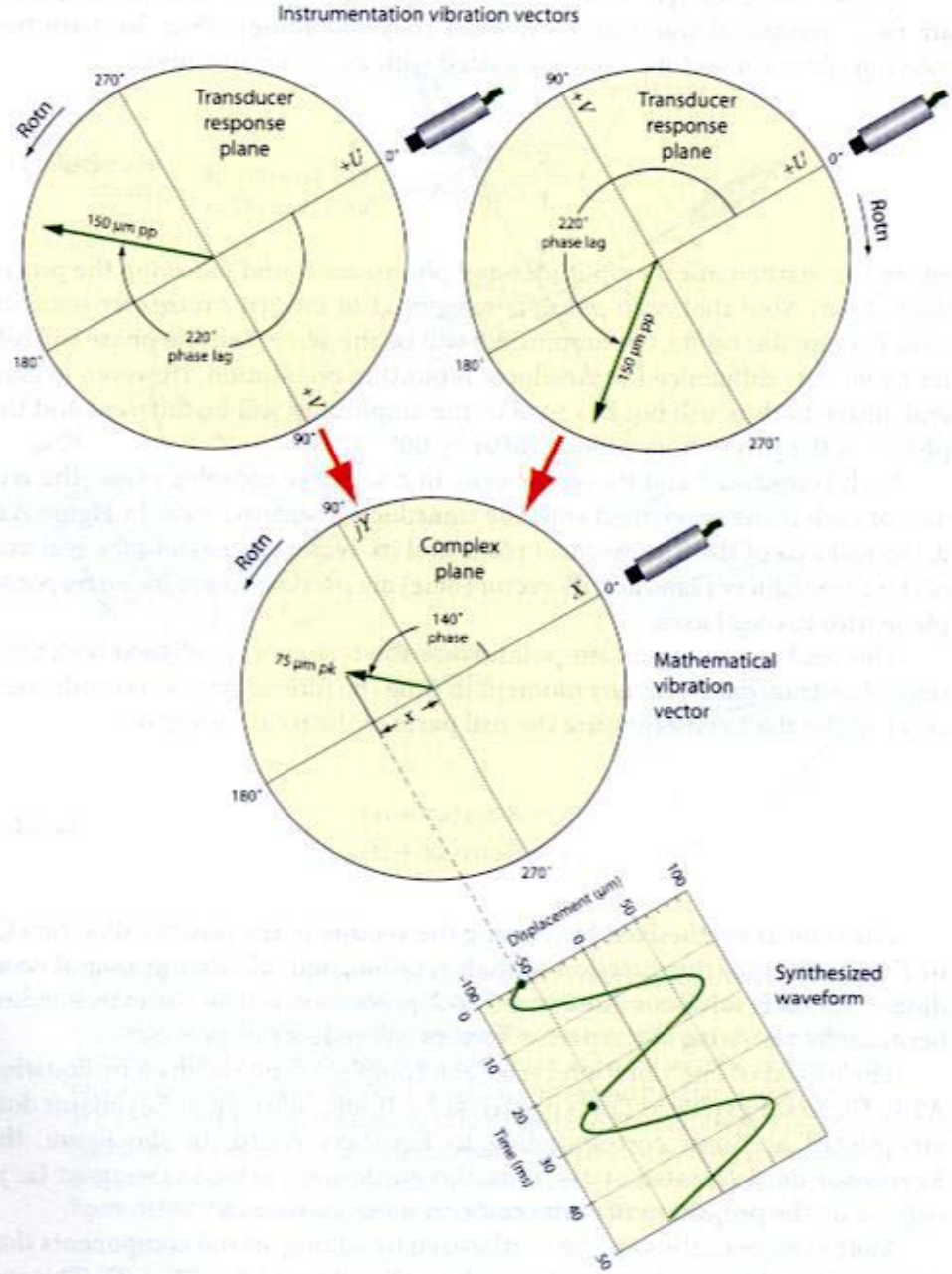
The Bently Nevada plot standard places the Keyphasor dot at the time of each Keyphasor event. Thus, for Equation A2-8, the Keyphasor dot would be plotted at the beginning of the plot ($t = 0$), and *once per revolution of the shaft* thereafter. The time (in seconds) to plot the $k$th Keyphasor dot is

$$t_k = \frac{60(k-1)}{\Omega(\text{rpm})} \tag{A2-10}$$

The blank is created by omitting several points before the Keyphasor dot.

**Figure A2-1.** Timebase synthesis from a vibration vector. A 1X displacement vibration vector (150 μm pp ∠220°) is plotted in the transducer response plane (top). The same instrumentation convention vector is plotted two ways, corresponding to different rotation directions. The location of the vectors shown corresponds to the Keyphasor event. Both instrumentation vectors convert to the same mathematical vibration vector, which is plotted in the complex plane (middle). The real part of the rotating vector, the projection of the vector on the transducer sensitive axis, produces the synthesized timebase waveform. Keyphasor dots are added at the beginning and at intervals of one shaft rotation, which, because this is a 1X vector, equals one cycle of vibration.

**Figure A2-1.**


Instrumentation vibration vectors
Transducer response plane
Rotn
270°
+U
0°
150 µm pp
220° phase lag
180°
+V
90°
Transducer response plane
90°
+V
+U
0°
220° phase lag
150 µm pp
180°
Rotn
270°
Complex plane
90°
jY
X
0°
Rotn
140° phase
75 µm pk
x
180°
270°
Mathematical vibration vector
Displacement (µm)
100
50
0
-50
-100
0
10
20
30
40
50
Time (ms)
Synthesized waveform

## Orbit Synthesis

Synthesis of a filtered orbit proceeds along similar lines. Now, however, there are two, orthogonal transducers (at least they should be). Thus, we have two rotating vibration vectors, each associated with its own transducer,

$$\begin{aligned} \boldsymbol{x} &= Ae^{j(\omega t+\alpha)} \\ \boldsymbol{y} &= Be^{j(\omega t+\beta)} \end{aligned} \tag{A2-11}$$

where the mathematical amplitudes and phases are found following the procedure above. Note that *each phase is referenced to its own transducer sensitive axis*. For circular orbits, the amplitudes will be the same, and the phase will differ by 90°, the difference in transducer mounting orientation. However, in general, filtered orbits will not be circular; the amplitudes will be different and the phases of the two vectors will not differ by 90°.

Each transducer and its vector exist in a separate complex plane; the real axis of each plane is oriented with the transducer's sensitive axis. In Figure A2-2, the real axis of the *X* transducer plane and its vector (green) and the real axis of the *Y* transducer plane and its vector (blue) are overlaid to produce a response plane with two real axes.

This leads to an important point. Since the real axes are aligned with their respective transducers, at any moment in time the filtered physical coordinates, $(x, y)$, of the shaft centerline are the real parts of the rotating vectors,

$$\begin{aligned} x &= A\cos(\omega t+\alpha) \\ y &= B\cos(\omega t+\beta) \end{aligned} \tag{A2-12}$$

The orbit is synthesized by rotating the vectors in the positive direction (*X* to *Y*), regardless of the direction of shaft rotation, and calculating a set of coordinates for each value of *t*. Note that *Y* to *X* precession will be correctly handled because the phase lag of the parent vectors will reflect this precession.

The period of one vibration cycle (one complete orbit) is given by Equation A2-9. The first Keyphasor dot is plotted at $t = 0$, and subsequent Keyphasor dots are plotted at times corresponding to Equation A2-10. In the figure, the Keyphasor dot is located at $t = 0$ on the synthesized orbit at the point $(x, y)$ defined by the projection of each vector on its sensitive axis.

More complex orbits can be synthesized by adding orbital components that are filtered to different frequencies, such as 1X + 2X, or 1X + 2X + 3X. This can allow construction of complex orbits when only vector data is available.

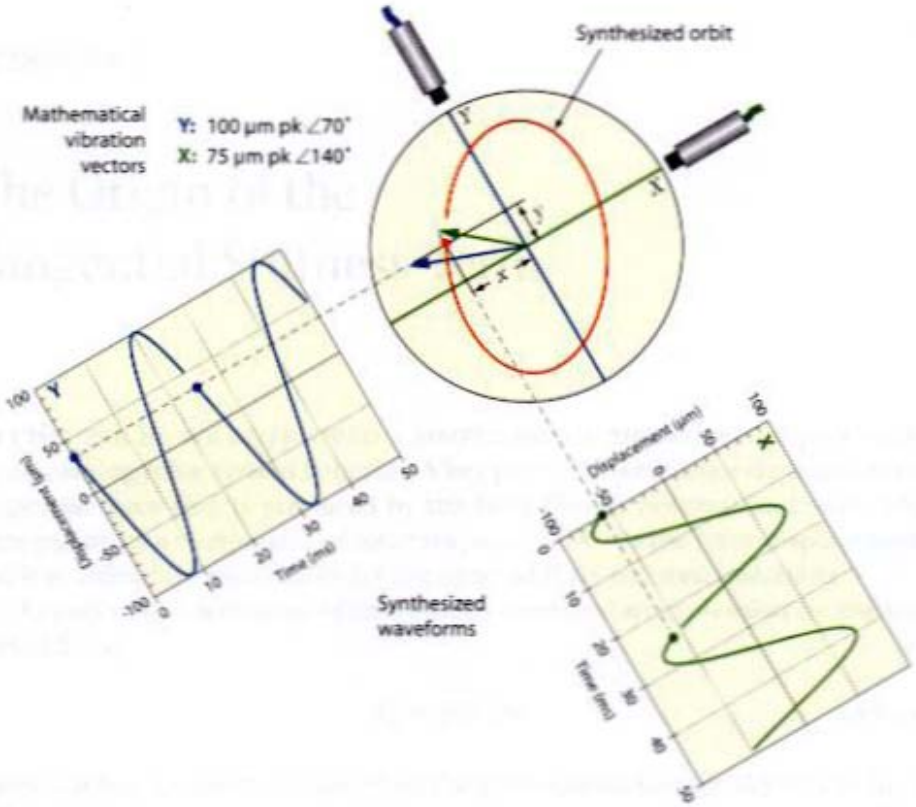



**Figure A2-2.** Filtered orbit synthesis from vibration vectors. Each transducer/vector combination exists in a separate complex response plane, which has its real axis oriented with its transducer's sensitive axis. The two response planes have been overlaid in the figure, producing real *X* and *Y* axes, and each vector is located at its phase with respect to its real axis. The orbit is synthesized by rotating the vectors at the filter frequency in the positive (*X* to *Y*) direction and calculating a set of coordinates for each value of *t*. The coordinates are defined by the instantaneous projection of each vector on its sensitive axis. The projections of the vectors in the figure define the coordinates of the Keyphasor dot on the orbit.

Appendix 3

# The Origin of the Tangential Stiffness Term

In chapter 10, we developed a simple rotor model that is quite useful for explaining rotor system behavior. A key part of this behavior depends on the tangential force that is produced by the fluid-film in bearings and seals. This force produces a component of rotor response at 90° to the rotor displacement, and it is ultimately responsible for the onset of fluid-induced instability.

As part of the derivation of that model, we stated an expression for the tangential force,

$$\boldsymbol{F}_T = jD\lambda\Omega\boldsymbol{r} \tag{A3-1}$$

where $D$ is the damping, $\lambda$ is the Fluid Circumferential Average Velocity Ratio, $\Omega$ is the rotor speed in rad/s, and $\boldsymbol{r}$ is the position vector of the rotor measured relative to the origin of the coordinate system, which was taken to be the center of the bearing. Finally, $j = \sqrt{-1}$, which, in our model, indicates that the force is directed at 90° from the displacement in the direction of rotation.

In this chapter, we will show how this expression can be derived from the concept of a rotating system of fluid forces.

## Modeling the Pressure Wedge

As we stated in Chapter 10, the physical source of the tangential force is the pressure distribution that forms when a rotating rotor is displaced from the center of a fluid-film bearing. The fluid, which has been dragged into motion by the rotor, now encounters a reduction in available clearance and slows down. The deceleration of this fluid produces an increase in the local pressure in the fluid near the area of minimum clearance between the rotor and the bearing surface

(Figure A3-1). The fluid produces a circumferential *pressure wedge* that pushes on the rotor and moves it to the side.

When the applied load is static, the rotor moves until the force produced by the pressure wedge exactly opposes the force applied to the rotor. This pressure wedge produces a spring-like force that is the primary means of rotor support in hydrodynamic journal bearings. When the rotor centerline moves in response to dynamic forces, the pressure wedge shifts position, orienting itself relative to the instantaneous position of the shaft. At the same time, the fluid produces damping forces that oppose rotor centerline motion.

Any mathematical model of the fluid force system must take these static and dynamic behaviors into account, and the resulting model behavior must be a good fit to observed rotor system behavior.

It is possible to model this complex fluid behavior with a relatively simple system of rotating springs and dashpots. The springs provide the spring stiffness, and the dashpots provide damping. These elements surround the rotor, which has angular velocity $\Omega$, and they rotate more slowly, at the average fluid angular velocity, $\lambda\Omega$. The elements maintain a radial orientation as they move around the clearance. The clearance at any location is a measure of the length of the springs and dashpots.

When the rotor is centered in the clearance (Figure A3-2, top), there is no change in the length of the springs and dashpots as they move around the rotor; thus, the forces due to the surrounding fluid are balanced around the rotor.

When the rotor is moved to a static position below the center (Figure A3-2, bottom), the springs change length as they move around the clearance. The

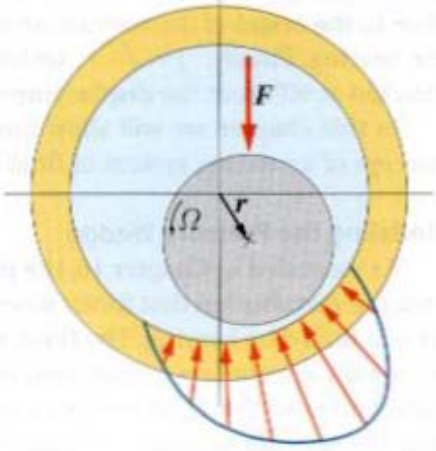



**Figure A3-1.** The fluid pressure wedge. When an applied force, $F$, moves a rotating rotor to an eccentric position in the bearing, the gap between the journal and bearing surfaces is partially closed. When the moving fluid encounters this restriction, the fluid pressure increases (red arrows). The pressure distribution creates a force that is equal and opposite to the applied force.

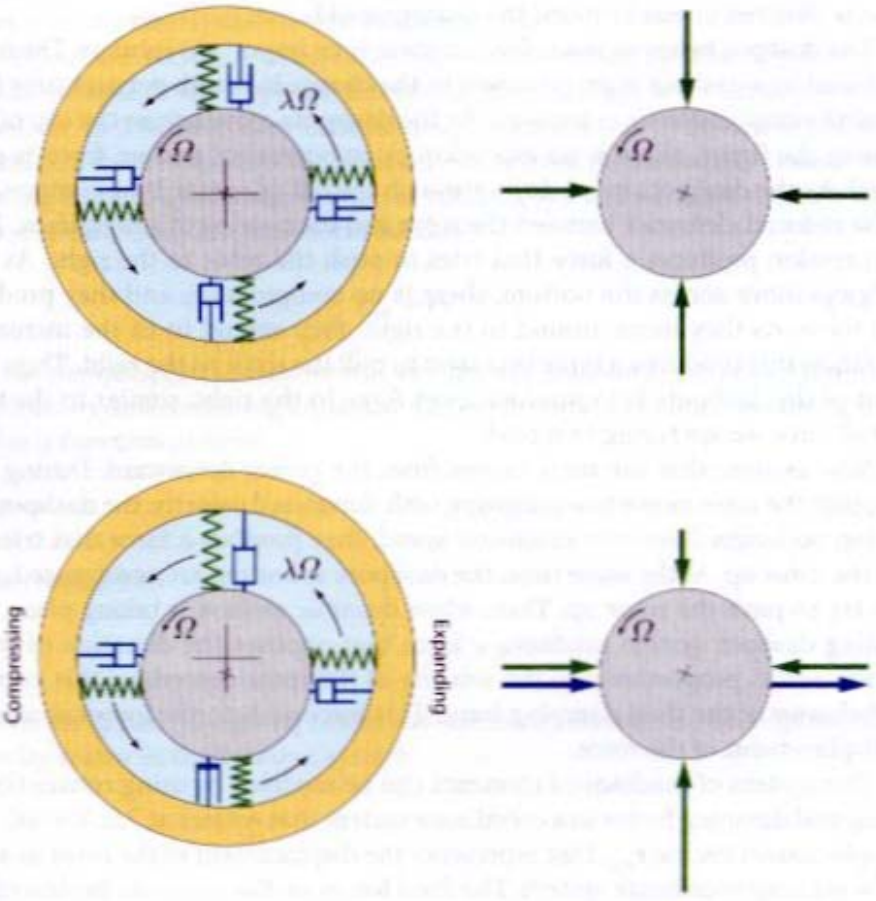



**Figure A3-2**. Spring and dashpot model of the fluid force system. The springs and the dashpots, which represent spring stiffness and damping, rotate at the average fluid angular velocity, $\lambda\Omega$, a fraction of rotor speed. The elements maintain a radial orientation to the shaft as they move around the clearance, which is greatly exaggerated for clarity. The length of the spring is equal to the available clearance at its location. When the rotor is centered in the clearance (top), the springs and dashpots do not change length as they move around the rotor; thus, the forces due to the surrounding fluid are balanced around the rotor. When the rotor is moved to a static position below the center (bottom), the elements change length as they move around the clearance, producing an unbalanced system of forces. See the text for more details.

force produced by a spring is proportional to its compression. As the springs reach the top, they are longer, producing a smaller compression force; when they reach the bottom, they are shorter, producing a larger force. Thus, the net spring force is directed upward toward the center, equilibrium position.

The dashpot behavior is similar, but there is an important subtlety. The force produced by a dashpot is proportional to the *change* in length per unit time (the speed of compression or extension). As the dashpots move across the top of the rotor in the figure, there is no extension or compression, and no force is produced. As the dashpots move down through the left of center, they compress to fit the reduced clearance between the rotor and the bearing or seal surface. This compression produces a force that tries to push the rotor to the right. As the dashpots move across the bottom, there is no compression, and they produce zero force. As they move around to the right, they extend to fit the increased clearance; this produces a force that tries to pull the rotor to the right. Thus, the effect of the dashpots is to produce a net force to the right, similar to the tangential force we are trying to model.

Now assume that the rotor moves from the center downward. During the time that the rotor centerline is moving with downward velocity, the dashpots at the top no longer have zero extension speed; they produce a force that tries to pull the rotor up. At the same time, the dashpots at bottom are compressed, and they try to push the rotor up. Thus, when dynamic motion is taking place, the rotating dashpot system produces a force that opposes the direction of rotor motion and is proportional to the velocity of the rotor centerline; this mimics the behavior of the fluid damping force. This force adds to the forces produced by displacement of the rotor.

This system of mechanical elements can be modeled by using conventional spring and damping forces in a coordinate system that rotates at $\lambda\Omega$. We will use a displacement vector, $\boldsymbol{r}_{rot}$, that represents the displacement of the rotor as seen in the *rotating* coordinate system. The fluid forces on the rotor can be described [1] in this rotating coordinate system as

$$\boldsymbol{F}_{rot} = -K\boldsymbol{r}_{rot} - D\dot{\boldsymbol{r}}_{rot} \tag{A3-2}$$

where $K$ is the overall fluid spring constant, and $D$ is the overall fluid damping coefficient. As the elements move around the available clearance at $\lambda\Omega$, this expression duplicates the behavior we have been discussing. This expression cannot be used directly in our rotor model because the model is derived in a stationary coordinate system; thus, we must transform the expression from the rotating coordinate system to the stationary system.

### Transformation to Stationary Coordinates

The rotating coordinate system, to which the spring and dashpot are attached, is rotating at angular velocity $\lambda\Omega$. The angle that it makes with the stationary coordinate system is $\lambda\Omega t$ (Figure A3-3).

A position vector, $\boldsymbol{r}$, when observed in the rotating system, appears to fall behind (rotate in the opposite direction). Even though we wish to transform Equation A3-3 to stationary coordinates, the easiest way to do this is to find an expression that transforms the position vector from the stationary system to the rotating system. To transform $\boldsymbol{r}$ to the rotating system, we must rotate it in the opposite direction. The transform becomes

$$\boldsymbol{r}_{rot} = \boldsymbol{r}e^{-j\lambda\Omega t} \tag{A3-3}$$

For the damping force term, we will also need to transform the velocity, which is obtained by differentiating Equation A3-3 with respect to time. Assuming that $\boldsymbol{r}$ is also a function of time,

$$\dot{\boldsymbol{r}}_{rot} = \dot{\boldsymbol{r}}e^{-j\lambda\Omega t} - j\lambda\Omega\boldsymbol{r}e^{-j\lambda\Omega t} \tag{A3-4}$$

which reduces to

$$\dot{\boldsymbol{r}}_{rot} = (\dot{\boldsymbol{r}} - j\lambda\Omega\boldsymbol{r})e^{-j\lambda\Omega t} \tag{A3-5}$$

The terms in parentheses represent the stationary coordinate description of the velocity vector in the rotating system.

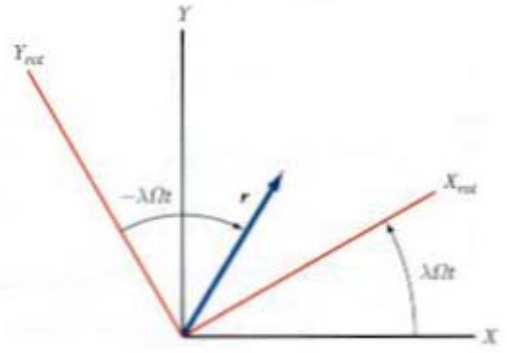



**Figure A3-3.** Fixed and rotating coordinate systems. The spring and dashpot are attached to the rotating coordinate system (red), which is rotating at angular velocity $\lambda\Omega$ relative to the stationary coordinate system (black). The angle that it makes with the stationary coordinate system is $\lambda\Omega t$. To an observer in the rotating system, the position vector, $\boldsymbol{r}$, will appear to rotate backwards.

We now substitute the stationary coordinate descriptions of the vectors into Equation A3-2 and obtain the expression for the force in stationary coordinates,

$$\boldsymbol{F} = -K\boldsymbol{r} + jD\lambda\Omega\boldsymbol{r} - D\dot{\boldsymbol{r}} \tag{A3-6}$$

The middle term is the tangential force expression used in our rotor model.

## Reference

1. Muszynska, A., and Bently, D. E., "Frequency-Swept Rotating Input Perturbation Techniques and Identification of the Fluid Force Models in Rotor/Bearing/Seal Systems and Fluid Handling Machines," *Journal of Sound and Vibration* 143, no. 1, (1990): pp. 103-124.

Appendix 4

# SAF Calculation

THE SYNCHRONOUS AMPLIFICATION FACTOR (SAF) is a measure of how much 1X vibration is amplified when the system passes through a resonance. Systems with low Quadrature Dynamic Stiffness have narrow resonances with high amplification of the unbalance response; those with high quadrature stiffness have relatively broad resonances with low amplification (Figure A4-1).

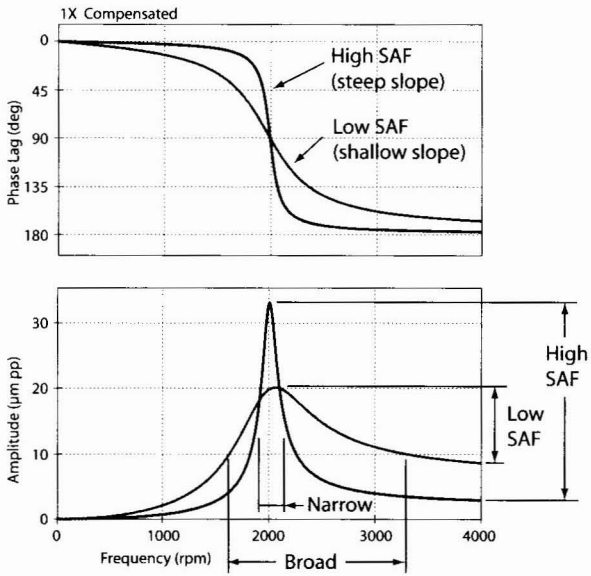



**Figure A4-1**. The Synchronous Amplification Factor (SAF). Systems with low Quadrature Dynamic Stiffness have a narrow resonance, high amplification of the unbalance response, and a rapid phase change (blue); those with high quadrature stiffness have a relatively broad resonance, low amplification, and relatively slow phase change (green).

There are four methods used to measure the SAF, three (Peak Ratio, Half-power Bandwidth, and Phase Slope) based on the Bode plot, and one (a variation of the Half-power Bandwidth method) based on the polar plot. The methods are summarized in Figure A4-2. Since the Half-power Bandwidth method was discussed in Chapter 7, in this chapter we will show how to calculate the SAF using the other three methods.

The SAF is based on the dynamic response of rotor systems to a rotating unbalance force. Because of this, *all methods for measuring the SAF use 1X, compensated, polar and Bode plots*. Slow roll compensation removes the part of the rotor response that is not due to unbalance. The following discussions assume that the 1X data has been compensated.

A word of caution is appropriate here. All methods for measuring the SAF are based on ideal, isotropic rotor behavior. Real rotor systems can (and usually do) have some degree of anisotropy in the support stiffness. This causes the measurement of SAF using *any* method to be sensitive to the mounting angle of the measurement probes. Be sure to see Chapter 13 before applying any of the SAF measurement methods to data from your machine.

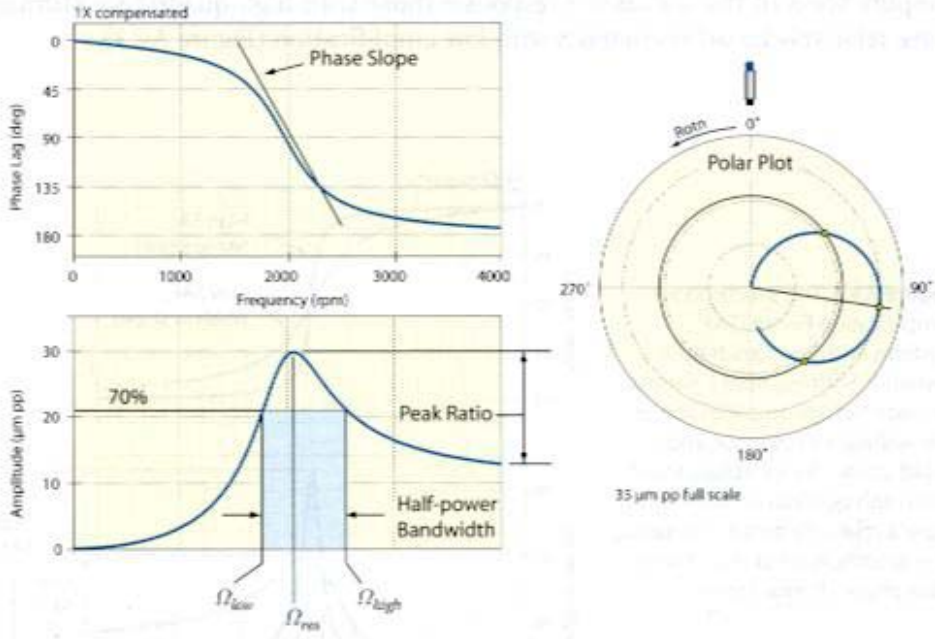



**Figure A4-2**. SAF graphical calculation methods. The Half-power Bandwidth and Peak Ratio methods utilize the Bode amplitude plot. The Phase Slope method uses the Bode phase plot, and the Polar Plot method uses the polar plot.

Because of the presence of anisotropic stiffness, all of these methods are approximate, and the precision of any SAF calculation is usually meaningful to only one or two significant figures.

### Peak Ratio Method

The *Peak Ratio* method is probably the most intuitive (Figure A4-3).

1. On the Bode plot, find the amplitude at the resonance peak and the amplitude at some speed well above resonance.
2. The SAF is the ratio of these two values:

$$SAF = \frac{A_{res}}{A_{fast}} \tag{A4-1}$$

For example, in the figure, $A_{res} = 30$ μm pp, and $A_{fast} = 13$ μm pp. Applying Equation A4-1,

$$SAF = \frac{30}{13} = 2.3$$

The rotor behavior shown in the figure is from the ideal, isotropic rotor model developed in Chapter 10. In practice, after the rotor passes through a resonance and experiences a decline in vibration amplitude, a higher mode will usually cause the vibration to increase again. For closely spaced modes, if the next mode amplitude rises before the previous mode amplitude falls to a limiting value, the SAF calculated using this method will not be accurate.

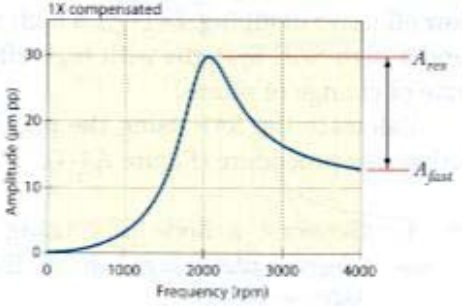



**Figure A4-3.** The Peak Ratio method. Identify the amplitude at the resonance peak, $A_{res}$, and the amplitude at a speed well above the resonance, $A_{fast}$. To calculate the SAF, divide $A_{res}$ by $A_{fast}$, Equation A4-1.

A variation of the SAF is sometimes used for estimating shaft relative alarm limits. Ideally, we want to shut down a machine so that it can safely pass through one or more resonances without rubbing. To estimate the vibration limit at running speed,

1. Determine the maximum permissible vibration amplitude at all resonances, using knowledge of the mode shapes of the machine and its internal clearances.

2. Calculate the ratio of the vibration amplitude at the most critical resonance peak to the amplitude *at running speed.*

3. Divide the maximum permissible resonance vibration by the ratio. This defines the maximum allowable vibration amplitude at running speed.

In most machines, unfiltered vibration is predominantly 1X, but may contain significant amounts of other frequency components. If alarms are based on unfiltered vibration amplitude, calculate the ratio from a plot of unfiltered vibration amplitude.

This method only provides one small view of the problem of setting alarms for machinery vibration, and it should be used cautiously. Vibration alarm levels should be set based on manufacturers' recommendations and personal experience with the particular type of machine and its application. This method provides additional perspective, but it is not the complete solution to the problem.

### Phase Slope Method

When a rotor system passes through a resonance, the phase lag increases. The slope of the phase change is usually steepest near the peak of the resonance. The *rate* of this change is related to the Quadrature Dynamic Stiffness, $jD(1-\lambda)\Omega$, of the system. Systems with low damping or high lambda will have low effective damping, $D(1-\lambda)$, a high rate of change of phase during resonance, and a high SAF. Systems with high effective damping will have a relatively low rate of change of phase.

Calculate the SAF using the maximum slope of the phase on a Bode plot using this procedure (Figure A4-4):

1. Generate a Bode plot using the Bently Nevada Convention (*degrees phase lag*), where all phase angles are between 0° and 360°.

2. Draw a line parallel to the steepest part of the phase plot in a convenient location.

3. Select two arbitrary points on the line, far enough apart to provide good resolution of the data.

4. For each point, identify the speed and phase, $\Omega_1$ $\alpha_1$ and $\Omega_2$ $\alpha_2$, where $\Omega_2 > \Omega_1$.

5. If $\alpha_2 < \alpha_1$, add 360° to $\alpha_2$. For example, if $\alpha_1 = 350°$ and $\alpha_2 = 10°$, then $\alpha_2 = 10° + 360° = 370°$.

6. Identify the speed at which the slope of the phase is steepest. For the simple rotor model, this is the speed where the high spot lags the heavy spot by 90°, $\Omega_{90}$.

7. Calculate the SAF:

$$SAF = \frac{\pi}{360^\circ}\Omega_{90}\left[\frac{\alpha_2 - \alpha_1}{\Omega_2 - \Omega_1}\right] \qquad \text{(A4-2)}$$

In the figure, $\Omega_1 = 1550$ rpm, $\alpha_1 = 0°$, $\Omega_2 = 2600$ rpm, $\alpha_2 = 180°$, $\Omega_{90} = 1900$ rpm, and

$$SAF = \frac{\pi}{360^\circ}(1900 \text{ rpm})\left[\frac{180^\circ - 0^\circ}{2600 \text{ rpm} - 1550 \text{ rpm}}\right] = 2.8$$

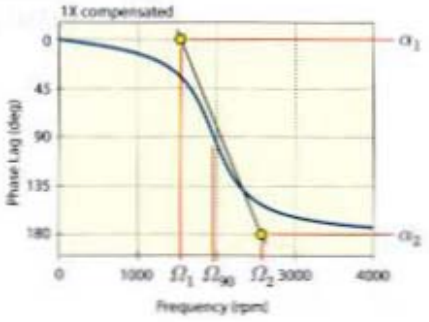



**Figure A4-4.** The Phase Slope method. Draw a line parallel to the steepest slope of the phase. Choose two arbitrary points and determine their phase and speed values. The steepest slope of the plot identifies the rotor speed where the high spot lags the heavy spot by 90° for that mode. Calculate the SAF using Equation A4-2.

### Polar Plot Method

The polar plot contains the same information as a Bode plot, so it is not surprising that it can be used to calculate the SAF. This method is actually a polar plot transformation of the Half-power Bandwidth method. For the ideal rotor model, on a polar plot, the half-power points are located 45° before and after the resonance peak. When rotor behavior deviates from ideal, the half-power points will not be at 45°. It is best to locate these points by finding the 70% amplitude levels. This will provide good agreement with the results obtained using a Bode plot.

The SAF is calculated using the following procedure (Figure A4-5):

1. Identify the resonance peak by locating the maximum amplitude on the plot. Draw a line from the origin through this peak. This identifies the resonance speed, $\Omega_{res}$.

2. Draw a circle on the plot centered on the origin with a radius equal to 70% of the peak amplitude. The half-power bandwidth speeds, $\Omega_{low}$ and $\Omega_{high}$, are found where the circle intersects the plot. These speeds may have to be interpolated between existing data points.

3. Calculate the SAF:

$$SAF = \frac{\Omega_{res}}{(\Omega_{high} - \Omega_{low})} \tag{A4-3}$$

In the figure, $\Omega_{res} = 2050$ rpm, $\Omega_{high} = 2550$ rpm, and $\Omega_{low} = 1700$ rpm, and

$$SAF = \frac{2050 \text{ rpm}}{(2550 \text{ rpm} - 1700 \text{ rpm})} = 2.4$$

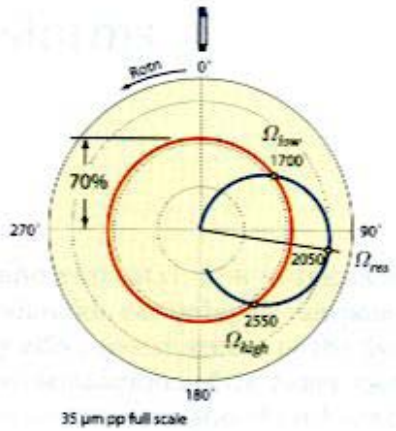



**Figure A4-5.** The Polar Plot method. The maximum amplitude identifies the resonance speed. Draw a circle (red) at 70% of the maximum amplitude to identify the speeds used to calculate the half-power bandwidth. Calculate the SAF using Equation A4-3.

Appendix 5

# Vector Transforms

In chapter 13, we showed that, for systems with anisotropic stiffness, measurement of rotor vibration can be highly viewpoint dependent. Anisotropic stiffness can adversely affect measurement of the Synchronous Amplification Factor (SAF) and make identification of the heavy spot ambiguous.

Two methods of vector transformation have been developed to improve our understanding of anisotropic rotor behavior: vector data can be transformed to what would be measured by a set of *virtual probes* mounted at any desired orientation, or vector data can be transformed to *forward* and *reverse* responses.

This chapter will present the mathematical algorithms used for these transforms. Both transforms operate on vibration vectors from a pair of orthogonal transducers. The transforms can be applied to a pair of vectors measured at one operating speed, or they can be applied to a full set of startup or shutdown vectors. The transforms are usually applied to displacement data, but they can be applied to velocity or acceleration data from orthogonal transducers.

## Virtual Probe Rotation

*Virtual probes* are imaginary orthogonal probes that we can mount at any angle we choose. Suppose a pair of probes are mounted on the machine at 0° and 90° R, and that, after taking vector data, we wish to know what vibration vectors would have been measured if the probes had been installed at some other orientation, perhaps 30° L and 60° R. We can mathematically transform the measured vector data to produce the vectors that would have been measured by these virtual probes.

As we discussed in Chapter 3, vibration vectors are complex numbers that can be expressed in polar or rectangular form. Most coordinate transforms

involve operations on scalar quantities, usually position coordinates. The special quality of this transform is that it operates directly on the complex vibration vectors.

We start with a pair of orthogonal ($XY$) transducers, which we will call the *physical* transducers. The transducers measure a pair of vibration vectors, $\boldsymbol{x}_{inst}$ and $\boldsymbol{y}_{inst}$, where

$$\begin{aligned} \boldsymbol{x}_{inst} &= A_{inst} \angle\alpha_{inst} \\ \boldsymbol{y}_{inst} &= B_{inst} \angle\beta_{inst} \end{aligned} \tag{A5-1}$$

The subscript *inst* refers to the instrumentation measurement convention (Appendix 1), $A_{inst}$ and $B_{inst}$ are the amplitudes, and $\alpha_{inst}$ and $\beta_{inst}$ are the measured phase lags. The phases are measured relative to the transducer axes: $\alpha_{inst}$ is measured relative to the $X$ transducer sensitive axis, and $\beta_{inst}$ is measured relative to the $Y$ transducer sensitive axis.

What vibration vectors would be measured by a pair of orthogonal transducers mounted at a different orientation (virtual probes)? To answer this question, we assume that the new, virtual probes are created by rotating the physical probes through an arbitrary angle, $\theta$, as shown in Figure A5-1.

In the figure, the vibration vectors, $\boldsymbol{x}$ and $\boldsymbol{y}$, are shown in a real $XY$ plane which is formed by the combination of the two complex transducer response planes. Each vector is complex, with its real part aligned with its transducer's sensitive axis.

First, the measured vectors are converted to the mathematical phase measurement convention (Appendix 1) and then to rectangular form,

$$\begin{aligned} \boldsymbol{x} &= x_d + jx_q \\ \boldsymbol{y} &= y_d + jy_q \end{aligned} \tag{A5-2}$$

where $x_d$ and $x_q$ are the *in-phase* (real part) and *quadrature* (imaginary part) components of $\boldsymbol{x}$, and $y_d$ and $y_q$ are the corresponding components of $\boldsymbol{y}$.

Then we apply a standard coordinate transform,

$$\begin{pmatrix} \boldsymbol{x}_R \\ \boldsymbol{y}_R \end{pmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{pmatrix} \boldsymbol{x} \\ \boldsymbol{y} \end{pmatrix} \tag{A5-3}$$

where $\boldsymbol{x}_R$ and $\boldsymbol{y}_R$ are the vibration vectors that would be measured by the virtual probes. Substituting Equations A5-2 and expanding,

$$\begin{aligned} \boldsymbol{x}_R &= (x_d \cos\theta + y_d \sin\theta) + j(x_q \cos\theta + y_q \sin\theta) \\ \boldsymbol{y}_R &= (y_d \cos\theta - x_d \sin\theta) + j(y_q \cos\theta - x_q \sin\theta) \end{aligned} \tag{A5-4}$$

The transformed vectors can be converted back to polar form and to the instrumentation convention for comparison to the original vectors.

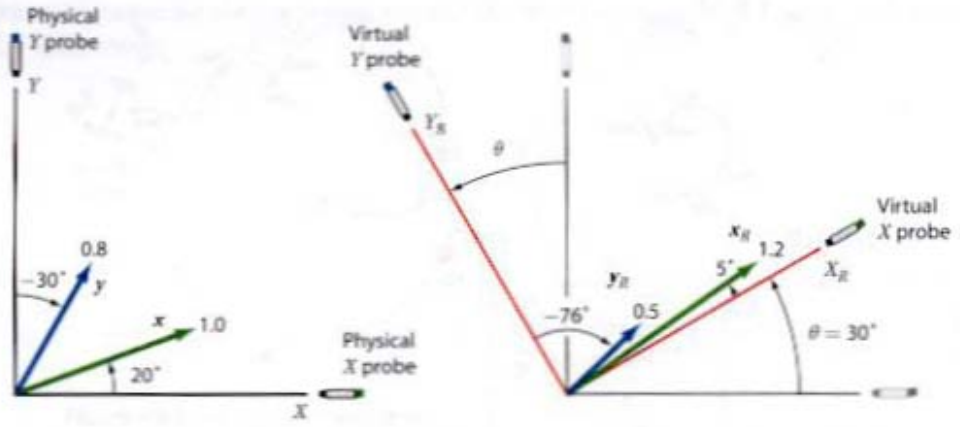



**Figure A5-1**. Physical and virtual probe coordinate axes. A set of physical *XY* probes are shown at left with their respective vibration vectors, $\boldsymbol{x}$ and $\boldsymbol{y}$. The vectors are plotted using the mathematical convention, and their amplitudes and phases are shown. A set of virtual probes, $X_R$ and $Y_R$ are created by mathematically rotating the physical probes through an angle, $\theta$. The new vibration vectors (right), $\boldsymbol{x}_R$ and $\boldsymbol{y}_R$, are what would have been measured by the virtual probes and will generally have different amplitudes and phases.

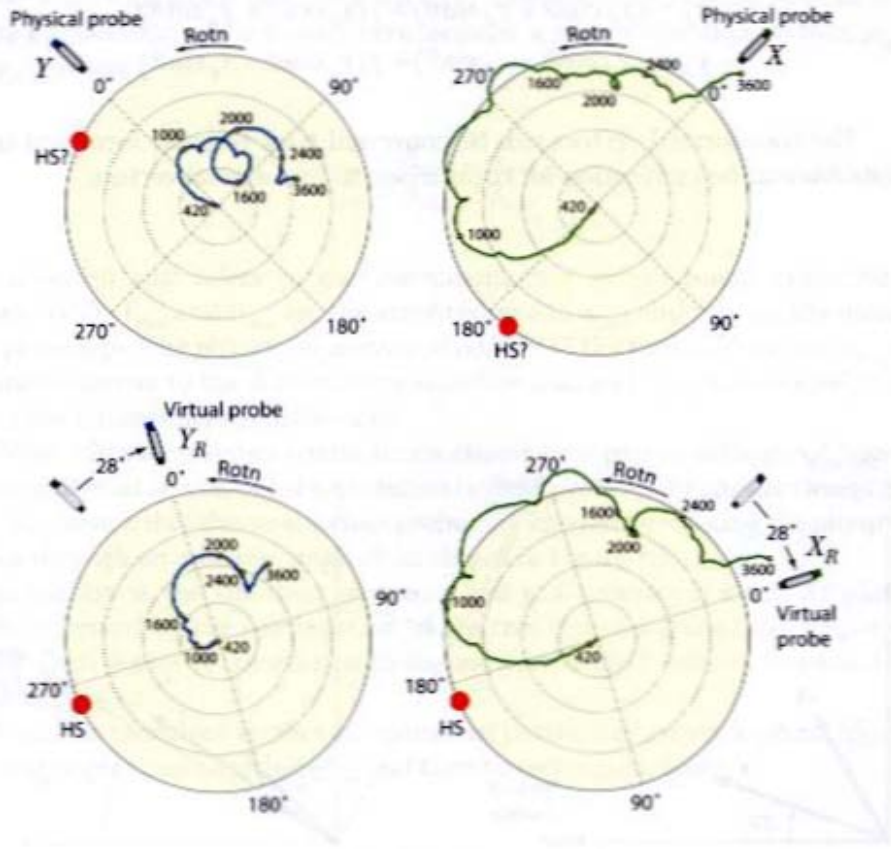



**Figure A5-2** Virtual probe data from a gas turbine generator set. This is a fluid-film bearing machine with probes mounted at ±45° from the vertical. On the physical probe polar plots (top), the indicated locations of the heavy spots disagree. On the virtual probe polar plots (bottom), where the probes have been rotated 28° clockwise, the heavy spot locations show good agreement.

The transformed vectors are plotted the same way as the vibration vectors from the physical probes: the phase lag starts at zero degrees *relative to the virtual probe* and increases in a direction opposite to rotation (Figure A5-2). In the figure, virtual probes have been constructed at 28° clockwise relative to the physical probes, and the data is plotted on a separate set of polar plots.

Polar plots generated for virtual probes are interpreted in the same way as for physical probes. For example, the indicated heavy spot in the figure is located about 260° relative to the *virtual*, $Y_R$, probe location. You must add 28° to the phase lag to arrive at its location relative to the physical probe.

### The Forward and Reverse Transform and Full Spectrum

In Appendix 2, we showed how a filtered orbit can be reconstructed from a pair of vibration vectors. The same orbit can be constructed from a pair of forward and reverse vectors that rotate in opposite directions at the filter frequency, $\omega$ (Figure A5-3). The forward vector rotates in the same direction as rotor rotation. Research [1] has shown that, at speeds well below the resonance, the forward response always points toward the heavy spot, making it a valuable tool for balancing.

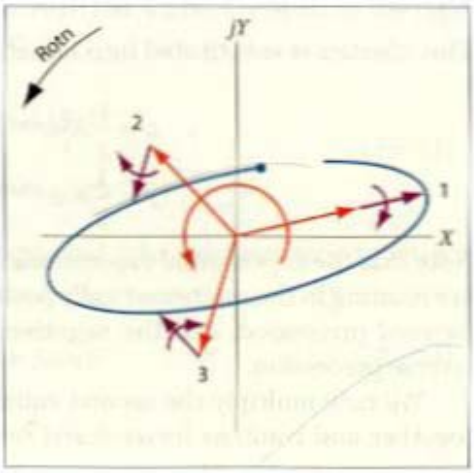



**Figure A5-3.** Forward and reverse vectors and the filtered orbit. Forward (red) and reverse (violet) rotating vectors are shown at three moments in the complex plane. At any instant in time, the vector sum locates the coordinates of the rotor centerline in the filtered orbit.

The forward/reverse transform starts with a pair of vibration vectors, $\boldsymbol{x}_{inst}$ and $\boldsymbol{y}_{inst}$, from orthogonal ($XY$) probes. The vibration vectors are first converted to the mathematical convention (positive phase lead and zero-to-peak amplitude) as discussed in Appendix 1. These vectors can be expressed in rotating form as

$$\begin{aligned} \boldsymbol{x} &= Ae^{j(\omega t+\alpha)} \\ \boldsymbol{y} &= Be^{j(\omega t+\beta)} \end{aligned} \tag{A5-5}$$

where $A$, $B$, $\alpha$, and $\beta$ are the amplitudes and phases of the vibration vectors in mathematical convention. $\alpha$ is measured relative to the $X$ transducer axis, and $\beta$ is measured relative to the $Y$ transducer axis. Thus, the physical coordinates of the rotor centerline on the filtered orbit are $(x, y)$, where

$$\begin{aligned} x &= A\cos(\omega t+\alpha) \\ y &= B\cos(\omega t+\beta) \end{aligned} \tag{A5-6}$$

The transform to forward and reverse uses the identity,

$$\cos\theta = \frac{e^{j\theta}+e^{-j\theta}}{2} \tag{A5-7}$$

This identity is substituted into Equation A5-6 to obtain

$$\begin{aligned} x &= \frac{1}{2}A\left[e^{j(\omega t+\alpha)}+e^{-j(\omega t+\alpha)}\right] \\ y &= \frac{1}{2}B\left[e^{j(\omega t+\beta)}+e^{-j(\omega t+\beta)}\right] \end{aligned} \tag{A5-8}$$

Note that the exponential expressions contain terms that represent vectors that are rotating in the mathematically positive direction, $e^{j\omega t}$, which is equivalent to *forward* precession, and the negative direction, $e^{-j\omega t}$, which is equivalent to *reverse* precession.

We now multiply the second equation in A5-8 by $j$, add the two equations together, and combine forward and reverse parts to obtain

$$x + jy = \frac{1}{2}\left[Ae^{j(\omega t+\alpha)} + jBe^{j(\omega t+\beta)}\right] + \frac{1}{2}\left[Ae^{-j(\omega t+\alpha)} + jBe^{-j(\omega t+\beta)}\right] \quad \text{(A5-9)}$$

The sum of these four complex, rotating vectors represents the instantaneous position of the rotor centerline in the filtered orbit. This orbit exists in what is now a complex plane ($X + jY$) with the real axis aligned with the $X$ transducer. The two vectors in the left bracket rotate in the forward direction ($+\omega$); the two in the right bracket are reverse ($-\omega$). Setting $t = 0$ gives us the filtered position at the Keyphasor event in terms of the measured vibration vectors,

$$(x + jy)_{t=0} = \frac{1}{2}(Ae^{j\alpha} + jBe^{j\beta}) + \frac{1}{2}(Ae^{-j\alpha} + jBe^{-j\beta}) \quad \text{(A5-10)}$$

Equations A5-9 and A5-10 can be used in a computer program (such as MATLAB) that supports complex numbers in this form.

However, we wish to find expressions for the amplitude and phase of the forward and reverse vectors. Equation A5-9 can be described by the sum of two forward and reverse rotating vectors,

$$x + jy = A_F e^{j(\omega t+\phi_F)} + A_R e^{j(-\omega t-\phi_R)} \quad \text{(A5-11)}$$

where $A_F$ and $A_R$ are the amplitudes of the forward and reverse vectors, and $\phi_F$ and $\phi_R$ are the phases, both measured relative to the $X$ axis. Comparing the right sides of Equations A5-9 and A5-11 at $t = 0$,

$$\begin{aligned} A_F e^{j\phi_F} &= \frac{1}{2}(Ae^{j\alpha} + jBe^{j\beta}) \\ A_R e^{-j\phi_R} &= \frac{1}{2}(Ae^{-j\alpha} + jBe^{-j\beta}) \end{aligned} \quad \text{(A5-12)}$$

The exponential functions can be expanded into trigonometric functions using Euler's identity,

$$e^{j\theta} = \cos\theta + j\sin\theta \quad \text{(A5-13)}$$

Applying Euler's identity to the right sides of Equations A5-12, and, after some algebra and a trigonometric identity or two, we obtain

$$A_F = \frac{1}{2}\left[A^2 + B^2 + 2AB\sin(\alpha - \beta)\right]^{0.5}$$
$$A_R = \frac{1}{2}\left[A^2 + B^2 - 2AB\sin(\alpha - \beta)\right]^{0.5} \qquad \text{(A5-14)}$$

and

$$\phi_F = \arctan\left(\frac{A\sin\alpha + B\cos\beta}{A\cos\alpha - B\sin\beta}\right)$$
$$\phi_R = \arctan\left(\frac{A\sin\alpha - B\cos\beta}{A\cos\alpha + B\sin\beta}\right) \qquad \text{(A5-15)}$$

The phase angles should be calculated using the *arctangent2* function to yield angles between ±180° (which can be reduced mod 360 if desired), and the data should be unwrapped to prevent jump discontinuities.

It is important to remember that these expressions are based on a mathematical convention where *both* of the phase angles are measured relative to the $X$ (real) axis, which is aligned with the $X$ transducer sensitive axis, and $\phi_F$ is measured in the positive (counterclockwise) direction. However, as a result of the phase angle definition in equation A5-8 for the reverse vector, positive $\phi_R$ is measured in the *clockwise* direction, the reverse of the usual convention.

Before plotting, the forward and reverse vectors should be converted to the instrumentation convention (the negative of the mathematical phase) to be consistent with other data plots.

When plotted on a polar plot (Figure A5-4), as the phase lag of the forward response increases (the mathematical phase decreases), it will move in a direction opposite to rotation (the same as standard vibration vectors); on a Bode plot, it will move downward.

As the phase lag of the *reverse* response increases, it will move in the *same* direction as rotation on the polar plot, and, if plotted on a Bode plot together with the forward response, it will move upward.

A full spectrum (Chapter 8) can be calculated using the same algorithm. Full spectrum uses the time delay between events on $X$ and $Y$ transducers to deduce the $X$ to $Y$ or $Y$ to $X$ precession direction information. For this reason, full spectrum computation requires that the unfiltered waveform data from an orthogonal ($XY$) pair of transducers be sampled simultaneously and with the same sam-

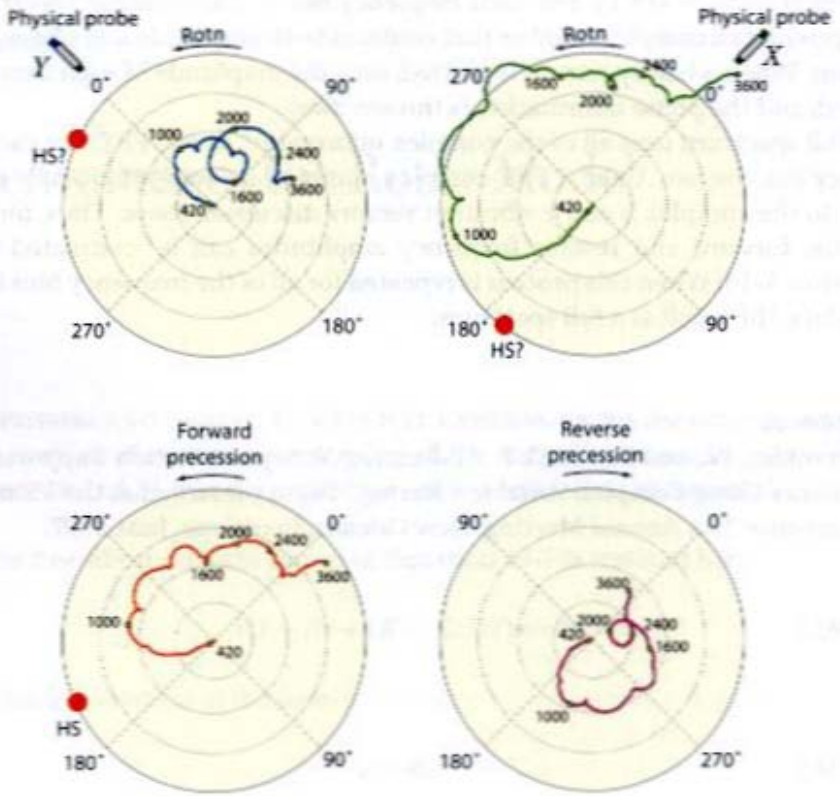



**Figure A5-4.** Forward and reverse polar plots. The polar plots (bottom) show the forward (left) and reverse (right) vectors transformed from the vibration vectors in the standard *XY* polar plots above. The phase is plotted in degrees lag relative to the direction of precession. The low speed phase of the forward plot points toward the heavy spot and agrees with the location found using virtual probe rotation in Figure A5-2. For any speed, the sum of the forward and reverse vectors will reconstruct the original filtered orbit, including its orientation. The low speed phase of the reverse polar plot does not, typically, point in the direction of the heavy spot.

pling frequency. The direction of shaft rotation is needed to determine whether the calculated precession direction is forward or reverse.

In a half spectrum, the $X$ and $Y$ waveforms are each subjected to a fast Fourier transform (FFT). For each frequency bin in the resulting spectra, the FFT produces a complex number that contains both amplitude and phase information. When a half spectrum is plotted, only the amplitude of each frequency is used, and the phase information is thrown away.

Full spectrum uses all of the complex information in the FFT. For each frequency bin, the two $X$ and $Y$ FFT complex numbers are computationally equivalent to the complex $\boldsymbol{x}$ and $\boldsymbol{y}$ vibration vectors discussed above. Thus, for each bin, the forward and reverse frequency amplitudes can be calculated using Equation A-14. When this process is repeated for all of the frequency bins in the FFT data, the result is a full spectrum.

## References

1. Franklin, W., and Bently, D. E., "Balancing Nonsymmetrically Supported Rotors Using Complex Variable Filtering." Paper presented at the Vibration Institute 21st Annual Meeting, New Orleans, Louisiana, June 1997.

Appendix 6

# Eigenvalues of the Rotor Model

It is possible to derive an explicit expression for the eigenvalues (also called *roots*, or in control theory, *poles*) of the simple rotor model presented in Chapters 10 and 14. The derivation we will use will follow Muszynska's [1].

The free vibration rotor model of Equation 14-7 is restated here:

$$M\ddot{\boldsymbol{r}} + D\dot{\boldsymbol{r}} + (K - jD\lambda\Omega)\boldsymbol{r} = 0 \tag{A6-1}$$

We assume a solution of the form

$$\boldsymbol{r} = \boldsymbol{R}e^{st} \tag{A6-2}$$

Solution leads to the *characteristic equation*,

$$Ms^2 + Ds + K - jD\lambda\Omega = 0 \tag{A6-3}$$

The eigenvalues, $s$, are the roots of this equation; because the equation is quadratic, there will be two. We find them using the standard solution to a quadratic equation,

$$s_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{A6-4}$$

The important feature here is that the $c$ term is a complex number. Applying Equation A6-4, we obtain

$$s_{1,2} = \frac{-D \pm \sqrt{D^2 - 4M(K - jD\lambda\Omega)}}{2M} \tag{A6-5}$$

which leads to

$$s_{1,2} = -\frac{D}{2M} \pm \sqrt{\left(\frac{D}{2M}\right)^2 - \frac{K}{M} + j\frac{D\lambda\Omega}{M}} \tag{A6-6}$$

Note that, if either $\Omega$ or $\lambda$ is zero, then the imaginary term under the radical disappears, and the eigenvalues are the same as those of a simple harmonic oscillator,

$$s_{1,2} = -\frac{D}{2M} \pm \sqrt{\left(\frac{D}{2M}\right)^2 - \frac{K}{M}} \tag{A6-7}$$

This situation would apply to a rotor that has negligible fluid interaction, such as an electric motor with rolling element bearings. It would also apply to a stopped rotor supported by externally pressurized (hydrostatic) bearings. For now, we will examine Equation A6-7 in detail; we will return to Equation A6-6 later.

There are three possible cases for the term under the radical: it is either zero, greater than zero, or less than zero.

If the term is equal to zero, then

$$\left(\frac{D}{2M}\right)^2 - \frac{K}{M} = 0 \tag{A6-8}$$

Solving for the damping, $D$, leads to the definition of *critical damping*, $D_{cr}$:

$$D = D_{cr} = 2\sqrt{MK} \tag{A6-9}$$

If we substitute this expression into Equation A6-7, Equation A6-2 becomes

$$\boldsymbol{r} = \boldsymbol{R}_1 e^{s_1 t} + \boldsymbol{R}_2 e^{s_2 t} \tag{A6-10}$$

where $\boldsymbol{R}_1$ and $\boldsymbol{R}_2$ depend on the initial conditions, and

$$s_1 = s_2 = \gamma_1 = \gamma_2 = -\sqrt{\frac{K}{M}} \tag{A6-11}$$

The eigenvalues are real, consisting only of two equal values of the growth/decay rate, $\gamma$; thus, for critical damping there is no natural frequency term, $\omega_d$, and no precession. For zero initial centerline velocity, the disturbed rotor system will move back to the equilibrium position *in a straight line* without overshooting or oscillation. On the root locus plot, the two eigenvalues will plot at the same point on the horizontal axis in the left half plane at the location given by Equation A6-11.

The expression for critical damping leads to the definition of the *damping factor*, $\zeta$, which is the ratio of the damping to the critical damping:

$$\zeta = \frac{D}{D_{cr}} = \frac{D}{2\sqrt{KM}} \tag{A6-12}$$

Thus, when $\zeta = 1$, the system is critically damped.

Returning to Equation A6-7, if the term under the radical is greater than zero, then both eigenvalues are real and *unequal.* Substituting the eigenvalues into Equation A6-2, we obtain the free response of the system,

$$\boldsymbol{r} = \boldsymbol{R}_1 e^{s_1 t} + \boldsymbol{R}_2 e^{s_2 t} \tag{A6-13}$$

where the values of $\boldsymbol{R}_1$ and $\boldsymbol{R}_2$ depend on the initial conditions, and

$$\begin{aligned} s_1 = \gamma_1 &= -\frac{D}{2M} + \sqrt{\left(\frac{D}{2M}\right)^2 - \frac{K}{M}} \\ s_2 = \gamma_2 &= -\frac{D}{2M} - \sqrt{\left(\frac{D}{2M}\right)^2 - \frac{K}{M}} \end{aligned} \tag{A6-14}$$

Again, the eigenvalues are real; when disturbed and released with zero initial velocity, the system will also return slowly back to the equilibrium position in a

straight line, without overshoot. For this case, $D > D_{cr}$, $\zeta > 1$, and the system is *supercritically damped*, or *overdamped*; the system will return to the equilibrium position more slowly than for the critically damped case. The two eigenvalues will be located at two points on the horizontal axis of the root locus plot; both points will be in the stable, left half plane.

In the third case, the term under the radical in Equation A6-7 is negative. For this case, $D < D_{cr}$, $\zeta < 1$, and the system is *subcritically damped*, or *underdamped*. Factoring out the −1, we obtain

$$s_{1,2} = -\frac{D}{2M} \pm j\sqrt{\frac{K}{M} - \left(\frac{D}{2M}\right)^2} \tag{A6-15}$$

The eigenvalues are now complex, of the form $\gamma \pm j\omega_d$, and the system will oscillate. The rotor response is given by

$$\boldsymbol{r} = \boldsymbol{R}_1 e^{\gamma_1 t} e^{j\omega_d t} + \boldsymbol{R}_2 e^{\gamma t} e^{-j\omega t} \tag{A6-16}$$

where

$$\gamma_1 = \gamma_2 = -\frac{D}{2M}$$

$$\omega_d = \sqrt{\frac{K}{M} - \left(\frac{D}{2M}\right)^2} \tag{A6-17}$$

The eigenvalues are complex conjugates and are located above and below each other on the root locus plot, with equal positive (forward precession) and negative (reverse precession) frequencies. The growth/decay rates are equal, and the two, counterrotating vectors produce a line orbit that decreases in amplitude with time. The decay in amplitude determined by the value of $\gamma$.

The natural frequency, $\omega_d$, is the classical definition of the *damped natural frequency*, which is lower than the *undamped natural frequency*, $\omega_n$,

$$\omega_n = \sqrt{\frac{K}{M}} \tag{A6-18}$$

We have been exploring the situation where either $\Omega$ or $\lambda$ is zero. When $\Omega$ and $\lambda$ are *both nonzero*, the rotor system will have eigenvalues given by Equation A6-6. The term under the radical is complex, and we must convert this equation to a form where $j$ is outside the radical. The conversion can be found in a good mathematics handbook [2]:

$$\sqrt{a+jb} = \sqrt{\frac{\sqrt{a^2+b^2}+a}{2}} \pm j\sqrt{\frac{\sqrt{a^2+b^2}-a}{2}} \tag{A6-19}$$

When this expression is applied to Equation A6-6, after some algebra, we obtain expressions for the two eigenvalues of the form

$$s_{1,2} = \gamma_{1,2} + j\omega_{r1,2} \tag{A6-20}$$

where

$$\gamma_1 = \left[ -\frac{D}{2M} + \frac{1}{\sqrt{2}}\sqrt{-\omega_d^2 + \sqrt{\omega_d^4 + \left(\frac{D\lambda\Omega}{M}\right)^2}} \right]$$

$$\gamma_2 = \left[ -\frac{D}{2M} - \frac{1}{\sqrt{2}}\sqrt{-\omega_d^2 + \sqrt{\omega_d^4 + \left(\frac{D\lambda\Omega}{M}\right)^2}} \right]$$

$$\omega_{r1} = j\left[ \frac{1}{\sqrt{2}}\sqrt{\omega_d^2 + \sqrt{\omega_d^4 + \left(\frac{D\lambda\Omega}{M}\right)^2}} \right]$$

$$\omega_{r2} = -j\left[ \frac{1}{\sqrt{2}}\sqrt{\omega_d^2 + \sqrt{\omega_d^4 + \left(\frac{D\lambda\Omega}{M}\right)^2}} \right] \tag{A6-21}$$

and $\omega_d$ is the damped natural frequency of Equation A6-17.

Here, we are using $\omega_r$ to indicate that this is the damped natural frequency of precession of the *rotating* rotor system with significant fluid interaction. (This subscript will only be used here to emphasize the distinction between the rotating and nonrotating natural frequencies. Elsewhere in the book, $\omega_d$ will be

used to indicate a damped natural frequency for either a rotating or nonrotating system.)

All of the terms contain the tangential stiffness, $D\lambda\Omega$; thus, all are functions of both lambda and rotor speed. There are two unequal values of $\gamma$; thus, the eigenvalues are not conjugate pairs. The damped natural frequency of precession, $\omega_r$, is a much more complicated expression than that of Equation A6-17, and the tangential stiffness term makes the natural frequency of the rotor system increase with increasing rotor speed or $\lambda$.

The only way the term under the radical in Equation A6-6 can be zero is if the real and imaginary parts of the term are both zero simultaneously. The straight line behavior of critically and supercritically damped systems derived from the fact that the term under the radical was either zero, or real and positive. For nonzero values of $\lambda$ and $\Omega$, this is no longer possible, which means that *a rotor system with fluid interaction and nonzero rotor speed cannot move in a straight line in free vibration.* Once the rotor starts turning, the tangential stiffness term will guarantee that, if disturbed from equilibrium, the rotor will move in some sort of spiral orbit, at a natural frequency, $\omega_r$.

For a rotor system with nonzero $\lambda$ and damping equal to or greater than $D_{cr}$, the eigenvalues will plot on the horizontal axis only when $\Omega = 0$. As soon as the rotor starts turning, the two eigenvalues will move off the horizontal axis, and the frequency of precession given by $\omega_r$ will be nonzero. Figure A6-1 summarizes the eigenvalue positions for each of the cases we have discussed.

### The Threshold of Instability

The Threshold of Instability speed, $\Omega_{th}$, is the speed at which the growth/decay rate, $\gamma$, is zero, and it can be found by setting the real parts of $\gamma_{1,2}$ equal to zero:

$$-\frac{D}{2M} \pm \frac{1}{\sqrt{2}} \sqrt{-\omega_d^2 + \sqrt{\omega_d^4 + \left(\frac{D\lambda\Omega_{th}}{M}\right)^2}} = 0 \qquad \text{(A6-22)}$$

which leads to

$$\Omega_{th} = \frac{1}{\lambda}\sqrt{\frac{K}{M}} \qquad \text{(A6-23)}$$

This is identical to Equation 14-5, which was derived from the Dynamic Stiffness.

## References

1. Muszynska, A, "One Lateral Mode Isotropic Rotor Response to Nonsynchronous Excitation," *Proceedings of the Course on Rotor Dynamics and Vibration in Turbomachinery*, von Karman Institute for Fluid Dynamics, Belgium, September 1992, pp. 21-25.
2. Tuma, Jan J., and Walsh, Ronald A. (contributor), *Engineering Mathematics Handbook*, 4th Ed. (New York: McGraw-Hill Book Company, 1997).

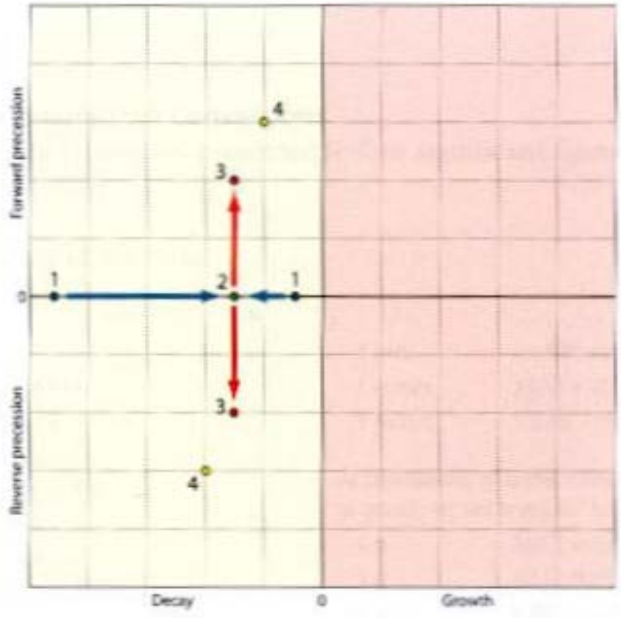



**Figure A6-1**. Eigenvalue locations on the root locus plot for different damping factors for a one-degree-of-freedom system. Points 1, 2, and 3, are for systems where either $\lambda$ or $\Omega = 0$. Eigenvalues of a supercritically damped system ($\zeta > 1$) will plot as two separate points on the horizontal axis (points labeled 1). Eigenvalues of a critically damped system ($\zeta = 1$) will plot as one point (2). Both of these systems have real eigenvalues. Subcritically damped systems ($\zeta < 1$) have complex eigenvalues that are conjugates and plot off the horizontal axis (3). Systems where both $\lambda$ and $\Omega$ are nonzero have complex eigenvalues that are not conjugate (4).

Appendix 7

# Units of Measurement

### Metric/US Customary Unit Conversions

All conversion factors are presented in four significant figures except when the conversion is exact.

All conversions are of the form

A = B

**Displacement**

| | |
|---|---|
| 1 in | 25.4 mm |
| 1 in | $25.4 \times 10^{-3}$ m |
| 1 m | 39.37 in |
| 1 mil | $1 \times 10^{-3}$ in |
| 1 mil | $25.4 \times 10^{-3}$ mm |
| 1 mil | 25.4 μm |
| 1 mm | $39.37 \times 10^{-3}$ in |
| 1 mm | 39.37 mil |
| 1 mm | $1 \times 10^{3}$ μm |
| 1 μm | $39.37 \times 10^{-6}$ in |
| 1 μm | $39.37 \times 10^{-3}$ mil |
| 1 μm | $1 \times 10^{-3}$ mm |

**Velocity**

| | |
|---|---|
| 1 in/s | 25.4 mm/s |
| 1 in/s | $25.4 \times 10^{-3}$ m/s |
| 1 m/s | 39.37 in/s |
| 1 m/s | $1 \times 10^{3}$ mm/s |
| 1 mm/s | $39.37 \times 10^{-3}$ in/s |
| 1 mm/s | $1 \times 10^{-3}$ m/s |

**Acceleration**; "g" is the standard acceleration of gravity at sea level, 45° latitude.

| | |
|---|---|
| 1 g | 386.1 in/s$^2$ |
| 1 g | 32.17 ft/s$^2$ |
| 1 g | 9.807 m/s$^2$ |
| 1 in/s$^2$ | $2.590 \times 10^{-3}$ g |
| 1 ft/s$^2$ | $31.08 \times 10^{-3}$ g |
| 1 m/s$^2$ | $102.0 \times 10^{-3}$ g |

**Displacement, velocity, and acceleration**: $d$ is the amplitude of displacement vibration, $v$ is velocity, and a is acceleration; $\omega$ is the frequency of vibration in rad/s, and $f$ is the frequency of vibration in hertz. These equations are applicable to sinusoidal motion (single frequency) only.

| | |
|---|---|
| $d$ | $v/\omega$ |
| $d$ | $v/2\pi f$ |
| $d$ | $a/\omega^2$ |
| $d$ | $a/(2\pi f)^2$ |
| $v$ | $\omega d$ |
| $v$ | $2\pi f d$ |
| $v$ | $a/\omega$ |
| $v$ | $a/2\pi f$ |
| $a$ | $\omega^2 d$ |
| $a$ | $(2\pi f)^2 d$ |
| $a$ | $\omega v$ |
| $a$ | $2\pi f v$ |

**Amplitude, $A$, peak-to-peak (pp), peak (pk), and rms.** The formulas for rms are for a sine wave (single frequency) only.

| | |
|---|---|
| $A_{pk}$ | $1/2\ A_{pp}$ |
| $A_{pp}$ | $2\ A_{pk}$ |
| $A_{rms}$ | $0.7071\ A_{pk}$ |
| $A_{pk}$ | $1.414\ A_{rms}$ |
| $A_{pp}$ | $2.828\ A_{rms}$ |

**Phase, $\Phi$, of displacement, velocity, acceleration.**

| | |
|---|---|
| $\Phi_{disp}$ | $\Phi_{vel} - 90°$ |
| $\Phi_{disp}$ | $\Phi_{accel} \pm 180°$ |
| $\Phi_{vel}$ | $\Phi_{disp} + 90°$ |
| $\Phi_{vel}$ | $\Phi_{accel} - 90°$ |
| $\Phi_{accel}$ | $\Phi_{disp} \pm 180°$ |
| $\Phi_{accel}$ | $\Phi_{vel} + 90°$ |

**Phase, $\Phi$, instrumentation and math conventions.** See Appendix 1.

| | |
|---|---|
| $\Phi_{math}$ | $-\Phi_{inst}$ |

**Force, weight, and mass.** See the last section for a discussion of force, weight, and mass in the US customary system. In the following, "g" is the symbol for gram.

| | |
|---|---|
| 1 g | $1 \times 10^{-3}$ kg |
| 1 g | $2.205 \times 10^{-3}$ lb |
| 1 g | $5.711 \times 10^{-6}$ lb · s$^2$/in |
| 1 g | $68.53 \times 10^{-6}$ lb · s$^2$/ft |
| 1 g | $35.27 \times 10^{-3}$ oz |
| 1 kg | $1 \times 10^{3}$ g |
| 1 kg | 2.205 lb |
| 1 kg | $5.711 \times 10^{-3}$ lb · s$^2$/in |
| 1 kg | $68.53 \times 10^{-3}$ lb · s$^2$/ft |
| 1 kg | 35.27 oz |
| 1 lb | 453.6 g |
| 1 lb | $453.6 \times 10^{-3}$ kg |
| 1 lb | $2.590 \times 10^{-3}$ lb · s$^2$/in |
| 1 lb | $31.08 \times 10^{-3}$ lb · s$^2$/ft |
| 1 lb | 4.448 N |
| 1 lb | 16 oz |
| 1 lb · s$^2$/in | $175.1 \times 10^{3}$ g |
| 1 lb · s$^2$/in | 175.1 kg |
| 1 lb · s$^2$/in | 386.1 lb |
| 1 lb · s$^2$/in | $6.178 \times 10^{3}$ oz |
| 1 lb · s$^2$/ft | $14.59 \times 10^{3}$ g |
| 1 lb · s$^2$/ft | 14.59 kg |
| 1 lb · s$^2$/ft | 32.17 lb |
| 1 lb · s$^2$/ft | 514.7 oz |
| 1 N | $224.8 \times 10^{-3}$ lb |
| 1 N | 3.597 oz |
| 1 oz | 28.35 g |
| 1 oz | $28.35 \times 10^{-3}$ kg |
| 1 oz | $62.50 \times 10^{-3}$ lb |
| 1 oz | $161.9 \times 10^{-6}$ lb · s$^2$/in |
| 1 oz | $1.943 \times 10^{-3}$ lb · s$^2$/ft |
| 1 oz | $278.0 \times 10^{-3}$ N |

**Stiffness**

| | |
|---|---|
| 1 lb/in | 175.1 N/m |
| 1 N/m | $5.710 \times 10^{-3}$ lb/in |

**Damping**

| | |
|---|---|
| 1 lb · s/in | 175.1 N · s/m |
| 1 N · s/m | 1 kg/s |
| 1 N · s/m | $5.710 \times 10^{-3}$ lb · s/in |

**Unbalance**

| | |
|---|---|
| 1 g · mm | $1.389 \times 10^{-3}$ in · oz |
| 1 in · oz | 720.1 g · mm |

**Speed and frequency**

| | |
|---|---|
| 1 cpm | $16.67 \times 10^{-3}$ Hz |
| 1 Hz | 60 cpm |
| 50 Hz | 3000 cpm |
| 60 Hz | 3600 cpm |
| 1 rad/s | $104.7 \times 10^{-3}$ rpm ($\pi$/30 rpm) |
| 1 rpm | 9.549 rad/s (30/$\pi$ rad/s) |

**Pressure**

| | |
|---|---|
| 1 atm | 1.013 bar |
| 1 atm | 29.92 in Hg |
| 1 atm | 101.3 kPa |
| 1 atm | 14.70 lb/in$^2$ |
| 1 atm | 760 mm Hg |
| 1 bar | $986.9 \times 10^{-3}$ atm |
| 1 bar | 29.53 in Hg |
| 1 bar | 100 kPa |
| 1 bar | 14.50 lb/in$^2$ |
| 1 bar | 750.1 mm Hg |
| 1 in Hg | $33.42 \times 10^{-3}$ atm |
| 1 in Hg | $33.86 \times 10^{-3}$ bar |
| 1 in Hg | 3.386 kPa |
| 1 in Hg | $491.2 \times 10^{-3}$ lb/in$^2$ |
| 1 in Hg | 25.4 mm Hg |
| 1 kPa | $9.869 \times 10^{-3}$ atm |
| 1 kPa | $10 \times 10^{-3}$ bar |
| 1 kPa | $295.2 \times 10^{-3}$ in Hg |
| 1 kPa | $145.0 \times 10^{-3}$ lb/in$^2$ |
| 1 kPa | 7.501 mm Hg |
| 1 lb/in$^2$ | $68.03 \times 10^{-3}$ atm |
| 1 lb/in$^2$ | $68.95 \times 10^{-3}$ bar |
| 1 lb/in$^2$ | 2.036 in Hg |
| 1 lb/in$^2$ | 6.895 kPa |
| 1 lb/in$^2$ | 51.71 mm Hg |
| 1 mm Hg | $1.316 \times 10^{-3}$ atm |
| 1 mm Hg | $1.333 \times 10^{-3}$ bar |
| 1 mm Hg | $39.37\ 10^{-3}$ in Hg |
| 1 mm Hg | $133.3 \times 10^{-3}$ kPa |
| 1 mm Hg | $19.34 \times 10^{-3}$ lb/in$^2$ |

**Temperature**

Use the following for temperature *differences* only.

| | |
|---|---|
| 1 °C | 5/9 °F |
| 1 °C | 5/9 °R |
| 1 °C | 1 K |
| 1 °F | 9/5 °C |
| 1 °F | 1 °R |
| 1 °F | 9/5 K |
| 1 °R | 9/5 °C |
| 1 °R | 1 °F |
| 1 K | 1 °C |
| 1 K | 5/9 °F |

Use the following for general temperature conversion.

| | |
|---|---|
| $X$ °C | 5/9 ($Y$ °F − 32°) |
| $Y$ °F | (9/5 $X$ °C) + 32° |
| $X$ K | $Y$ °C + 273.15 |
| $X$ °R | $Y$ °F + 459.67 |
| 1 °R | 9/5 K |
| 1 K | 5/9 °R |

**Torque**

| | |
|---|---|
| 1 lb · in | $113.0 \times 10^{-3}$ N · m |
| 1 lb · ft | 12 lb · in |
| 1 lb · ft | 1.356 N · m |
| 1 N · m | $737.6 \times 10^{-3}$ lb · ft |
| 1 N · m | 8.851 lb · in |

**Mass moment of inertia**

| | |
|---|---|
| 1 in · lb · s$^2$ | $113.0 \times 10^{-3}$ kg · m$^2$ |
| 1 kg · m$^2$ | 8.852 in · lb · s$^2$ |

**Torsional stiffness**

| | |
|---|---|
| 1 lb · ft/deg | 1.356 N · m/deg |
| 1 N · m/deg | 737.6 x $10^{-3}$ lb · ft/deg |

**Torsional damping**

| | |
|---|---|
| 1 lb · ft · s/deg | 1.356 N · m · s/deg |
| 1 N · m · s/deg | 737.6 x $10^{-3}$ lb · ft · s/deg |

**Power**

| | |
|---|---|
| 1 hp | 745.7 x $10^{-3}$ kW |
| 1 hp | 550 ft · lb/s |
| 1 kW | 1.341 hp |
| 1 kW | 1 kJ/s |
| 1 kW | 737.6 ft · lb/s |

## Unit Prefixes

| Multiple | Prefix | Symbol |
|---|---|---|
| $10^{-18}$ | atto | a |
| $10^{-15}$ | femto | f |
| $10^{-12}$ | pico | p |
| $10^{-9}$ | nano | n |
| $10^{-6}$ | micro | μ |
| $10^{-3}$ | milli | m |
| $10^{-2}$ | centi | c |
| $10^{-1}$ | deci | d |
| $10^{1}$ | deka | da |
| $10^{2}$ | hecto | h |
| $10^{3}$ | kilo | k |
| $10^{6}$ | mega | M |
| $10^{9}$ | giga | G |
| $10^{12}$ | tera | T |
| $10^{15}$ | peta | P |
| $10^{18}$ | exa | E |

## Unit Abbreviations

| Abbreviation | Meaning | Comments |
|---|---|---|
| bhp | brake horsepower | |
| cpm | cycle per minute | frequency, but not shaft speed |
| deg | degree | |
| ft · lb | foot pound | work or energy, but not torque |
| g | acceleration of gravity | |
| g | gram | |
| hp | horsepower | |
| Hz | hertz (cycles per second) | |
| in · lb | inch pound | work or energy, but not torque |
| in/s | inch per second | |
| kcpm | thousands of cycles per minute | |
| kHz | kilohertz | |
| krpm | thousands of revolutions per minute | |
| kW | kilowatt | |
| lb | pound | |
| lb · in | pound inch | torque or moment, but not work |
| lb · ft | pound foot | torque or moment, but not work |
| mil | 0.001 inch | |
| mm/s | millimeter per second | |
| mV/(in/s) | millivolt per in/s | velocity sensor scale factor |
| mV/(m/s$^2$) | millivolt per m/s$^2$ | accelerometer scale factor |
| mV/(mm/s) | millivolt per mm/s | velocity sensor scale factor |
| mV/μm | millivolt per micrometer | displacement sensor scale factor |
| mV/g | millivolt per g | accelerometer scale factor |
| mV/mil | millivolt per mil | displacement sensor scale factor |
| MW | megawatt | |
| μm | micrometer | |
| N · m | newton meter | torque, moment, or work |
| N · m/s | newton meter per second | power; equal to a watt |
| oz | ounce | |
| pk | peak (zero-to-peak) | |
| pp | peak-to-peak | |
| rad | radian | |
| rad/s | radian per second | |
| rms | root-mean-square | |
| rpm | revolution per minute | shaft rotative speed |
| shp | shaft horsepower | |
| V | volt | |
| Vac | volt alternating current | |
| Vdc | volt direct current | |
| W | watt | |

### Force, Weight, and Mass in the US Customary System

Often, units of force and mass are written as *lbf* and *lbm*, for the pound, or *ozf* and *oz* for the ounce. Pounds and ounces are, fundamentally, units of *force*; thus, throughout this book, the units *lb* and *oz* are used with the understanding that force is implied. We find it more convenient to express US customary units of *mass* as $lb \cdot s^2/in$ or $lb \cdot s^2/ft$. (1 lb · $s^2$/ft is also called a *slug*, a derived unit of mass weighing 32.17 lb, but this is not a convenient unit for calculations which involve dimensional bookkeeping.)

According to Newton's Law, when a mass is subjected to an acceleration, in this case due to the earth's gravity, it produces a force, $W$:

$$W = mg \tag{A7-1}$$

We perceive (and measure) this force as a *weight*. Using this equation, a balance weight that has a mass of 162 x $10^{-6}$ lb · $s^2$/in, will weigh

$$W = \left(162 \times 10^{-6} \ \frac{\text{lb} \cdot \text{s}^2}{\text{in}}\right)\left(386.1 \ \frac{\text{in}}{\text{s}^2}\right)$$

$$W = 0.0626 \text{ lb}\left(16 \ \frac{\text{oz}}{\text{lb}}\right) = 1 \text{ oz}$$

This balance weight exerts a force, $W$, on your hand of 1 oz when held stationary in our 1 g gravitational field. However, when installed in a 3600 rpm rotor at a radius of 24 in, the force produced by this weight will be very different. This force can be found by substituting the centripetal acceleration, $a$, of the rotor for the acceleration due to gravity:

$$F = \left(\frac{W}{g}\right)a \tag{A7-2}$$

Here $a = r_u\Omega^2$, where $r_u$ is the radius of attachment, and $\Omega$ is the rotor speed in rad/s. Thus, Equation A7-2 becomes

$$F = \left(\frac{W}{g}\right)r_u\Omega^2 = mr_u\Omega^2 \tag{A7-3}$$

where $m$ is the mass of the weight, $W/g$.

We can calculate the force, $F$, using the middle part of Equation A7-3,

$$F = \left[\frac{1 \text{ oz}}{386.1 \text{ in/s}^2}\right]\left\{(24 \text{ in})\left[(3600 \text{ rpm})\left(\frac{\pi}{30}\frac{\text{rad/s}}{\text{rpm}}\right)\right]^2\right\}$$

$$F = 8800 \text{ oz}\left(\frac{1 \text{ lb}}{16 \text{ oz}}\right) = 550 \text{ lb}$$

Or, using the right side of Equation A7-3, we can apply the conversion factor from a previous section,

$$F = \left[(1 \text{ oz})\left(162 \text{ x } 10^{-6} \ \frac{\text{lb} \cdot \text{s}^2/\text{in}}{\text{oz}}\right)\right](24 \text{ in})\left[(3600 \text{ rpm})\left(\frac{\pi}{30}\frac{\text{rad/s}}{\text{rpm}}\right)\right]^2$$

$$F = 550 \text{ lb}$$

Appendix 8

# Nomenclature

THROUGHOUT THIS BOOK, SCALARS (magnitude only) are displayed as *italic*; vectors (magnitude and direction) are displayed as ***bold italic***.

**Upper case Roman**

| | |
|---|---|
| $A$ | amplitude |
| $A_F$ | amplitude of the forward rotating vector |
| $A_{inst}$ | amplitude of a vibration vector in the instrumentation reference frame |
| $A_R$ | amplitude of the reverse rotating vector |
| $B_{inst}$ | amplitude of a vibration vector in the instrumentation reference frame |
| $D$ | damping constant |
| $D_{cr}$ | critical damping |
| $D_E$ | effective damping |
| $D_S$ | system damping |
| $D_T$ | torsional damping |
| $E$ | modulus of elasticity (Young's modulus) |
| $\boldsymbol{F}$ | force vector |
| $\boldsymbol{F}_B$ | radial force component |
| $\boldsymbol{F}_D$ | damping force |
| $\boldsymbol{F}_P$ | perturbation force |
| $\boldsymbol{F}_S$ | spring force |
| $\boldsymbol{F}_T$ | tangential force component |
| $G$ | shear modulus |
| HS | heavy spot |
| $I$ | (mass or section) moment of inertia |
| $J$ | polar moment of inertia |

| | |
|---|---|
| $K$ | spring stiffness constant |
| $K_{strong}$ | strong spring stiffness in anisotropic systems |
| $K_{weak}$ | weak spring stiffness in anisotropic systems |
| $K_B$ | bearing spring stiffness |
| $K_D$ | Direct Dynamic Stiffness |
| $\boldsymbol{K}_N$ | nonsynchronous Dynamic Stiffness |
| $K_Q$ | Quadrature Dynamic Stiffness |
| $\boldsymbol{K}_S$ | synchronous Dynamic Stiffness |
| $K_T$ | torsional spring stiffness |
| $K_x$ | spring stiffness in $X$ direction |
| $K_y$ | spring stiffness in $Y$ direction |
| $L$ | length of shaft section |
| L | transducer orientation angle *left* of reference direction |
| $M$ | rotor mass |
| $P$ | power |
| $Q$ | Synchronous Amplification Factor |
| $\boldsymbol{R}$ | Arbitrary constant displacement vector |
| R | transducer orientation angle *right* of reference direction |
| $T$ | period of a vibration signal |
| $T$ | torque |
| $U$ | transducer response plane axis that is aligned with the transducer |
| $U$ | unbalance |
| $V$ | transducer response plane axis that is perpendicular to the $U$ axis |
| $W$ | partial weight of the rotor supported by a particular bearing |
| X | multiple (order) of rotor speed (1X, 2X, ...) |
| $X$ | axis of rectangular coordinate system |
| $XY$ | coplanar, orthogonal transducers |
| $X_R$ | rotated virtual $X$ probe |
| $Y$ | axis of rectangular coordinate system |
| $Y_R$ | rotated virtual $Y$ probe |
| $Z$ | axial coordinate axis of rectangular coordinate system |
| $Z$ | peak amplitude of response in axial direction |

### Lower case Roman

| | |
|---|---|
| $a$ | acceleration |
| $c$ | radial clearance |
| $d$ | displacement |
| $d$ | differential, as in $dt$ |
| $e$ | base for natural logarithm, 2.718281828... |
| $e$ | eccentricity |

| | |
|---|---|
| $f$ | frequency |
| $f(\ )$ | function |
| $i$ | element counter, as in the $i$th position, $x_i$ |
| $j$ | $\sqrt{-1}$ |
| $j$ | element counter, as in the $j$th position, $x_j$ |
| $k$ | integer |
| $l$ | unsupported rotor length |
| $m$ | unbalance mass |
| n | multiplier of rotor speed (nX) |
| $r$ | radius of outer surface of shaft |
| $\boldsymbol{r}$ | shaft position or vibration vector |
| $\boldsymbol{r}_s$ | shaft absolute vibration vector |
| $\boldsymbol{r}_c$ | casing absolute vibration vector |
| $\boldsymbol{r}_{sr}$ | shaft relative vibration vector |
| $\dot{\boldsymbol{r}}$ | shaft velocity vector |
| $\ddot{\boldsymbol{r}}$ | shaft acceleration vector |
| $r_e$ | distance between mass center and geometric center of shaft |
| $r_u$ | distance of unbalance mass from geometric center of shaft; typically, the radius of the balance ring |
| $s_{1,2}$ | eigenvalues (roots) of the characteristic equation |
| $t$ | time |
| $\boldsymbol{u}$ | dummy vector variable |
| $u$ | position in the direction of the $U$ coordinate axis in the transducer response plane |
| $v$ | linear velocity |
| $v_{avg}$ | average angular velocity of swirling fluid |
| $v$ | position in the direction of the $V$ coordinate axis in the transducer response plane |
| $x$ | position in the direction of the $X$ coordinate axis |
| $\boldsymbol{x}$ | vibration vector measured by $X$ probe |
| $x_d$ | in-phase (real) component of the complex vibration vector, $\boldsymbol{x}$ |
| $\boldsymbol{x}_{inst}$ | vibration vector measured by the $X$ transducer, using instrumentation phase |
| $x_q$ | quadrature (imaginary) component of the complex vibration vector, $\boldsymbol{x}$ |
| $\boldsymbol{x}_R$ | vibration vector measured by virtual $X$ probe |
| $y$ | position in the direction of the $Y$ coordinate axis |
| $\boldsymbol{y}$ | vibration vector measured by $Y$ probe |
| $y_d$ | in-phase (real) component of the complex vibration vector, $\boldsymbol{y}$ |
| $\boldsymbol{y}_{inst}$ | vibration vector measured by the $Y$ transducer, using instrumentation phase |

| | | |
|---|---|---|
| $y_q$ | | quadrature (imaginary) component of the complex vibration vector, $\boldsymbol{y}$ |
| $\boldsymbol{y}_R$ | | vibration vector measured by virtual $Y$ probe |
| $z$ | | position in the directions of the $Z$ coordinate axis |

**Upper case Greek**

| | | |
|---|---|---|
| Ø | Phi | used in KØ to symbolize the Keyphasor probe or event |
| $\Phi$ | Phi | general phase angle |
| $\Phi$ | | amplitude of torsional vibration |
| $\Psi$ | Psi | attitude angle |
| $\Omega$ | Omega | shaft rotative speed |
| $\Omega_{res}$ | | resonance speed |
| $\Omega_{th}$ | | Threshold of Instability |

**Lower case Greek**

| | | |
|---|---|---|
| $\alpha$ | alpha | phase of $\boldsymbol{r}$ with respect to measurement transducer at $t = 0$ (KØ event), expressed in the mathematical measurement convention |
| $\alpha_{inst}$ | | phase lag angle in the instrumentation measurement convention |
| $\beta_{inst}$ | beta | phase lag angle in the instrumentation measurement convention |
| $\delta$ | delta | angular location of the force vector at $t = 0$ (KØ event) |
| $\delta$ | | logarithmic decrement |
| $\gamma$ | gamma | growth/decay rate |
| $\varepsilon$ | epsilon | eccentricity ratio |
| $\zeta$ | zeta | damping factor |
| $\theta$ | theta | angle of $\boldsymbol{r}$ with respect to the measurement axis in the complex plane |
| $\theta$ | | angular deflection of a shaft |
| $\theta_K$ | | angle of the Keyphasor transducer with respect to a reference direction |
| $\theta_T$ | | angle of the vibration transducer with respect to a reference direction |
| $\lambda$ | lambda | Fluid Circumferential Average Velocity Ratio, a measure of fluid circulation |
| $\pi$ | pi | the ratio of the circumference of a circle to its diameter, 3.141592654... |
| $\tau$ | tau | torsional shear stress |
| $\phi$ | phi | relative angular displacement (angle of twist) of a shaft, measured between two axial locations |

| | | |
|---|---|---|
| $\dot{\phi}$ | | relative angular velocity of a shaft, measured between two axial locations |
| $\ddot{\phi}$ | | relative angular acceleration of a shaft, measured between two axial locations |
| $\phi_F$ | | phase of the forward vector in the mathematical measurement convention |
| $\phi_R$ | | phase of the reverse vector in the mathematical measurement convention |
| $\omega$ | omega | any nonsynchronous frequency |
| $\omega_d$ | | damped natural frequency of vibration |
| $\omega_n$ | | undamped natural frequency of vibration |

# Glossary

**⅓X, ⅖X, ½X, ⅘X, etc.**
In a complex vibration signal, signal components having frequencies equal to fractions of rotative speed. Also called subsynchronous and, when the numerator is 1, subharmonic.

**1X**
In a complex vibration signal, the signal component that occurs at the same frequency as rotative speed (1 times rotor speed). Also called synchronous.

**2X, 3X, etc.**
In a complex vibration signal, signal components having frequencies equal to integer multiples of shaft rotative speed. Also called harmonic, superharmonic, and supersynchronous.

**Absolute phase**
The timing difference between a once-per-turn reference event (Keyphasor event) and the first positive peak of a filtered vibration signal (filtered to a harmonic of running speed), expressed as degrees phase lag.

**Absolute vibration**
Vibration of an object measured relative to an inertial (fixed) reference frame. Accelerometers and velocity transducers measure absolute vibration, typically of machine housings or structures; they are often referred to as seismic or inertial transducers.

**Acceleration**
The rate of change of velocity with time. Acceleration is the first derivative of velocity and the second derivative of displacement. Acceleration is a zero-to-peak (pk) measurement.

**Accelerometer**
An inertial transducer that converts acceleration into an electrical signal.

**Acceptance region**
On a 1X or 2X polar APHT plot, a region of acceptable steady state response defined by amplitude and phase boundaries. Based on normal historical data for the machine, the user defines one or more acceptance regions for each radial vibration measurement and operating condition. Acceptance regions can also be defined in terms of the normal average shaft centerline position.

**Accuracy**
The degree of conformity of a calculation or measurement to a standard or true value. It is often expressed as a ratio, in percent, of the error to the full-scale meter reading, or of the error to the actual input value. Accuracy is not the same as precision (repeatability).

**Aeroderivative**
Aircraft jet (gas turbine) engines that have been modified for industrial use.

**Aerodynamic instability**
An abnormal aerodynamic flow in a compressor. Stall and surge are caused by flow separation in blades or impellers; choke is caused by supersonic flow. Aerodynamic instabilities can couple to the rotor dynamics of the machine.

**Alarm**
See: Alert

**Alert (point)**
The parameter value that activates a warning of a condition requiring corrective action or investigation. Synonymous with *alarm*.

**Aliasing**
In measurements, false indication of frequency components caused by sampling a dynamic signal at too low a sampling frequency. According to the Nyquist criterion, the sampling frequency must be at least twice the highest frequency component in the signal. Aliasing can be eliminated by adjusting the sampling frequency or by using a low-pass filter on the signal prior to sampling (antialias filter). The primary disadvantage of antialias filtering is that it causes phase and amplitude errors (as is the case with virtually any type of filtering).

**Alignment**
A machine is in perfect internal alignment when the average shaft centerline position is in the center of each interstage diaphragm or seal and the average shaft centerline of the rotor operates at the design position in the bearing. Two machines are in perfect external alignment when the shaft centerlines are collinear at the coupling and operate in their design axial positions within each machine. Also, the process of positioning machine components such as bearings, diaphragms, rotors, casing, foundation, piping, etc., with respect to each other so that they are aligned within an acceptable tolerance under normal operating conditions.

**Amplification Factor, Nonsynchronous**
See: Nonsynchronous Amplification Factor

**Amplification Factor, Synchronous**
See: Synchronous Amplification Factor

**Amplitude**
The magnitude of periodic dynamic motion (vibration) or position. Amplitude is typically expressed in terms of peak-to-peak, zero-to-peak, root-mean-square, or average.

**Amplitude modulation**
The process whereby variation in the amplitude of a vibration signal results in a variation of the amplitude of a carrier signal. AM is used when high frequency signal recordings are needed (for example, for gear mesh frequencies). AM (also called direct) tape recorders have a finite lower frequency response above zero hertz (dc). They capture dynamic data above the lower response frequency, but not the average shaft position data (dc voltage), which is also available from a proximity probe signal.

**Anisotropy**
A system or material property such as stiffness, damping, elasticity, density, composition, conductivity, etc., that varies with respect to the direction it is measured.

**Antialias filter**
A low-pass filter which is used to prevent false frequency components from appearing in the spectrum of a digitally sampled signal.

**Antinode**
A location on a mode shape where the vibration amplitude is maximum.

**Antiswirl**
A technique used in the bearings and seals of pumps and compressors to improve rotor stability. To decrease or prevent fluid circumferential flow around the rotor, fluid is injected into the clearance in the direction opposing the fluid dragged into rotation.

**Antiswirl injection**
The injection of fluid tangentially into a bearing or seal, against rotation, to control a fluid-induced instability problem. Antiswirl injection increases the Threshold of Instability by decreasing $\lambda$.

**APHT plot (Amplitude-PHase-Time)**
A trend plot of filtered vibration amplitude and phase lag angle data in a rectangular or polar format. Commonly used for 1X, 2X and nX vibration data.

**API**
The American Petroleum Institute is a national trade organization representing the petroleum industry. API was formed in 1919 to standardize specifications for oil drilling and production equipment.

**Apparent power**
The product of the applied rms voltage times the rms current.

**Asymmetry**
A geometric property of an object such that when it is divided by a plane, the pieces are not similar, or when it is subjected to a certain operation (rotation, inversion, etc.), it does not appear the same unless the operation returns it to its original condition.

**Asymmetric rotor**
A rotor whose cross section, when rotated at any angle about its geometric center, does not appear the same unless returned to its original orientation; for example, an elliptical cross section, a rotor with a keyway, or a rotor section with a crack.

**Asynchronous**
Not synchronized with a timing signal. Two waveforms sampled at slightly different times are said to be asynchronous waveforms.

**Attenuation**
A reduction in amplitude of a signal. Also, the decrease in amplitude that results from the transmission of vibration energy from one machine part to another (for example shaft to fluid-film bearing housing).

**Attitude angle**
In a bearing, the angle between the resultant of the radial loads acting on a rotor and a line connecting the bearing and shaft centers, measured in the direction of rotation. Sometimes confused with rotor position angle.

**Auxiliary machines**
Machines that assist process machines. Auxiliary machines include lube oil systems, sealing systems, fuel delivery systems, cooling systems, and, in power plants, condensate feedwater systems. Lube oil systems supply fluid-film bearings with clean lubricant at the correct pressure, flow rate, and temperature. Sealing systems, which can be quite complex, provide buffer oil or gas where needed. Cooling systems range from elaborate cooling tower systems in power plants, to interstage cooling systems, to lube oil cooling systems.

**Average amplitude**
An amplitude measurement rarely used for vibration signals; the half cycle average amplitude of a sine wave is 0.637 x zero-to-peak (pk) amplitude. This formula applies to sinusoidal (single frequency) signals only.

**Average shaft centerline plot**
A transient data or trend plot of the average shaft centerline position, presented in an *XY* (rectangular) format.

**Average shaft position**
The average position of the shaft relative to the displacement probe and its mounting. The most common application is rotor axial (thrust) position relative to the thrust bearing. Another important application is average shaft radial position. These measurements are made using the dc (position) component of the proximity probe signal. Two proximity probes mounted in an *XY* configuration are required for a two-dimensional, radial position measurement.

**Axial**
In the same direction as the shaft centerline (the axis of shaft rotation).

**Axial Dynamic Stiffness**
The ratio of the dynamic axial force to the dynamic axial response of a rotor system.

**Axial position**
The average position or change in position of a rotor in the axial direction with respect to some fixed reference. Typically, the reference is the thrust bearing support structure or other casing member to which the displacement probe is mounted. The probe may observe the thrust collar or some other integral, axial shaft surface. Also called thrust position.

**Axial vibration**
Vibration of the rotor shaft or machine casing in the axial direction.

**Balance-of-plant machinery**
All machinery in a plant that is considered neither critical nor essential. Many of these machines have tandem or spared installations (redundant installations).

**Balance plane**
A specific axial location in a machine at which balance weights may be installed.

**Balance resonance speed**
The shaft rotative speed which approximately coincides with a rotor system natural frequency. The vibration characteristics are a peak in the 1X amplitude together with a significant increase in the 1X vibration phase lag angle.

**Balancing**
The act of adjusting the radial mass distribution of a rotor at one or more balance planes so that the center of mass (principal axis of inertia) is within an acceptable distance from the rotor geometric center. This reduces the 1X lateral vibration of the rotor.

**Band-pass filter**
A filter that has a single transmission band extending from a lower corner

frequency to an upper corner frequency. The amplitude at the corner frequencies is 70% (−3 dB) of the amplitude at the center frequency.

**Bandwidth**

The span between the corner frequencies of a band-pass filter. Normally expressed in terms of frequency for constant bandwidth filters and as a percent of the center frequency for constant $Q$ (constant percentage) filters.

**Baseline data**

A set of reference data acquired when a machine is in acceptable condition, usually after installation, overhaul, or turnaround.

**Baseplate**

A cast or fabricated steel support for a machine. The baseplate would normally include the pedestals for all the machines in the machine train. The baseplate can be directly grouted and bolted to a foundation. Also called a soleplate.

**Bearing**

Any low friction structure which supports the rotor and provides dynamic constraint in the radial (lateral) and/or axial (thrust) directions. The main categories are fluid-film bearings, rolling element bearings, magnetic bearings, and foil bearings.

**BEP (Best efficiency point)**

The flow at which a pump operates with maximum efficiency (the minimum amount of losses). The BEP is important for two reasons: operating costs are minimized and radial (side) loads are minimized.

**Blade passing frequency**

The number of blades (on a disk or stage) times shaft rotative speed. The blade passing frequency is a potential vibration frequency on any machine with blades (a turbine, axial compressor, fan, propeller, etc.).

**Bode plot**

A pair of graphs in rectangular (Cartesian) format displaying the amplitude and phase of the 1X vibration vector as a function of shaft rotative speed. The $Y$ axis of the top graph represents 1X phase, measured as positive phase lag, increasing downward; the $Y$ axis of the bottom graph represents 1X amplitude. Both $X$ axes represent shaft rotative speed or perturbation frequency. Sometimes called an unbalance response plot. Also used for 2X, 3X, etc. vibration vectors.

**Bow**

A condition of plastic deformation of a shaft which results in a bent geometric shaft centerline. Often the centerline is bent in a single plane, due to gravity sag, thermal warping, etc.; however, the bow may be three

dimensional (corkscrew). Shaft bow can be detected by measuring the shaft relative displacement (the eccentricity) with a proximity probe at rotor slow roll speed.

**Cage frequency**

See: Fundamental train frequency

**Calibration weight**

A weight of known magnitude which is placed on the rotor at a known axial and angular location, in order to measure the resulting change in machine 1X vibration response at known operating conditions. Sometimes called a trial weight.

**Campbell diagram**

A plot of rotor system natural frequencies (vertical axis) versus excitation frequencies (horizontal axis). The plot allows a designer to detect when frequencies produced by the machine, including the harmonics of running speed, are close to a natural frequency of the system. Diagonal lines represent the frequencies that could be produced by the rotor system. Horizontal lines represent natural frequencies as they exist in the machine at operating speed.

**Cartesian format**

See: XY plot

**Cascade plot**

See: Spectrum cascade plot

**Casing expansion**

A measurement of the axial position of the machine casing relative to a fixed reference, usually the foundation. The measurement is typically made with a linear variable differential transformer (LVDT) installed on the foundation at the opposite end of the machine from the point where the casing is attached to the foundation. Changes in casing axial position are the result of thermal expansion and contraction of the casing during start-up and shutdown.

**Casing transducer**

A velocity or acceleration transducer that measures the absolute vibration of the machine housing or structure. These transducers measure absolute vibration relative to an internal inertial reference and are referred to as seismic or inertial transducers.

**CCW**

Counterclockwise. See: *X* to *Y*

**Center frequency**
For band-pass filters, the arithmetic center of a constant bandwidth filter or the geometric center (midpoint on a logarithmic scale) of a constant percentage filter.

**Centerline position**
See: Average shaft position

**Channel**
A single transducer and the instrumentation hardware path to display or record its output signal.

**Characteristic equation**
A polynomial in $s$ that describes the free vibration behavior of a system. For a second order system, the characteristic equation is a quadratic polynomial in $s$. The values of $s$ that satisfy this equation are the roots of the equation and are also called the eigenvalues, characteristic values, or poles.

**Characteristic values**
The roots, or eigenvalues, of the characteristic equation.

**Clearance circle**
The circle or ellipse traced out by the shaft centerline as the shaft walks around in contact with the bearing or seal. The clearance circle should be documented on the average shaft centerline plot.

**Clearance, diametral**
The diameter of the available clearance in which the shaft can move. The diameter of the clearance circle.

**Clearance, radial**
The distance from the bearing or seal center to the shaft centerline when the shaft contacts the bearing or seal. The radius of the clearance circle.

**Coastdown**
A machine speed change from higher to lower rpm during shutdown; typically, when a machine goes from operating speed to zero, or near zero, speed.

**Complex conjugate**
A complex number formed by changing the sign of the imaginary (quadrature) term of a complex number. The result of multiplying a number by its complex conjugate is a real number equal to the sum of the squares of the two terms.

**Cold water stands**
An arrangement of piping and brackets installed along a machine foundation for hot alignment measurements. Proximity probes observe exposed shaft areas or targets on the machine casing from brackets through which

water circulates. This provides a thermally stable reference for the alignment measurement.

**Compensation**
See: Runout compensation

**Complex number**
A number of the form $x + jy$, where $j = \sqrt{-1}$. A complex number is equivalent to a point that is plotted in the complex plane. In mathematics, a complex number has a real part and an imaginary part. In machinery applications, complex numbers are used for rotor modeling and to represent vibration vectors. Dynamic Stiffness is a complex number; the two parts are referred to as direct and quadrature, respectively. Complex numbers can be added, subtracted, multiplied, and divided.

**Conditional baseline data**
Reference data sets used for a short term comparison of correlation, typically under different operating conditions.

**Constant bandwidth filter**
A band-pass filter having a fixed frequency bandwidth regardless of the value of the center frequency.

**Constant percentage filter**
A band-pass filter whose bandwidth is a fixed percentage of the center frequency. Also called a constant Q filter.

**Constant Q filter**
See: Constant percentage filter

**Critical**
Defined by the API [7] as any balance resonance frequency with a Synchronous Amplification Factor > 2.5.

**Critically damped**
See: Damping factor

**Critical machinery**
Machinery that represents such large business risks, including economic, safety, government compliance, or production interruption, that failures cannot be tolerated.

**Critical speed**
In general, any shaft rotative speed which is associated with high vibration amplitudes. When it corresponds to the resonance frequency of a rotor radial vibration mode that is excited by rotor unbalance, it is more correctly called the *balance resonance speed.*

**Critical speed map**
An *XY* (Cartesian) diagram used in rotating machinery design as a tool to approximate the effect of changes in bearings, supports, and pedestal

designs on system natural frequencies. The $X$ axis represents bearing stiffness and the $Y$ axis represents rotor system natural frequency.

**Cross-axis sensitivity**

The ratio of change in the signal output to an incremental change in the input along any axis perpendicular to the sensitive axis of an inertial (seismic) transducer.

**Crosstalk**

Interference or noise in a transducer signal or channel which originates in another transducer or channel. Crosstalk can occur when the tips of two proximity probes are too close together. The interaction of the probes' electromagnetic fields causes a noise component in each of the transducers' output signals. The frequency of the noise component is the difference (beat frequency) of the two Proximitor oscillator frequencies.

**CW**

Clockwise. See: $Y$ to $X$

**Cycle**

One complete sequence of values of a periodic quantity. A cycle of vibration represents one complete set of signal values.

**Cylinder expansion**

See: Casing expansion

**Damped natural frequency**

The natural frequency of a mechanical system that includes the effects of damping. Damping shifts the natural frequency of a system to slightly below the undamped natural frequency.

**Damping factor**

The damping factor, $\zeta$, is the ratio of the damping to the critical damping, a nondimensional number that defines the behavior of a freely vibrating mechanical system. If $\zeta < 1$, the system is *subcritically damped* (underdamped) and, when disturbed, will oscillate with decreasing amplitude. The larger (closer to 1) the damping factor, the more quickly the vibration will die away. When $\zeta = 1$, the system is *critically damped* and will not vibrate, but will return to the equilibrium position in the shortest possible time without overshooting (oscillating). When $\zeta > 1$, the system is *supercritically damped* (overdamped) and will return relatively slowly to the equilibrium position without any overshoot or oscillation.

**Damping force**

A force that is proportional to the speed of a body and acts in a direction opposite to the velocity: $\boldsymbol{F} = -D\dot{\boldsymbol{r}}$ , where $D$ is the damping constant, and $\dot{\boldsymbol{r}}$ is the velocity of the body. The damping force is produced by the viscosity of the fluid through which the body is moving.

**Damping stiffness**
In the simple rotor model, the dynamic stiffness term produced by the damping force. The damping stiffness is $jD\omega$, where $D$ is the damping, $\omega$ is the frequency of vibration, and $j = \sqrt{-1}$ . The $j$ indicates that the damping stiffness acts at 90° to the direction of the applied dynamic force. Like the mass stiffness, the damping stiffness is a dynamic stiffness, and it only appears when the rotor centerline has nonzero velocity.

**Danger (point)**
A parameter value at which immediate action is required, up to and including automatic or manual shutdown of a system.

**Decibel (dB)**
The logarithm of the ratio of the power ($P$) or voltage ($V$) levels of electrical signals. $\text{dB} = 10 \log P_1 / P_2 = 20 \log V_1 / V_2$.

**Degree(s) of freedom**
In rotor system modeling, the number of independent displacement variables. For example, a rotor system modeled with one rotor mass using the complex variable $\boldsymbol{r} = x + jy$, has one complex degree of freedom (1-CDOF), but the same rotor system modeled treating the two real variables, $x$ and $y$, independently has two real degrees of freedom (2-DOF or 2-RDOF).

**Detector**
See: Transducer

**Diagnostics**
The process of finding an answer or solution to a problematic situation. Diagnostics is inherently reactive in nature. It requires a problem to exist before an answer is sought.

**Differential expansion**
The measurement of the axial position of the rotor with respect to the machine casing at some distance from the thrust bearing. Changes in axial position relative to the casing affect axial clearances and are usually the result of different rotor and casing thermal expansion rates during startup and shutdown. The measurement is typically made with a proximity probe transducer mounted to the machine casing and observing an axial surface (for example, collar) of the rotor. The measurement is usually incorporated as part of a Turbine Supervisory Instrumentation system.

**Differential phase**
A technique which measures the phase difference between vibration signals filtered to the same frequency at different axial locations on a rotor system. It is primarily used for determining the location of the source of an instability. The vibration signal whose phase leads all others usually indi-

cates the transducer location which is closest to the source of the instability.

**Digital runout compensation**
The process of digitally sampling the unfiltered signal from the proximity transducer at slow roll speed and subtracting it from the signal at any other speed. It requires a phase reference such as a Keyphasor signal.

**Digital vector filter**
See: Vector filter

**Direct**
In Dynamic Stiffness, acting in the same or opposite direction to the applied dynamic force. See: Dynamic Stiffness, Direct.

**Direct data**
Data or a signal that represents the original transducer signal. Sometimes called unfiltered, raw, all pass, or overall data.

**Disk**
A wheel, usually solid and axially slim, on which the mechanical work of a rotating machine is performed or from which work is extracted. Disks may be integral to or mounted on the shaft. Examples of disks are a turbine disk, compressor wheel, and pump impeller.

**Displacement**
The change in position of an object relative to a reference. Machinery displacement vibration is typically a peak-to-peak measurement of the observed oscillating motion and is usually expressed in units of mil pp or μm pp. Proximity probes measure displacement directly.

**Dominant end**
The end of a machine where most of the unbalance is concentrated. For a two-mode machine, the indicated heavy spots for the two modes will be approximately in the same direction on the polar plot from the dominant end.

**Driven**
Most driven machines take input in the form of rotating kinetic energy and convert it to useful work. Driven machines include blowers, fans, generators, exciters, pumps, and compressors. These machines increase the energy state of the working fluid or product.

**Driver**
Drivers are machines that act as prime movers; most convert some form of energy to the kinetic energy of rotation. Internal combustion engines and gas turbines utilize the chemical energy of fuel, steam turbines use the potential and kinetic energy of high pressure and high temperature steam, and electric motors use the energy of electricity. Two types of drivers con-

vert the kinetic energy of one substance to the kinetic energy of rotation: wind turbines use moving air, and hydro turbines use moving water, which was previously stored as gravitational potential energy.

**Dual probe**

A pair of transducers, consisting of a proximity probe and velocity transducer, installed at the same point (usually in a common junction box on the machine bearing housing) with the same radial orientation. Four separate measurements are provided by this transducer system: shaft relative radial position within the bearing clearance, shaft dynamic motion relative to the bearing, machine casing absolute vibration measured by the velocity transducer, and shaft absolute motion represented by the summation of the velocity signal integrated to displacement and the shaft relative signal.

**Dwell time**

During a rub, the period of time during which the rotor maintains contact with the stationary part.

**Dynamic data**

Data, such as a waveform, which is rapidly changing. Dynamic data usually mimics the raw data in a way that allows further signal processing, such as for direct orbit and timebase displays, and spectrum. From dynamic data, it is possible to derive static data signal parameters, such as direct amplitude, filtered amplitude, and phase lag angle. See: Static data

**Dynamic equilibrium**

While moving in an orbit, a rotor is in dynamic equilibrium if, when it is subjected to some temporary, disturbing force, it eventually returns to the original orbit; the growth/decay rate, $\gamma$, is less than zero.

**Dynamic motion**

See: Vibration

**Dynamic Stiffness**

The ratio of the applied dynamic force to the dynamic response of a mechanical system. Dynamic Stiffness consists of the spring stiffness of the mechanical system complemented by the dynamic effects of mass and damping. Dynamic Stiffness is a characteristic of a system, and it opposes an applied dynamic force to limit vibration response.

**Dynamic Stiffness, Direct**

The component of complex Dynamic Stiffness, consisting of spring stiffness and mass stiffness, which acts at 0° or 180° to (collinear with) the applied dynamic force. In mathematical terms, it is called the real component.

**Dynamic Stiffness, Quadrature**

The component of complex Dynamic Stiffness, which acts at 90° to the

applied dynamic force. In mathematical terms, it is the imaginary part of Dynamic Stiffness, which is indicated by $j$ in the equation.

**Dynamic transducer**

A transducer that is capable of capturing high-frequency transient events. For example, a pressure transducer with a frequency response in the kHz range would be considered to be a dynamic transducer. This is in contrast to a transducer with a very slow time response that essentially produces static (averaged) data, such as a thermocouple imbedded in bearing backing.

**Eccentricity**

The radial displacement of the rotor journal centerline from the geometric center of a fluid-film bearing.

**Eccentricity, peak-to-peak**

The difference between the positive and negative extremes of the slow roll rotor bow at the measurement location, usually near the free end of a shaft. The shaft bow may be due to permanent mechanical bow or temporary thermal bow.

**Eccentricity position (obsolete)**

See: Average shaft position

**Eccentricity ratio**

A dimensionless quantity representing the average position of the shaft within the bearing compared to the available clearance. It is calculated by dividing the distance between the bearing center and the shaft centerline position by the available radial clearance. The eccentricity ratio is zero when the shaft is at the center of the bearing, and one when the shaft is in contact with the bearing wall. The *average* eccentricity ratio is obtained by using the distance between the average position of the shaft centerline and the bearing (seal) centerline. The *dynamic* eccentricity ratio is obtained by using the *instantaneous* position of the shaft within the bearing.

**Eccentricity, slow roll**

See: Eccentricity peak-to-peak

**Eddy current**

A localized electric current induced in a conductive material which experiences a changing magnetic field. The operating principle of the eddy current proximity probe.

**Effective damping**

In synchronous rotor behavior, the damping after modification by the effects of fluid circulation. The effective damping, $D_E$, of the synchronous rotor system is $D_E = D(1-\lambda)$, where $D$ is the damping, and $\lambda$ is the Fluid

Circumferential Average Velocity Ratio. Because $\lambda$ is usually a positive number less than 0.5, the effective damping for a system is usually less than the actual damping constant, $D$. The effective damping is the slope of the synchronous Quadrature Dynamic Stiffness.

**Eigenvalue**

A complex number that contains information about the free vibration behavior of a system. It is also called a root of the characteristic equation, a characteristic value, or a pole of the system. The real part of the eigenvalue is the growth/decay rate; the imaginary part is the damped natural frequency of the system.

**Elastic bow**

Rotor bow due to static radial loads that do not exceed the yield strength of the rotor. They can be caused, for example, by process fluid loads in pumps, internal misalignment, and partial arc steam admission in steam turbines, but most commonly, in a horizontal machine, by the weight of the rotor.

**Electrical runout**

A noise component in the output signal of an eddy current transducer system due to nonuniform properties of electrical conductivity or magnetic permeability, or local (spot) magnetic fields on the circumference of the observed shaft surface. Runout produces a change in the output signal which does not result from a probe gap change. The error repeats exactly with each shaft revolution as long as the shaft does not change axial position.

**Element Passage Frequency (EPx)**

In rolling element bearings, the frequency at which rolling elements pass a fixed point on either the inner or outer race. Harmonics of the element passage frequency are then indicated as 2EPx, 3EPx, ..., nEPx.

**Ellipticity**

The difference between the major and minor axes of an ellipse, divided by the major axis: 0 denotes a circle and 1 denotes a straight line.

**Engineering units**

Units of measurement; typically, for vibration, μm, mil, mm/s, in/s, g, or similar.

**Error**

The difference between the indicated value and the true value of the measured variable. It is often expressed as relative error, that is, as a percent of the output reading of the transducer.

**Essential machinery**
Machinery that can cause partial production interruption or some other form of business loss (such as an emissions violation) if it fails or runs at reduced capacity.

**Externally pressurized bearing**
A fully lubricated, fluid-film bearing that derives its primary support force from hydrostatic pressure supplied by a source external to the bearing. Also called a hydrostatic bearing.

**Filter**
Electronic circuitry designed to pass or reject a specific frequency band of a signal.

**Flexible rotor**
Rotors that experience significant bending modes. The first mode of a flexible rotor with its mass supported between bearings is usually a simple, predominantly in-phase bending mode. The second mode of a flexible rotor is usually an out-of-phase, S-shaped bending mode. Machines with fluid-film bearings can have rigid body behavior in the very lowest modes and transition to flexible rotor behavior at higher modes.

**Fluid Circumferential Average Velocity Ratio, (lambda, $\lambda$)**
The dimensionless ratio of the average angular velocity of a fluid in a bearing, seal, or other rotor-to-stator clearance, divided by shaft angular velocity, $\Omega$. $\lambda\Omega$ is the average angular velocity of the circulating fluid.

**Fluid-film bearing**
A bearing which supports the shaft on a thin film of oil. The fluid-film layer may be generated primarily by journal rotation (an internally pressurized, or hydrodynamic bearing) or by external pressure (an externally pressurized, or hydrostatic bearing). Film thickness is normally in the range of several micrometers to tens of micrometers.

**Fluid-induced instability**
Self-excited rotor lateral vibration caused by interaction between the rotor and a surrounding fluid.

**Foil bearing**
A gas bearing comprised of a cylindrical shell lined with corrugated bump foils topped with a thin flat foil. The foil complies to the hydrodynamic pressure distribution of the gas inside the bearing, resulting in a larger gap than would be present for an identically loaded, rigid gas bearing.

**Forced mode shape**
The deflection shape of the rotor at a particular operating speed. It is the sum of several free vibration (natural) mode shapes, each of which is excited to varying degrees by the unbalance distribution of the system.

**Forced vibration**
The response vibration of a mechanical system due to a forcing function (exciting force). Typically, forced vibration has the same frequency as that of the exciting force.

**Forward component**
See: Forward and reverse vectors

**Forward and reverse vectors**
A pair of counterrotating vectors that, when added, reconstruct a filtered orbit. Both vectors rotate at the same frequency; the ratio of the amplitudes determines both the ellipticity and direction of precession of the orbit.

**Forward precession**
Precession of a shaft in the same direction as its rotation.

**Fracture toughness**
A measure of a material's resistance to fast fracture. It is a function of the alloy, heat treatment, temperature, and rate of loading.

**Free vibration**
Vibration response of a mechanical system following an initial disturbance. If the system is underdamped (subcritically damped), the mechanical system vibrates at one or more of its natural frequencies with decaying amplitude. If it is critically damped or overdamped (supercritically damped), the system will not vibrate, but will return slowly to the equilibrium position with no oscillation (overshoot).

**Frequency**
The repetition rate of a periodic vibration per unit of time. Frequency is typically expressed in units of cycles per second (Hz), cycles per minute (cpm), or radians per second (rad/s).

**Frequency component**
The amplitude, frequency, and phase characteristics of a dynamic signal after it has been filtered to a single frequency.

**Frequency modulation**
Variation of the frequency of a carrier signal caused by variation in the amplitude of a vibration signal. FM tape recordings have a frequency response down to zero hertz (dc). This allows recording of proximity probe dc gap voltages, which represent average shaft position.

**Friction instability**
An instability in rotors caused by friction between disks and the rotor or by internal material friction. It manifests itself as large amplitude, subsynchronous vibration. Elastic bow in built up rotors has been known to cause friction instability.

**Full annular rub**
Rub that maintains continuous contact over all of the vibration cycle. It may occur with either forward or reverse precession.

**Full spectrum**
A spectrum of an orbit, calculated from the simultaneously sampled waveforms from a pair of *XY* transducers. It displays the amplitudes versus frequency of precession of the forward and reverse components of the orbit, with forward precession to the right and reverse precession to the left.

**Full spectrum cascade plot**
See: Spectrum cascade plot

**Full spectrum waterfall plot**
See: Spectrum waterfall plot

**Fundamental**
The lowest frequency of a series in frequency analysis. Integer (1, 2, 3, ...) multiples of this frequency are called harmonics. Usually, in rotor systems, the fundamental frequency is rotative speed, 1X. Higher harmonics are 2X, 3X, etc.

**Fundamental train frequency**
In rolling element bearings, the frequency at which the cage and element assembly completes one revolution of the bearing. For bearings with fixed outer races, the fundamental train frequency is near ½X.

**g**
The standard acceleration due to earth's gravity at 45° latitude and sea level. By international agreement, g = 9.807 $m/s^2$ = 386.1 $in/s^2$ = 32.17 $ft/s^2$

**Gain**
See: Signal gain

**Gap voltage**
The voltage output of a proximity transducer system indicating the distance (gap) between the transducer and the object observed.

**Gear mesh forces**
The contact forces that result from tooth contact in gears. The lateral gear mesh force can be resolved into a tangential force and a radial separation force as a function of the pressure angle. The tangential force is directly related to the torque delivery of the gears, the product of the tangential force and the pitch radius of the gear. The radial separation force attempts to separate the gears; for a typical 20° pressure angle, it is about 37% of the tangential force. Single helical gearing can also produce an axial force.

**Gear mesh frequency**
The number of gear teeth times shaft rotative speed.

**General purpose machinery**
See: Balance of plant machinery

**Glitch**
The combination of mechanical and electrical runout in the proximity probe viewing area. Glitch produces a signal variation that does not represent vibration.

**Grout**
An epoxy-based material used to fill all voids between a soleplate, or baseplate, and the foundation to improve the load transfer between the machine and the ground.

**Growth/decay rate, $\gamma$**
The real part of the complex eigenvalue that describes how free vibration amplitude changes over time. The magnitude and sign of $\gamma$ control how fast the vibration amplitude increases or decreases.

**Half-power Bandwidth method**
A method, adopted by API, to measure the Synchronous Amplification Factor. It is the dimensionless ratio of the rpm of the resonance peak to the half-power bandwidth (at 70% of resonance amplitude).

**Half spectrum**
A display of the amplitude as a function of frequency of a signal's components calculated from the waveform from a single transducer. See: Full spectrum

**Half spectrum cascade plot**
See: Spectrum cascade plot

**Half spectrum waterfall plot**
See: Spectrum waterfall plot

**Harmonic**
A vibration frequency that is an integer multiple of the fundamental, or lowest frequency in a series.

**Harmonic series**
A series of vibration signal components whose frequencies are integer multiples of the fundamental, or lowest frequency, vibration component; for example, 1X, 2X, 3X, etc., or ½X, 1X, 3/2X, etc.

**Harmonic vibration**
Sinusoidal vibration with a single frequency component.

**Heavy spot (HS)**
The angular location of the unbalance at a specific axial location on a rotor.

**Hertz**
Unit of frequency measurement in cycles per second (Hz).

**High-eccentricity natural frequency**
The natural frequency of a rotor system operating at a high eccentricity ratio in a fluid-film bearing. The high-eccentricity natural frequency is usually close to the nameplate resonance.

**High frequency**
A frequency range, typically above 5 kHz, used to measure the very high vibration frequencies generated in structures by rolling element bearing components with microscopic faults.

**High-pass filter**
A filter having a transmission band extending from a lower corner frequency (where amplitude is attenuated by 3 dB) to the upper frequency response limit of the transducer or instrument.

**High spot**
In 1X vibration, the location on the shaft directly under the vibration probe when the shaft makes its closest approach to that probe. It corresponds to the positive peak of the vibration waveform.

**Homogeneous equation**
A differential equation in which the forcing function is zero.

**HS**
Abbreviation for heavy spot.

**Hydraulic load**
A fluid radial load which appears in pumps because of asymmetries in the design of the volute or operation of the pump off its best efficiency point (BEP). Also called a preload or sideload.

**Hydrodynamic bearing**
See: Internally pressurized bearing

**Hydrostatic bearing**
See: Externally pressurized bearing

**Hysteretic damping**
A form of internal material loss that will remove energy from vibrating systems. The hysteretic damping force is directly proportional to the velocity and inversely proportional to the frequency of the vibration.

**Impulse response**
The response to a well-defined, short duration, impulsive force such as produced by a hammer. The force is considered to be of such short duration that no significant system changes occur during that time.

**Inboard**
When identifying the location of an object on a single machine, it is the side toward the inside of the machine case; for example, the seal is *inboard* of the bearing. When identifying the location of two similar components of

a machine case, it is the one toward the driver coupling; for example, the *inboard* bearing of the HP compressor. When there is a chance of ambiguity, the components should be defined and a written record kept.

**In phase**
A response in the same direction as the applied force vector. The magnitude of the response vector times the cosine of the angle between the force and response vectors.

**Inertially referenced**
Referenced to a mass element whose inertia is presumed to keep it stationary; a transducer with an internal inertial reference mass.

**Influence coefficient**
See: Influence vector

**Influence vector**
A complex number that describes how the 1X amplitude and phase of vibration change when a balance weight is added to the machine. Each influence vector is specific to a particular combination of weight and measurement planes and to a particular set of operating conditions, including speed, load, temperature, and alignment state.

**Influence vector, direct**
An influence vector where the measured vibration vector and the unbalance force vector are in the same or closely associated planes along the rotor axis.

**Influence vector, inverse**
An influence vector whose magnitude is the reciprocal of the magnitude of a standard influence vector.

**Influence vector, longitudinal**
An influence vector where the measured vibration vector and the unbalance force vector are in distinctly different planes along the rotor axis.

**Inner race ball pass frequency**
In rolling element bearings, the frequency at which rolling elements pass a fixed point on the inner race.

**Instability**
See: Fluid-induced instability; Stability

**Integrator**
An electronic circuit that converts a velocity signal to a displacement signal or converts an acceleration signal to a velocity signal.

**Interference diagram**
See: Campbell diagram

**Internally pressurized bearing**
A fluid-film bearing that, in normal conditions, is partially lubricated and

derives its primary support force from dynamic pressure of the lubricating fluid produced by rotation of the journal. Also called a hydrodynamic bearing.

**ISO**

International Organization for Standardization.

**Isotropic supports**

Rotor support systems that provide uniform Dynamic Stiffness in all radial directions.

**Isotropy**

A system or material property such as stiffness, damping, elasticity, density, composition, etc., which is constant in all directions.

***j***

The square root of negative 1 ($j = \sqrt{-1}$ ).

**Jeffcott rotor**

An early, simple rotor model developed by H. H. Jeffcott. It was introduced in his paper, "The Lateral Vibration of Loaded Shafts in the Neighbourhood of a Whirling Speed.—The Effect of Want of Balance," Philosophical Magazine 6, vol. 37 (1919): pp. 304-314.

**Journal**

The portion of a shaft inside the fluid-film bearing. It transmits the rotor load through the fluid to the bearing supports.

**Keyphasor event**

The sudden voltage change caused by a once-per-turn shaft marker passing a Keyphasor transducer. This signal provides a reference for measuring vibration phase lag angle and also serves to measure shaft rotative speed. The period between successive Keyphasor events represents one revolution of the shaft.

**Keyphasor pulse**

See: Keyphasor event

**Keyphasor transducer**

A transducer that produces a once-per-turn voltage pulse, called the Keyphasor event. The Keyphasor transducer is typically a proximity probe, but can be an optical pickup or a magnetic pickup.

**Lambda ($\lambda$)**

See: Fluid Circumferential Average Velocity Ratio

**Lateral vibration**

See: Radial vibration

**Lift test**

A method to quickly check the bearing clearance on smaller, horizontal machines when the machine is stopped. A dial indicator is used to record

position changes when the shaft is moved from one side of the bearing to another. A correction factor must be applied for tilting pad bearings, because pad movement can give the false impression of greater diametral clearance.

**Linear system**

A mechanical system with constant parameters (such as mass, stiffness, and damping) that can be modeled by linear differential equations. Linear systems have output that is proportional to input and output frequency equal to input frequency.

**Lissajous curve**

The plot of two, simple-harmonic-motion (sine wave) signals versus each other on a rectangular format. The shape of the curve identifies the relative frequency and phase characteristics of an unknown signal versus a known signal. An orbit is a form of Lissajous curve.

**Load**

The resultant of all forces (static and dynamic) applied to the rotor system. These forces can be either axial or radial and can be due to either internal or external mechanisms. Because the load can affect the shaft centerline position in a fluid-film bearing or seal, it can act to stabilize or destabilize the dynamic condition of the machine. Static load is sometimes referred to as preload.

Also, the output power being delivered by a rotating machine; for example, the electrical load of a generator.

**Load zone**

A region in a rolling element bearing where, when a radial or axial force is applied to the shaft, there is solid contact between the shaft, inner race, elements, and outer race of the bearing. The angular size of the load zone depends on the tolerance class of the bearing, with close tolerance bearings having wide load zones.

**Low-eccentricity natural frequency**

The natural frequency of a rotor system operating at a low eccentricity ratio in a fluid-film bearing. The system stiffness, the series combination of the bearing stiffness and the shaft stiffness, approaches its lower limit when the average and dynamic eccentricity ratio of the rotor approach zero.

**Low-pass filter**

A filter having a transmission band extending from zero frequency (or the lower frequency response limit of the transducer or instrument) to some upper corner frequency (where the amplitude is attenuated by 3 dB).

**LVDT**
: Linear variable differential transformer. A contacting displacement transducer consisting of a movable core and a stationary transformer. Typically, the core is attached to the moving part and the transformer is attached to a fixed reference. A change in core position causes a change in output voltage.

**Machinery management**
: The use of products to provide data or information that is interpreted and applied by humans to assess the condition of their machinery and correctly operate, maintain, and improve its operation.

**Machinery protection**
: The use of products to provide automatic shutdown of a machine or return it to a safe or nondestructive mode of operation without human intervention

**Magnetic bearing**
: A rotor support system that uses strong magnetic fields to suspend the rotor.

**Margin of Stability**
: The frequency range between the zero values of the Direct Dynamic Stiffness and the nonsynchronous Quadrature Dynamic Stiffness.

**Mass center**
: The principal axis of inertia of a rotor. At any axial location, the mass center is the point where all the mass of the body appears to be located. On a rotor, the mass centerline connects the mass centers of adjacent cross sections of the rotor.

**Mass stiffness**
: In the simple rotor model, the negative dynamic stiffness term produced by the inertia of the rotor: $-M\omega^2$, where $M$ is the mass, and $\omega$ is the frequency of vibration. The mass stiffness only appears when the rotor centerline vibrates.

**Mechanical runout**
: A noise component in the output signal of a transducer system due to geometric imperfections. In proximity probes, this produces a probe gap change that is not caused by a shaft centerline position change. Common sources include an out-of-round shaft, scratch, chain mark, dent, rust, conductive buildup on the shaft, stencil mark, flat spot, and engraving. An out of round shaft can also transmit a noise signal to velocity and acceleration transducers.

**Microinch**
: A unit of length or displacement equal to $10^{-6}$ inches or $10^{-3}$ mils.

**Micrometer**
A unit of length or displacement equal to $10^{-6}$ meters. Also called micron (obsolete).

**Mil**
A unit of length or displacement equal to 0.001 inch. One mil equals 25.4 micrometers.

**Misalignment**
See: Alignment

**Modal damping**
The actual damping force available to the system when mode shape effects are included. The actual damping force depends on the damping constants of the individual bearings and seals and the local shaft lateral velocities in those bearings and seals. For example, rigid body modes produce large amplitudes of vibration in fluid-film bearings, larger damping forces, and higher modal damping, while bending modes tend to produce smaller amplitudes of vibration in the bearings, smaller damping forces, and lower modal damping.

**Modal mass**
The effective dynamic mass of the system. Modal mass stores the kinetic energy associated with vibration. The modal mass will be different for each mode, and, except when an entire rotor moves rigidly, the modal mass of the rotor will usually be less than the static weight of the rotor. Successively higher modes have more nodal points that prevent shaft motion and reduce modal mass.

**Modal stiffness**
The effective stiffness of a system in dynamic motion. The rotor shaft behaves like a beam with bending stiffness related to the unsupported length. Successively higher modes have more nodal points that act as pinned supports from a stiffness perspective, decreasing the effective unsupported length and increasing the modal stiffness. Modal stiffness includes the series/parallel combination of all stiffness elements in the rotor and support system.

**Mode identification probes**
Probes that are installed to help with the identification of the shaft mode shape. Mode shapes must normally be interpolated between a small set of measurement points, which do not always provide enough information to measure complicated, higher-order mode shapes. Additional mode identification probes can be installed to provide more information.

**Model**
A mathematical representation of a system that is designed to mimic cer-

tain important features to help us understand system behavior. The rotor model is an attempt to describe the function of the black box that transforms dynamic forces into vibration. Applying a model beyond the limits expressed in the assumptions will usually lead to error.

**Mode shape**

The three-dimensional vibration pattern of a rotor system in free vibration (*free* mode shape) or due to a dynamic force distribution (*forced* mode shape). For large, distributed systems, the vibration amplitude and phase will change continuously along the rotor, and the machine casing will participate in the vibration in some complicated way. The forced mode shape changes with rotor speed.

**Moment of inertia, mass**

The sum of the products of each differential mass element times the square of its distance from the rotational axis

**NARF**

Acronym for Natural Axial Resonant Frequency; usually refers to axially compliant couplings.

**Nameplate**

A placard on a machine that contains basic information about the machine, such as the manufacturer, model number, rated power, power factor, operating speed, resonance speeds, pressures, temperatures, etc.

**Nameplate resonance**

Normal resonance speed(s) for a particular machine as indicated by the manufacturer on the nameplate. Typically the high-eccentricity natural frequency.

**Natural frequency**

The frequency of free vibration of a mechanical system. Each natural frequency is associated with a particular vibration pattern, called a mode shape, and each mode shape is independent of all the other mode shapes of the system.

See: Damped natural frequency; Undamped natural frequency

**Natural mode**

See: Natural frequency

**Nodal point (node)**

A point of minimum (or zero) shaft deflection in a specific mode shape. Nodal points can change axial location due to changes in speed or Dynamic Stiffness. Vibration on opposite sides of a node is usually 180° out of phase.

**Noise**
Any component of a transducer signal which does not represent the variable intended to be measured.

**Nonlinear function**
A function which cannot be expressed as $a_1x_1 + a_2x_2 + \ldots a_nx_n = y$. Examples of nonlinear functions are of a parabola, $y = x^2$, and a trigonometric function, $y = \sin x$.

**Nonlinear system**
A system with dynamic behavior characteristics that cannot be modeled by linear differential equations. The mathematical model must include nonlinear functions of displacements and/or velocities, which result in a set of nonlinear differential equations. Nonlinear systems can have output that is not proportional to input and output frequencies that include harmonics of the input frequency.
See: Linear system

**Nonsymmetric rotor**
See: Asymmetric rotor

**Nonsynchronous**
A vibration frequency component which is different than shaft rotative speed.

**Nonsynchronous Amplification Factor**
A measure of the susceptibility of rotor vibration response to a *nonsynchronous* harmonic force at a rotor system natural frequency. Typically, this is the ratio of the response amplitude at resonance to the response amplitude well away from any resonance.

**Nonsynchronous Dynamic Stiffness**
The generalized form of Dynamic Stiffness, in which the perturbation frequency is different than the rotative speed. See: Dynamic Stiffness.

**Normal-loose**
A behavior where a rotor, because of high vibration, moves to a position where the average stiffness decreases significantly. A worn bearing or loose support structure can allow normal-loose behavior.

**Normal-tight**
A behavior where a rotor contacts a bearing wall, seal, or some other part for a portion of the vibration cycle (a partial rub), increasing the average stiffness significantly.

**Not-1X**
Vibration data from which the 1X response has been removed, by either notch filtering or waveform subtraction.

**Notch filter**
A filter which has a rejection band extending from a lower cutoff frequency to an upper cutoff frequency. Frequencies within the rejection band are eliminated or attenuated while frequencies outside the rejection band are retained. The opposite of a band-pass filter.

**Nyquist criterion**
See: Aliasing

**Nyquist limit**
The highest frequency that can be reproduced in a digitally sampled waveform without aliasing. The Nyquist limit is one-half of the sampling frequency; thus, to ensure the proper frequency content in a sampled waveform, the sampling rate is typically set to some number greater than twice (for example, 2.2 times) the highest desired frequency.

**Observed damping**
See: Effective damping

**Octave**
A 2 to 1 ratio between two frequencies. An octave higher than a particular frequency is twice the frequency; an octave lower is one-half the frequency.

**Offset balancing**
Balancing used to compensate a repeatable thermal- or strain-induced unbalance. Even generators in new condition can have small asymmetries caused by slots, windings, and cooling gas flow that produce uneven heating and thermal bow. Manufacturers will often offset balance a generator so that it will run smoothly at full load. However, changes in load or excitation can produce temperature changes that lead to a different thermal bow, causing an increase in 1X vibration.

**Oil whip**
See: Whip

**Oil whirl**
See: Whirl

**Optical pickup**
A noncontacting transducer with an internal LED, typically infrared, and a phototransistor that detects the level of reflected light. It is most commonly used as a temporary Keyphasor transducer, observing a once-per-turn change in shaft reflectivity (a dark or light paint spot or a small strip of highly reflective tape on the shaft).

**Orbit**
The two dimensional path of the shaft centerline. The orbit can be observed when *XY* vibration transducers are connected to an oscilloscope

in $X$ versus $Y$ mode. A Keyphasor dot is usually added to provide phase and precession information. An orbit is a specific form of a Lissajous curve.

**Order lines**

Diagonal lines starting at the origin on a spectrum cascade plot or vertical lines on a waterfall plot that indicate frequencies that maintain a fixed ratio to running speed; for example, 1X, 2X, 0.47X, etc.

**Orientation**

See: Probe orientation

**Orthogonal probes**

Probes whose sensitive axes intersect at right angles and which are in a plane at right angles to the rotor axis.

**Oscillator-demodulator**

See: Proximitor sensor

**Outboard**

When identifying the location of an object on a single machine, it is the side toward the outside of the machine case; for example, the probe is *outboard* of the bearing. When identifying the location of two similar components of a machine case, it is the one away from the driver coupling; for example, the *outboard* bearing of the HP compressor. When there is a chance of ambiguity, the components should be defined and a written record kept.

**Outer race**

In rolling element bearings, the track for the rolling elements in the outer ring of the bearing.

**Outer race ball pass frequency**

In rolling element bearings, the frequency at which rolling elements pass a fixed point on the outer race.

**Out of phase**

Having a 180°, or essentially 180°, phase relationship. Occasionally used to mean not exactly in phase.

**Overdamped**

See: Damping factor

**Parameter**

A rotor system parameter is a property of the system that affects system response. Mass, stiffness, damping, and $\lambda$ are examples of rotor system parameters.

**Partial rub**

Rub in which contact between the rotor and stator is maintained for only part of the vibration cycle.

**Peak**
An amplitude measurement. For a sine wave signal, the amplitude is measured from the average signal level (the middle) to the peak value. For complex signals containing many frequencies, the peak value is found by taking one-half of the peak-to-peak amplitude. Peak is abbreviated pk.

**Peak-to-peak**
An amplitude measurement based on the difference between positive and negative peaks of an electronic signal or dynamic motion. Abbreviated pp.

**Period**
The time required for one complete oscillation or for a single cycle of events. The reciprocal of frequency.

**Periodic vibration**
Oscillatory motion that repeats itself identically at regular intervals.

**Permanent bow**
A bending deformation of the rotor that is not self-reversing without special intervention. For this to be true, the stresses in some part of the rotor must have exceeded the yield strength of the material.

**Perturbation**
Use of an external or internal (such as unbalance) device to apply a known forcing function to a system to study the system characteristics.

**Perturbator**
A mechanism that produces a variable frequency, synchronous or nonsynchronous perturbation of a shaft or casing. The perturbator may apply the force unidirectionally or circularly. A Keyphasor mark on the perturbator provides a phase and speed reference.

**Phase lag angle**
The timing relationship between two vibration signals, such as a Keyphasor pulse and a vibration signal; also, the phase difference between two signals, such as an input force signal and an output response signal. Phase lag is expressed in degrees of the vibration cycle, where the number of degrees is a positive number between 0° and 360°.

**Pickup**
See: Transducer

**Pipe strain**
Deformation of a machine casing caused by excessive force applied to the casing by the attached piping. This force may be due to poor piping fit or a malfunctioning support system.

**Pitch radius**
In a gear, the radius of the pitch circle, the imaginary circle that rolls without slipping on the pitch circle of the mating gear.

**Pivotal mode shape**
A mode shape where the principal node is more or less in the middle and each end is out of phase. For a rigid body mode, the rotor traces out the shape of two coaxial cones attached at their points, the nodal point. Also used to describe the S-shaped bending mode of a rotor.

**pk**
See: Peak

**pp**
See: Peak-to-peak

**Polar plot**
A graphical format consisting of a center reference point surrounded by concentric circles for plotting the amplitude and absolute phase of a set of vibration vectors. Amplitude is represented by the distance from the center of the plot and phase by the angular position relative to the transducer orientation. A polar plot can be constructed from steady state or transient data.

**Poles**
In control theory, the eigenvalues of the system. They are also known as the characteristic values or roots of the characteristic equation. In an electric motor or generator, locations on opposite sides of a rotor equivalent to the north and south ends of a bar magnet.

**Position angle**
The angle between an arbitrary reference line drawn through the center of a bearing (typically vertical down in a horizontal machine) and the line connecting the bearing and shaft centers, measured in the direction of rotor rotation.

**Power factor**
The sine of the phase angle between the instantaneous voltage and the resulting current. The apparent power multiplied by the power factor equals the actual power.

**Precession**
The lateral, two-dimensional motion or vibration of the rotor geometric center in the *XY* plane, which is perpendicular to the axis of the rotor. Precession is also known as orbiting or vibration. Precession is completely independent of rotation.

**Precision**
A measure of the random error of a repeated measurement. Assuming no change in the parameter being measured, a measuring instrument is said to have high precision (repeatability) if repeated measurements of the parameter are clustered closely together.

**Pressure angle**
In gears, the angle between the normal to the tooth profile at the pitch circle and the tangent to the pitch circle at that point.

**Pressure wedge**
A region of local high pressure that forms in a fluid-film bearing or seal, produced when a shaft rotates at an eccentric position. The shaft draws the circulating fluid into an area of reduced clearance; some of the fluid escapes axially, but the fluid near the axial center of the bearing produces a circumferential pressure distribution that supports the rotor and moves it to the side. This pressure distribution is the primary means of rotor support in internally pressurized (hydrodynamic) journal bearings.

**Preswirl**
The injection of a fluid into a bearing or seal, intentionally or unintentionally, with a tangential angular velocity component in the direction of rotor rotation. In this case, the value of $\lambda$ can be considerably greater than 0.5.

**Prime Spike**
In the operation of rolling element bearings, a frequency range that encompasses, as a minimum, the ball pass frequencies and their harmonics.

**Probe**
Typically, an eddy current proximity transducer, although often used to describe any transducer.

**Probe gap**
The physical distance (gap) between the face of a proximity probe tip and the observed surface. The distance can be expressed in terms of displacement (mils, micrometers) or voltage (millivolts).

**Probe orientation**
The angular location of a probe, in a plane perpendicular to the rotor centerline, with respect to a unique reference direction. Typically, 0° is at top dead center (vertical or up) for a horizontal machine train and at a compass point or some significant feature for a vertical machine. The angle is measured from 0° to 180° to the left or the right.

**Protection**
See: Machinery protection

**Proximitor sensor**
A Bently Nevada signal conditioning device which sends a radio frequency signal to an eddy current proximity probe, demodulates the response, and provides an output signal proportional to probe gap distance. Also called an oscillator-demodulator.

**Proximity probe**
A noncontacting probe, most commonly an eddy current probe, which measures the displacement of an observed surface relative to the probe mounting location.

**Q**
See: Synchronous Amplification Factor

**Quadrature**
In Dynamic Stiffness, the component of a response that lies in a direction perpendicular to the applied force vector. See: Dynamic Stiffness, Quadrature.

**Radial**
A direction on a machine which is perpendicular to the shaft centerline.

**Radial position**
See: Average shaft position

**Radial vibration**
Shaft dynamic motion or casing vibration, sometimes called lateral vibration, measured in a direction perpendicular to the shaft axis.

**REBAM**
The acronym for Rolling Element Bearing Activity Monitor, which is a Bently Nevada system for monitoring and analyzing the performance of rolling element bearings using eddy current transducers and MicroPROX sensors.

**Reference data**
Recorded machine measurements, at defined conditions, for comparison and correlation with measurements taken at a later time or at different operating conditions.

**Relative phase**
The timing difference between equivalent events (such as positive peaks or zero crossings) on two signals filtered to the same frequency. Relative phase is expressed in degrees of the vibration cycle, "Signal A leads (or lags) signal B (the reference signal, in this case) by (the number of) degrees."

**Relative vibration**
Vibration measured relative to a chosen reference, which may also be vibrating. Proximity probes measure shaft dynamic motion and position relative to the probe mounting, usually the bearing or bearing housing.

**Repeatability**
The quality of a transducer or readout instrument to produce measurements of the same variable, under the same conditions, with small disper-

sion (random error). Such a device is said to have high repeatability (precision).

**Resolution**
The smallest change in a signal that will produce a detectable change in an instrument output. In a spectrum, the line width, equal to the span divided by the number of lines in the spectrum.

**Resonance**
The increase in amplitude and change in phase that occurs when the frequency of an applied periodic force coincides with a natural frequency of the system. During startup or shutdown, a resonance typically is identified by an amplitude peak, accompanied by a significant change of phase.

**Restoring force**
A force that acts in the direction of the original position, attempting to restore the system to equilibrium.

**Reverse component**
See: Forward and reverse vectors

**Reverse (backward) precession**
Precession of a shaft in the opposite direction of its rotation.

**Rigid rotor**
A rotor where the bending of the rotor is relatively small compared to the motion of the rotor in the bearings, produced by a high ratio of rotor stiffness to support stiffness. In many cases, the very lowest modes are an in-phase (cylindrical) or an out-of-phase (pivotal) rigid body mode.

**rms**
See: Root-mean-square

**Rolling element bearing**
A bearing in which the low friction property derives from mechanical rolling (usually with fluid lubrication), using ball or roller elements between two constraining rings.

**Rolling elements**
In a rolling element bearing, the rollers or balls that support the rotating load of a shaft.

**Root locus**
A graphical presentation of the eigenvalues (roots) of a system as a system parameter is changed. It gives us a qualitative description of the system's transient performance and stability and provides a quantitative measure of the damped natural frequency, Threshold of Instability, Synchronous Amplification Factor, and Margin of Stability.

**Root-mean-square (rms)**
A measure of amplitude that is the square root of the arithmetic mean of a

set of squared instantaneous values. For sine waves only, rms = $1/\sqrt{2}$ x (pk amplitude) = 0.707 x (pk amplitude) = 0.354 x (pp amplitude).

**Roots**

The characteristic values, eigenvalues or, in control theory, poles of a mechanical system.

**Rotating stall**

Partial flow separation in a compressor that creates one or more regions of impaired flow. These regions rotate around the rotor disk or diffuser at a subsynchronous frequency.

**Rotation**

Rotation is the angular motion of the rotor about its axial geometric center, or shaft centerline.

**Rotative Speed**

See: Shaft rotative speed

**Rotor**

The rotating part of a machine. A rotor usually consists of an assembly of components attached to a shaft or it can consist of a massive shaft.

**Rotor mode**

The state of rotor vibration identified by several characteristics: natural frequency, hierarchy (first, second, third, etc.), deflection shape (rigid or flexible), or end-to-end relative phase (in phase or out of phase).

**Rotor position angle**

See: Position angle

**Rotor system**

The rotor system includes all parts of the machine that are involved with vibration. This includes the shaft with any attached disks, the bearings that support the shaft, the structures that support the bearings, the machine casing, the foundation system, coupled machines, and attached piping systems or unsecured cabling. The rotor system can also include all of the plant equipment that is involved in the process in which the machine is imbedded.

**Rotor vibration region**

A frequency range which includes the principal frequency components of rotor vibration: typically, from ¼X to approximately 3X.

**RTD**

The abbreviation for Resistance Temperature Detector; a sensor which measures temperature through a change in resistance.

**Rub**

Contact between the rotating and stationary parts of the machine. Rub is

always a secondary effect caused by a malfunction that reduces the available clearance between the rotor and the stator.

**Runout**

See: Electrical runout, Mechanical runout

**Runout compensation**

Correction of a transducer output signal for the error resulting from mechanical or electrical runout.

**SAF**

See: Synchronous Amplification Factor

**Scalar**

A parameter or variable possessing only magnitude. Unlike a vector, a scalar has no direction information. For example, voltage, temperature, distance, and speed are scalars.

**Scale factor**

The change in output per change in input of a transducer. Also, the factor by which a signal is amplified or attenuated to meet the input requirements of an instrument. Also called sensitivity.

**Seismic transducer**

A vibration transducer that measures the absolute vibration of an object relative to an internal inertial reference. Examples are accelerometers and velocity transducers.

**Self-excited vibration**

The periodic vibration of a system at a natural frequency that occurs when an internal feedback mechanism of the system converts a supply of external energy, which may not be oscillatory in nature, to an oscillatory force.

**Sensitivity**

The ratio of the change in the output to a change in the input. A typical sensitivity for a proximity probe transducer is 200 millivolts per mil (7.84 volts per millimeter). Also called scale factor.

**Service factor**

An additional load capability that a machine (typically electric motors and gearboxes) can tolerate for brief periods without damage. The rated power is multiplied by the service factor to arrive at an allowable, brief overload capacity.

**Shaft**

A rod or a beam, usually with circular cross section, on which a disk may be carried, either integrally or by shrink fitting. Shafts rotate about their long axis.

**Shaft absolute vibration**

The radial vibration of a rotor shaft measured relative to an inertial, or

fixed, reference frame. If a machine casing were absolutely rigid and motionless, the shaft vibration measured by an eddy current transducer mounted to the case would be shaft absolute. When significant casing vibration exists, the eddy current measurement must be added to the integrated casing vibration to obtain shaft absolute.

**Shaft average centerline plot**

See: Average shaft centerline plot

**Shaft centerline**

The axial geometric centerline of the rotor.

**Shaft relative transducer**

Typically, a proximity probe, mounted on a bearing or bearing housing, that observes shaft motion. The output signal includes the effects of the shaft motion and the transducer motion.

**Shaft relative vibration**

Shaft radial vibration that is measured by a transducer mounted on a structure that can vibrate. The vibration signal will include the effects of transducer vibration.

**Shaft rotative speed**

The frequency at which a shaft is rotating, usually expressed in units of revolutions per minute (rpm). It may also be expressed in radians per second (rad/s).

**Shear**

The parallelogram distortion produced by the relative parallel displacement of two parallel planes in a material.

**Shim**

A metal spacer, preferably stainless steel, placed between the machine foot and the foundation to correct the height of the machine at that location

**Signal conditioner**

A device placed between a signal source and a readout instrument to change the signal. Examples include attenuators, preamplifiers, filters, and signal converters (for changing voltage to current or analog to digital, etc.).

**Signal gain**

The relative attenuation or amplification of a signal, usually presented in dB. Also, the amplification of small electronic signals to match the scale range on instruments, such as FM tape recorders, to improve the resolution of the measurement.

**Signal-to-noise ratio**

The number formed by dividing the magnitude of the signal by the magnitude of the noise present in the signal. A low noise signal has a high signal-to-noise ratio, while a high noise signal has a low signal-to-noise ratio.

**Signature**
All the information necessary to uniquely identify a state. The term is sometimes applied to a vibration frequency spectrum that is specific to a particular machine, component, system, or subsystem at a specific point in time, under specific machine operating conditions. It is used for historical comparison of mechanical condition over the operating life of the machine.

**Sine wave**
The waveform that results from filtering a vibration signal to a single frequency. It is described by the function $y = A\sin(\omega t)$, where $y$ is the instantaneous value of the signal, $A$ is the peak or peak-to-peak amplitude, $\omega$ is the circular frequency in rad/s, and $t$ is time.

**Single amplitude**
(Obsolete) See: Peak

**Skid**
A term usually used to identify a massive, fabricated support made of structural members. On small machines, it can be used without a foundation; on large machines, it may be bolted and grouted to the foundation.

**Slip frequency**
In an induction motor, a measure of how fast the loaded rotor is turning compared to the no load ideal. Slip $= 2(f/p) - \Omega$, where $f$ is the line frequency, $p$ is the number of poles, and $\Omega$ is the rotor speed.

**Slow roll compensation**
Slow roll compensation is the subtraction of slow roll data from vibration data taken from the same measurement point in order to remove the effects of mechanical and electrical runout. The resultant vibration data will only reflect the dynamic response of the rotor. Compensation can be performed using vectors or waveforms.

**Slow roll runout**
A combination of electrical and mechanical runout measured at a slow speed where the dynamic effects associated with rotation are negligible. Normally it can only be measured with a proximity probe because of the low frequency of the signal.

**Slow roll speed**
A rotor speed low enough that the dynamic effects of unbalance are negligible. Typically, slow roll speed is less than 10% of the first balance resonance speed.

**Slow roll vector**
A constant or very slowly varying component of the vibration vector that represents nondynamic action observed by a transducer. It consists of

thermal, electrical, and mechanical effects, such as a bowed rotor, mechanical or electrical runout, or a coupling problem. The slow roll vector will be different for each measurement transducer.

**Soft foot**

Machine feet that are not coplanar with the soleplate (baseplate) or foundation. When feet are not properly supported, tightening them down will warp the machine casing, changing the alignment of machine components.

**Soleplate**

A steel plate grouted and bolted to a foundation. If a soleplate is used, the machine is bolted to the soleplate. Also called a baseplate.

**Spall**

In rolling element bearings, a flake or chip of metal from a bearing race or rolling element.

**Span**

The range of displayed frequencies in a spectrum.

**Spectrum**

See: Half spectrum

**Spectrum cascade plot**

A series of half or full spectrum plots taken over a range of speeds of a machine, such as during a startup or shutdown, usually at set speed intervals.

**Spectrum waterfall plot**

A series of half or full spectrum plots taken over a range of time, usually during constant speed operation.

***s*-plane**

The *XY* plane of the root locus plot; the coordinates of the plot represent the components, $\gamma$ and $\omega_d$, of the complex eigenvalues, or roots, $s$, of the characteristic equation.

**Split resonance**

Two balance resonances that have a similar mode shape, but are separated in frequency.

**Spring stiffness**

The series and parallel combination of all the springlike elements of the rotor system, including shaft bending stiffness, bearing fluid-film direct stiffnesses, and any support and foundation stiffnesses; usually represented by the symbol $K$. When a system is displaced from the equilibrium position, spring stiffness produces a force that is proportional to displacement and directed toward the equilibrium position.

**Stability (Liapunoff definition, stability "in the small")**
The dynamic regime of a physical system is stable if an external, small perturbation of its steady-state regime causes a transient response that decays to the same steady state regime. A system is unstable if an external, small perturbation results in an increasing response.

**Stability (practical definition)**
A system is stable if a transient perturbation does not result in a system response with amplitudes exceeding an acceptable level.

**Stability threshold**
See: Threshold of Instability

**State space matrix**
The matrix resulting from the reduction of a second order system to a system of first order differential equations. The state space matrix produces a set of velocities from a corresponding set of input displacements. The eigenvalues of the system can be calculated from the state space matrix.

**Static data**
Data which represents a processed value of some measured parameter. Examples of static data include vibration amplitude, phase lag angle, frequency, average shaft position, shaft rotative speed, time, date, and monitor alarm and OK status. See: Dynamic data

**Stator**
The stationary part of the machine in which the rotor rotates.

**Steady state**
Operation at a constant shaft rotative speed.

**Steam whip**
See: Whip

**Stiffness axes**
The resolution of an anisotropic, radial stiffness distribution into orthogonal major and minor stiffnesses. The orientation of the major and minor stiffnesses identify the directions of the stiffness axes. The major axis of the low-speed, 1X orbit tends to align with the minor stiffness axis.

**Strain gauge**
A transducer attached to a deformable solid which reacts to changes in strain, typically through changes in resistance.

**Stress**
The force acting on a body per unit area, usually measured in terms of $lb/in^2$ or $N/m^2$.

**Structural resonance**
Excitation of a natural frequency of an attached structural component, often by rotor vibration. A structural resonance can appear as a small loop on a polar plot.

**Subcritical damping**
See: Damping factor

**Subharmonic**
A component of a vibration signal that is a submultiple (integer fraction) of a fundamental frequency, for example ½, ⅓ etc.

**Subsynchronous**
A component of a vibration signal which has a frequency less than shaft rotative speed.

**Supercritical damping**
See: Damping factor

**Superharmonic**
A component of a vibration signal that is an integer multiple, $n > 1$, of a fundamental frequency.

**Supersynchronous**
A component of a vibration signal which has a frequency greater than shaft rotative speed.

**Support structure**
The entire support system for a machine, usually including a foundation, grout, soleplate, pedestals, shims, and bolts. This structure transmits machinery and related loads to the ground and may transmit external loads from the ground to the machine.

**Surface-to-core thermal gradient**
A temperature gradient across a rotor cross section that appears because of heating or cooling of the outer surface of the rotor.

**Sweep frequency filter**
A type of band-pass filter which is automatically swept (tuned) through a frequency range of interest. An instrument which incorporates this type of filter can be used to generate a vibration frequency spectrum.

**Swirl ratio**
See: Fluid Circumferential Average Velocity Ratio

**Symmetric rotor**
A rotor whose cross section, when rotated about its geometric center, appears the same at all angular orientations; for example, a circular cross section.

**Symmetry**
A geometric property of an object such that, when it is subjected to certain operations (rotation, inversion, etc.), the object appears the same as before the operation.

**Synchronous**
The component of a vibration signal that has a frequency equal to the shaft rotative speed (1X).

**Synchronous Amplification Factor (SAF)**
A measure of the sensitivity of a rotor system to unbalance when shaft rotative speed is equal to a rotor system natural frequency. A high Synchronous Amplification Factor produces relatively high vibration at a resonance and indicates low system Quadrature Dynamic Stiffness; a low amplification factor produces relatively low vibration at a resonance and indicates high system Quadrature Dynamic Stiffness. Also called Q.

**Synchronous Dynamic Stiffness**
The ratio of the applied, synchronous dynamic force to the dynamic response of a mechanical system. In rotor systems, the synchronous force is usually rotating unbalance. See: Dynamic Stiffness

**Synchronous precession**
Rotor precession with frequency equal to rotor speed (1X).

**Synthetic gas (Syngas)**
An intermediate product consisting primarily of methane.

**System**
The interacting combination of rotors, bearings, casing, and surrounding fluid. The system responds as an entity to dynamic excitation.

**System mode shape**
The complicated vibration pattern produced by all the elements of a vibrating system.

**Tangential stiffness**
A springlike stiffness produced by the converging fluid pressure wedge when the rotor is at an eccentric position in a bearing or seal. It can be modeled as $+jD\lambda\Omega$, where $D$ is the damping, $\lambda$ is the Fluid Circumferential Average Velocity Ratio, $\Omega$ is rotor speed, and $j=\sqrt{-1}$. The tangential stiffness acts at 90° to the displacement vector, in the direction of rotation.

**Temperature stratification**
A temperature gradient that forms in the casing of a steam turbine after shutdown, with higher temperatures at the top of the casing.

**Temporary bow**
A rotor bow that does not involve plastic deformation of the rotor material; the rotor returns to its original shape when the cause of the bow disappears.

**Thermal bow**
A rotor bow that forms in response to a temperature gradient in the rotor. When the temperature returns to equilibrium, the bow disappears.

**Thermal fatigue**
Fatigue cracks that form as a result of thermal shock during startup or shutdown.

**Thermal shock**
A sudden temperature change on the outer surface of an object that produces surface temperatures significantly different from those in the interior. The thermal expansion or contraction of the surface layers are constrained by the underlying material; thus, thermal shock can generate very large stresses capable of producing cracks. Sudden surface heating will produce tensile stresses in the interior; sudden surface cooling will produce tensile stresses on the surface.

**Thermocouple**
A temperature transducer comprised of two dissimilar metal wires which, when heated or cooled, produce a proportional change in electrical potential at the point where they join.

**Threshold**
The level at which a trigger or other function is initiated.

**Threshold of Instability**
The speed, $\Omega_{th}$, at or above which a machine will experience a fluid-induced instability.

**Thrust position**
See: Axial position

**Timebase Plot**
A presentation of the instantaneous values of a signal as a function of time: a waveform. The vibration waveform can be observed on an oscilloscope in the time domain.

**Torque**
A measure of the tendency of a force to cause rotation; a force couple, equal to the force multiplied by the perpendicular distance between the line of action of the force and the center of rotation.

**Torque, average**
The average value of the torque applied to a rotor in order to sustain rotational speed, angular acceleration, or load requirements.

**Torque, dynamic**
The instantaneous value of the time-varying component of the torque applied to a rotor, typically resulting from a variation in driving load or torque.

**Torque, static**
The force times the perpendicular distance between the line of action of the force and the center of rotation (moment) as applied to a structure (nonrotating).

**Torsional vibration**
Oscillation of the angle of twist of a shaft, typically measured in tenths of degrees peak-to-peak.

**Track**
To maintain a specific frequency ratio with another vibration component, usually 1X, which is changing frequency. For example, ½X rub can *track* running speed over a limited speed range during startup or shutdown.

**Tracking filter**
See: Vector filter

**Transducer**
A device for converting the magnitude of one quantity into another quantity for the purpose of measurement. The second quantity often has units of measure different from the first. For example, displacement vibration transducers convert mechanical position into a voltage signal proportional to displacement.

**Transient**
Changing; commonly, when machine speed is changing, usually during startup or shutdown. Mathematically, the dynamic response (*transient* response) associated with free vibration that usually decays with time. See: Growth/decay rate

**Transient vibration**
The temporary vibration of a mechanical system, such as a machine, associated with instantaneous changes in machine condition, such as speed, load, etc.

**Translational mode shape**
See: Mode shape

**Transmission device**
A machine or machine part that connects a driver to a driven machine. Transmission devices include gearboxes, couplings, and clutches.

**Transverse sensitivity**
See: Cross-axis sensitivity

**Trend data**
Periodically sampled data that is used for the purpose of observing changes in machine behavior as a function of time.

**Trend interval**
The time period between consecutive data points on a trend plot.

**Trend period**
The time frame (beginning to end of data) of a trend plot.

**Trend plot**
A presentation in rectangular (Cartesian) or polar format of a measured variable versus time.

**Trigger**
Any event which is used as an initial timing reference. A trigger signal for a digital vector filter is a Keyphasor pulse, which provides a reference for measuring the amplitude and phase lag angle.

**Trip**
The shut down, often automatic, of a machine, based on predetermined levels of measured parameters or observed conditions.

**Trip multiplier**
In a monitor system, the function that temporarily increases the alarm (Alert and Danger) setpoint values by a specific multiple (usually two or three). This function is normally applied by manual (operator) action during startup to allow a machine to pass through high vibration speed ranges without monitor alarm indications. Also called setpoint multiplier.

**TSI**
Abbreviation for Turbine Supervisory Instrumentation. A TSI system is a continuous monitoring system generally used on turbogenerator sets. The TSI system consists of measurement transducers, monitors, interconnecting wiring, and a microprocessor-based monitoring, data acquisition, and processing system.

**Unbalance**
Unequal radial mass distribution in a rotor; a shaft condition where the mass centerline (principal axis of inertia) does not coincide with the geometric centerline.

**Unbalance response**
The amplitude and phase of rotor synchronous precession at a given speed, caused by the dynamic forcing action of rotating unbalance.

**Undamped natural frequency**
The natural frequency of a mechanical system without the effects of damping.

**Underdamped**
See: Damping factor

**Unfiltered Data**
See: Direct data

**Unstable**
See: Stability

**Valve position**
A measurement of the position of the process inlet valves on a machine, usually expressed as a percentage of the valve opening; zero percent is fully closed, 100 percent is fully open. The measurement is usually made with an LVDT or rotary transducer as part of a Turbine Supervisory Instrumentation system.

**Vane passing frequency**
A potential vibration frequency on vaned-impeller compressors, pumps, and other machines with vaned rotating elements. It is the number of vanes (on an impeller or stage) multiplied by shaft rotative speed.

**Vector**
A quantity which has both magnitude and angular orientation.
See: Vibration vector

**Vector filter**
An electronic instrument that measures the amplitude and phase lag angle of, primarily, 1X and 2X components of a vibration signal.

**Velocity**
The rate of change of displacement with time. Typical units for velocity are in/s or mm/s peak. Velocity measurements are used to evaluate machine housing and other structural response characteristics. Integration of a velocity signal yields dynamic displacement, but not average position.

**Velocity transducer**
An electromechanical transducer, typically with an internal inertial reference, with an output in units of velocity.

**Velocity profile**
A collection of velocity vectors that describe the velocity distribution of fluid flow in a defined region; for example, the velocity distribution in the lubricating fluid between a rotating journal and the bearing wall.

**Vibration**
The oscillatory (back and forth) motion of a physical object.

**Vibration form**
The appearance of the vibration signal when displayed in various formats. While form can also be seen in timebase and spectrum formats, for rotor shafts, it is best seen in the orbit, which combines the vibration character-

istics of amplitude, phase, and frequency in a two dimensional display; the form is determined by the shape and direction of the dynamic path of the shaft centerline.

**Vibration vector**
For displacement, velocity, or acceleration, a vector that represents the amplitude and absolute phase of a filtered vibration signal. A vibration vector is actually a complex number and can be expressed in rectangular or polar form.

**Virtual probe rotation**
A mathematical transform of vector data from an existing *XY* pair of vibration transducers to produce vectors that would be measured by *XY* probes mounted at some other, arbitrary orientation.

**Waterfall plot**
See: Spectrum waterfall plot

**Waveform**
The instantaneous value of a signal displayed as a function of time. A vibration waveform can be observed on a timebase plot or on an oscilloscope in the timebase mode.

**Waveform compensation**
Point by point subtraction of a slow roll waveform from an unfiltered waveform measured by the same transducer at some other speed.

**Waveform plot**
See: Timebase plot

**Whip**
Subsynchronous, fluid-induced instability vibration that is locked to a bending mode natural frequency of the rotor system. When the source of the instability is in a fluid-film bearing, it is sometimes called *oil whip*; *steam whip* is a fluid-induced instability caused by steam interaction with the rotor. All of these are different manifestations of fluid-induced instability.

**Whirl**
Also called *oil whirl*; a subsynchronous, fluid-induced instability vibration that tracks rotor speed. Typically, whirl occurs at frequencies below ½X. Also, an obsolete term used to describe an orbit.

**Window function**
A digital signal processing technique applied to the sample record to reduce noise in the calculated spectrum. Typically, it gradually and smoothly forces the signal values to zero at the beginning and end of the record. In the frequency domain, a window appears as a band-pass filter.

*X* to *Y*
: The direction of rotation using the rectangular coordinate system as a reference. *X* to *Y* is equivalent to counterclockwise (CCW).

*XY*
: Orthogonal (mutually perpendicular) axes in a rectangular (Cartesian) coordinate system. Usually used to indicate orthogonal radial vibration transducers. Mathematically, *Y* is the vertical axis, and *X* is the horizontal axis, but the axes can be aligned with the *X* and *Y* measurement transducers for convenience.

**XY plot**
: A rectangular graphical format consisting of a vertical (*Y*) axis and a horizontal (*X*) axis. This format is used to graph the results of one variable as a function of another; for example, vibration amplitude versus time (trend), or amplitude versus frequency.

***XY* probes**
: Two radial probes, in the same plane, oriented 90 degrees from each other.

*Y* to *X*
: The direction of rotation using the rectangular coordinate system as a reference. *Y* to *X* is equivalent to clockwise (CW).

**Zero-to-peak**
: See: Peak

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FOR REFERENCE ONLY

# About the Authors

DONALD E. BENTLY is the former Chairman of the Board and Chief Executive Officer of Bently Nevada Corporation, which was sold to GE Power Systems in 2002. He is now Chairman and Chief Executive Officer of Bently Pressurized Beariing Company. Mr. Bently pioneered the successful commercial use of the eddy current proximity transducer to measure vibration and other critical parameters in rotating machinery. His visionary work in this area gave rise to an entire industry surrounding the use of vibration instrumentation to protect and diagnose machinery. His active research of rotor dynamics has allowed him to make significant theoretical and practical contributions in this field. He is a globally recognized authority on these subjects and has authored or co-authored more than 140 papers. He has received numerous awards for his work. Mr. Bently earned his Bachelor's degree in Electrical Engineering (with Distinction) in 1949, followed by his Master's degree in 1950 from the University of Iowa. In 1987 he was awarded an honorary Doctorate in Engineering from the University of Nevada, Reno, and an honorary A.A. degree from Western Nevada Community College in 1998.

CHARLES T. HATCH joined Bently Nevada Corporation in 1989. He worked in Custom Products Engineering for several years before joining Bently Rotor Dynamics Research Corporation, where he performed research in Dynamic Stiffness, Root Locus, and Virtual Probe Rotation. He also served in the Technical Training Department as an instructor in the Machinery Diagnostics and Advanced Machinery Dynamics seminars. He holds B.S. and M.S. degrees in Mechanical Engineering from the University of California at Berkeley.

BOB GRISSOM joined Bently Nevada Corporation in 1978. He was manager of the Customer Training Department for several years before joining Bently Rotor Dynamics Research Corporation (BRDRC) as Technical Editor. Bob was also the Technical Editor for the ORBIT magazine for three years. Bob earned a B.A. in Physics from the University of California, Irvine, and is a member of Sigma Pi Sigma and IEEE.

# About Bently Nevada and Bently Pressurized Bearing Company

BENTLY NEVADA CORPORATION BEGAN in 1955 as Bently Scientific Company in Berkeley, California. Don Bently formed his own company to design, build, and sell his eddy current "distance detectors" via mail order. His conversion of the basic transducer system design from vacuum tubes to transistors made it robust and cost-effective for industrial applications.

In 1961, Bently Scientific moved to its present-day location in Minden, Nevada, a small community 45 miles south of Reno, and changed its name to reflect its new home. The proximity probe soon found its niche as a sensor for directly observing the position and vibration of rotating shafts inside industrial machinery, and Bently Nevada's business grew exponentially during this period.

In the mid-1960s, Bently Nevada began offering monitoring systems as well as transducers. The monitors accepted inputs from Bently transducers, provided continuous monitoring of vibration and position, and compared monitored parameters against user-established alarm limits. In the years since, the number of monitored parameters has grown to encompass rotative speed, phase, temperature, thermal expansion, valve position, pressure, and numerous other machine-related conditions.

In the 1970s, Bently Nevada developed a services organization to diagnose machinery problems as well as install, calibrate, and repair instrumentation. Bently also began offering training to customers on the fundamentals of vibration, the use and calibration of instruments, and machinery diagnostics.

The 1980s saw the establishment of Bently Rotor Dynamics Research Corporation, founded to conduct fundamental research, led by Dr. Agnes Muszynska for many years. Bently Nevada also introduced computerized, online, condition monitoring software as part of a growing emphasis on machinery protection and proactive management of machinery through the use of real-time data.

The 1990s brought further refinements in Bently Nevada's capabilities and scope, adding thermodynamic condition monitoring capabilities. At the same time, a significant growth in its services organization took place to address the shrinking staff of machinery engineers in many customers' plants and to provide system integration and turnkey project management expertise.

In 2002, Bently Nevada Corporation was purchased in a friendly acquisition by GE Power Systems. The purchase of Bently Nevada enabled Donald E. Bently to focus on another area of rotor dynamics that he believes is fertile ground for development. That area is bearings. Mr. Bently formed Bently Pressurized Bearing Company in 2002 in order to develop new bearing technology that gives machinery engineers greater ability to control machine condition and behavior.

The new millennium sees Mr. Bently continuing his leadership position in pioneering innovations to improve rotating machinery performance.

PT. EMOMI
DOC. ROOM

# Fundamentals of Rotating Machinery Diagnostics

DONALD E. BENTLY WITH CHARLES T. HATCH; Edited by BOB GRISSOM

A practical course in the fundamentals of machinery diagnostics for anyone who works with rotating machinery, from operator to manager, from design engineer to machinery diagnostician. This comprehensive book thoroughly explains and demystifies important concepts needed for effective machinery malfunction diagnosis:

VIBRATION FUNDAMENTALS: vibration, phase, and vibration vectors.

DATA PLOTS: timebase, average shaft centerline, polar, Bode, APHT, spectrum, trend, XY, and the orbit.

ROTOR DYNAMICS: the rotor model, dynamic stiffness, modes of vibration, anisotropic (asymmetric) stiffness, stability analysis, torsional and axial vibration, and basic balancing. Modern root locus methods, (pioneered by Walter R. Evans) are used throughout this book.

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